Chapter 1

BASIC CONCEPTS IN SET THEORY

1.1 Sets and Elements

Set theory is a fundamental concept of modern mathematics. All other notions of mathematics can be built up based on the notion of set. A set is a well defined collection of objects. Examples are; a set of vowels, a set of integer, a set of 100 level students of Federal University, Oye Ekiti, Ekiti State in a particular session, a set of approved secondary schools in Kwara State, a set of consonants, and so on. The objects making up the set are called the elements, or members of the set. The characteristics under consideration must be common to the elements of a particular set. That is, it should be straightforward to determine whether or not an object is a member of a given set. Sets are generally denoted with upper case(capital) letters **A, B, C,...** while the arbitrary elements are denoted with lower case (small) letters **a, b, c, ...** inside a special bracket called curly bracket. Two successive elements of a set are separated with a comma. For example, set of vowels is denoted as $V = \{a, e, i, o, u\}$. If **X** is a set and **x** is its element, we say that **x** belongs to **X**, and we write $x \in X$. If x does not belong to X, we write $x \notin X$. A set **X** is said to be finite if it has no elements, or if it contains countable number of distinct elements and the process of counting stops at a particular number, say m. The number of elements in **X** is called the cardinality of the set, and it is denoted as $n(\mathbf{X}) = m$. When $m = 0$, set **X** is said to be empty and **X** is called an empty or a null or a void set. This is denoted as $\{\ \}$ or $\emptyset$. A set with only one member is called a singleton set. A set whose elements can not be counted is called an infinite set.

Here are some frequently used sets in mathematics;

(i) $N = \{x : x = 1, 2, 3, 4, ...\}$ - Set of Natural Numbers.

(ii) $Z = \{x : x = ... -4, -3, -2, -1, 0, 1, 2, 3, 4, ..\}$ - Set of Integers

(iii) $Z^+ = \{x : x = ...1, 2, 3, 4, ..\}$ - Set of Positive Integers

(iv) $Z^- = \{x : x = ... -4, -3, -2, -1\}$ -Set of Negative Integers

(v) $Q = \{x : x = \frac{a}{b} : a,\ b, \in Z\}$ -Set of Rational Numbers

(vi) $Q^+ = \{x : x = \frac{a}{b} : a \text{ and } b, \in Z^+,\ \text{or } a \text{ and } b, \in Z^-\}$ -Set of Positive Rational Numbers

(vii) $Q^- = \{x : x = \frac{a}{b} : a \text{ or } b, \in Z^-\}$ -Set of Negative Rational Numbers

(viii) $R = \{x : x = \text{is a real number }\}$ -Set of Real Numbers

(ix) $C = \{x : x = a + ib, a, b \in R, i = \sqrt{2}\}$ -Set of Complex Numbers.

1.2 Sets Specifications

The four major ways of specifying sets are as follow:

(i) List notation

(ii) Abridge list notation

(iii) Predicate notation

(iv) Recursive rules notation

(i) **List Notation**: List notation is suitable for finite set of small cardinality. The elements are listed with two successive elements separated by a comma in an enclosed braces: Examples: $\{a, e, i, o, u\}$, $\{1, 3, 7, 11, 17\}$, $\{Ade, Ada, Paulsteven, Musa, Uche\}$.

(ii) **Abridge List Notation**: Abridge list notation is suitable for both finite set with large elements and an infinite set. For finite sets with large elements, few of the elements are listed with two successive elements separated by a comma. The last element is also indicated. Dots, usually three, are put between the listed elements and the last element in an enclosed braces to indicate that there are other elements not listed. For example $A = \{1, 2, ..., 5000\}$.
For infinite sets, the procedure is the same except that the last element is not known and can not be indicated. For example, $N = \{1, 2, ...\}$.

(iii) **Predicate Notation**: In Predicate notation, property(properties) that is(are) common to the elements of a set are stated. For example,
$P = \{x : x \text{ is a an approved private university in } Nigeria\}$
$M = \{x : x \text{ is a positive integer less that } 40\}$

(iv) **Recursive Rules Notation**: In recursive rule notations, rules(usually three) are given that generate the elements of a set. The first rule gives the basis of recursion, the second generates new element while the third terminates.
For example the set M of all multiples of 5 greater than 10:
(a) $10 \in M$
(b) if $x \in M$, then $x + 5 \in M$.
(c) nothing else belongs to M.
The first rule is the basis of recursion, the second one generates new elements from the elements defined before and the third rule restricts the defined set to the elements generated by rules (a) and (b). (The third rule, though optional, is necessary to be included).

1.3 Subsets

A set A is said to be a subset of another set B if every element in A is also an element in B. This is written as $A \subseteq B$ if and only if $x \in A \Rightarrow x \in B$. If A is a subset of B and $A \neq B$, we say that A is a proper subset of B written as $A \subset B$.
For example, let $A = \{1, 3, 7, 11, 17\}$ and $B = \{1, 2, 3, 47, 11, 17, 18, 19\}$. It is easy to see that every element in A is also an element in B. A is a subset of B. In fact, A is a proper subset of B since B contains some other elements apart from the ones in A. Therefore, $A = \{1, 3, 7, 11, 17\} \subset B = \{1, 2, 3, 4, 7, 11, 17, 18, 19\}$.
If A is a subset of B, then B is a superset of A. This is denoted by $B \supseteq A$.
A is not a subset of B if not every element in A is an element in B. This is written as $A \not\subseteq B$.
Two sets A and B are equal if and only if $A \subseteq B$ and $B \subseteq A$. That is two sets are equal if they contain the same elements.

1.4 Power Set

The power set of a set A, denoted as $P(A)$ is the set of all possible subsets of A. For example, consider the set $A = \{1, 2, 3, 4\}$, $P(A) = \{\{\}, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}, \{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}, \{2, 3, 4\}, A\}$
Note that if A has n elements, power set of A has 2^n elements.

1.5 Universal Set and Complement of a Set

Sometimes, it is necessary to consider one or more sets under consideration as belonging to a larger set. For example, if we are talking of set of students in secondary schools in Nigeria, set of students in polytechnics in Nigeria, set of students in Universities in Nigeria. Set of students in Nigeria is the largest set that contains all these mentioned sets.
A set that contains all other sets under consideration is called a **Universal Set**. This is denoted by U or E.
If A is any set, with some universal set U defined, the complement of A, normally written as A' or A^c, is defined as to be all those elements that are not members A but are present in the universal set U.
For example, if universal set $U = \{1, 2, 3, 4, 7, 11, 13, 17, 19\}$ and $A = \{1, 3, 7, 11, 17\}$ then $A' = \{2, 4, 13, 19\}$.

1.6 Set Operations

There are two basic operations defined on sets. These are set union and set intersection. Definition of other operations that make use of these basic operations are possible.

(i) **Union of Sets** : The Union of two sets A and B written as $A \cup B$ is defined as:

$$A \cup B = \{x : x \in A \text{ or } x \in B\}$$

For example, if A and B are sets given as;

$A = \{1, 3, 4, 5\}$ and $B = \{o, p, q, 3, 4\}$ then $A \cup B = \{1, 3, 4, 5, o, p, q\}$.

$A \cup B$ contains the elements that are either in set A or set B or in both sets A and B. The union of three or more sets can also be defined accordingly.

(ii) **Intersection of Sets** : The Intersection of two sets A and B written as $A \cap B$ is defined as:

$$A \cap B = \{x : x \in A \text{ and } x \in B\}$$

$A \cap B$ contains the elements that are in both sets A and B.

For example, if A and B are sets given as;

$A = \{1, 3, 4, 5\}$ and $B = \{o, p, q, 3, 4\}$ then $A \cap B = \{3, 4\}$

Note that union and intersection of more than two sets are defined in similar ways. That is for sets A, B, C, D, ...

$$A \cup B \cup C \cup D \cup ..... = \{x : x \in A \text{ or } x \in B \text{ or } x \in C \text{ or } x \in D.....\}$$

and

$$A \cap B \cap C \cap D \cap ..... = \{x : x \in A \text{ and } x \in B \text{ and } x \in C \text{ and } x \in D.....\}$$

(iii) **Sets Difference**: Set difference is another operation defined on sets that could otherwise be defined by making use of basic operation of intersection and compliment of a set. The set difference of a set A relative to another set B denoted by $A - B$, read as A difference B is defined as

$$A - B = \{x : x \in A \text{ and } x \notin B\}.$$

equivalently,

$$A - B = \{x : x \in A \cap B^c\}.$$

This operation involves removing from set A those elements it has in common with set B to get $A - B$. . It is otherwise called the relative complement of A with respect to B. For example. if $A = \{l, m, n, o, p, q, r, s\}$ and $B = \{h, i, o, p, q, r, s\}$. then $A - B = \{l, m, n, \}$ and $B - A = \{h, i, \}$.

(iv) **Symmetric Difference**: The symmetric difference of two sets A and B is the set of elements that are in one and only one of the sets. The symmetric difference of A and B is written $A \triangle B$. This is defined in mathematical notation as:

$$A \triangle B = \{(A - B) \cup (B - A)\}$$

For example, if $A = \{l, m, n, o, p, q, r, s\}$ and $B = \{h, i, o, p, q, r, s\}$, then $A - B = \{l, m, n, \}$ and $B - A = \{h, i, \}$.
Therefore,

$$A \triangle B = \{(A - B) \cup (B - A)\} = \{l, m, n, \} \cup \{h, i, \} = \{l, m, n, h, i\}$$

Notice: Recall that the cardinality of a set A, $n(A)$, is the number of elements in A.

If A, B, and C are subsets of a universal set U, then,

$$n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(A \cap C) - n(B \cap C) + n(A \cap B \cap C)$$

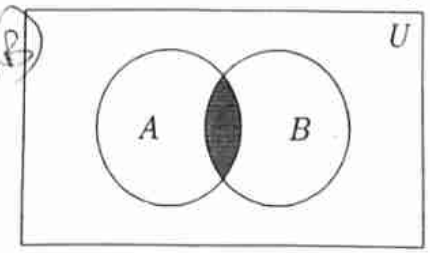
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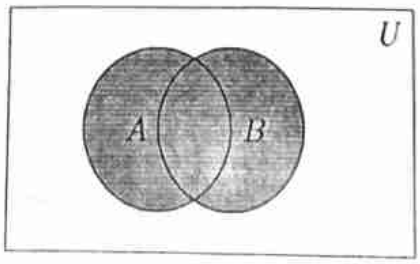
$$n(U) = n(A \cup B \cup C) + n(A \cup B \cup C)^c.$$

Note also that $n(A - B) = n(A) - n(A \cap B)$.

1.7 Venn Diagrams

A Venn diagram is the use of diagram to represent set. It is a way of depicting the relationship between sets. Each set is shown as a circle and circles overlap if the sets intersect.
Venn diagrams for the intersection and union of two sets are shown below





The shaded parts of the diagrams represent the intersection and union respectively. The rectangles enclosing the diagrams indicate that A and B are subsets of the Universal set U.

Class Activity

Let A, B, C be subsets of a reference set U, use Venn diagram to indicate the following sets:

(a.) $A \cup B \cup C$.

(b.) $A \cap B \cap C$.

(c.) $A \cap B \cap C^c$

(d.) $A \cap B \cap C$

1.8 Properties of Set Operations

The following properties are satisfied by set operations;

(1.) (a) $X \cup X = X$ (b) $X \cap X = X$ - Idempotent Laws

(2.) (a) $X \cup Y = Y \cup X$ (b) $X \cap Y = Y \cap X$ - Commutative Laws

(3.) (a) $(X \cup Y) \cup Z = X \cup (Y \cup Z)$ (b) $(X \cap Y) \cap Z = X \cap (Y \cap Z)$ -Associative Laws

(4.) (a) $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$ (b) $X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$ - Distributive Laws

(5.) (a) $X \cup \emptyset = X$ (b) $X \cap \emptyset = \emptyset$ (c) $X \cup U = U$ (d) $X \cap U = X$ - Identity Laws

(6.) (a) $X \cup X^c = U$ (b) $X \cap X^c = \emptyset$ (c) $(X^c)^c = X$ (d) $X - Y = X \cap Y^c$ - Complement Laws

(7.) (a) $(X \cup Y)^c = X^c \cap Y^c$ (b) $(X \cap Y)^c = X^c \cup Y^c$ – De-Morgans Laws

1.9 Solved Examples

Example 1
Let the universal set $U = \{1,2,3,4,5,6,7,8,9\}$. A, B , C are subsets of the universal sets defined as;

$A = \{1,2,3,4\}$, $B = \{1,3,5,7\}$, and $C = \{3,7,9\}$.

Find; (a)(i) $A \cup B \cup C$, (ii) $A \cap B \cap C$ (iii) B^c (iv) $(A \cup B \cup C)^c$ (v) $(A \cap B \cap C)^c$

(b) What is the cardinality of $(A \cup B \cup C)^c$

Solution
(a) (i) $A \cup B \cup C = \{1,2,3,4,5,7,9\}$,

(ii) $A \cap B \cap C = \{3\}$

(iii) $B^c = \{2,4,6,8,9\}$

(iv) $(A \cup B \cup C)^c = \{6,8\}$

(v) $(A \cap B \cap C)^c = \{1,2,4,5,6,7,8,9\}$.

(b) $n(A \cup B \cup C)^c = 2$.

Example 2

Let P, Q, S be subsets of $\Re$ that define the sets of all possible values of x as follows;

$P = \{x : (x^2 - 4)(x^2 - 9) = 0\}$;

$Q = \{x : (x - 2)(x + 3)(x - 10) = 0\}$;

$S = \{x : (x^2 - 1)(x + 2)(x - 3) = 0\}$.

(a) What types of notations are used for P, Q, and R?

(b) List the elements of the sets

(i) $P \cup Q \cup S$ (ii) $P \cap Q \cap S$ (iii) $P - (Q - S)$; (iv) $(P - Q) - S$

Solution

(a) Predicate notations.

(b) $P = \{-3, -2, 2, 3\}$

$Q = \{-3, 2, 10\}$

$S = \{-2, -1, 1, 3\}$

(i) $P \cup Q \cup S = \{-3, -2, -1, 1, 2, 3, 10\}$

(ii) $P \cap Q \cap S = \{\ \}$

(iii) $P - (Q - S) = \{-3, 2, 10\} - \{-2, -1, 1, 2\} = \{-2, 3\}$

(iv) $(P - Q) - S = \{-2, 3\} - \{-2, -1, 1, 3\} = \{\}$

Example 3

In a group of 190 male students of a Faculty, 100 students play football, 60 play basket ball and 80 play volley ball. If 22 of the students play both football and basket ball, 17 play both football and volley ball, 18 play both basket ball and volley ball. If 7 of the students play non of the three games, how many students play:

(i) all the three games ?

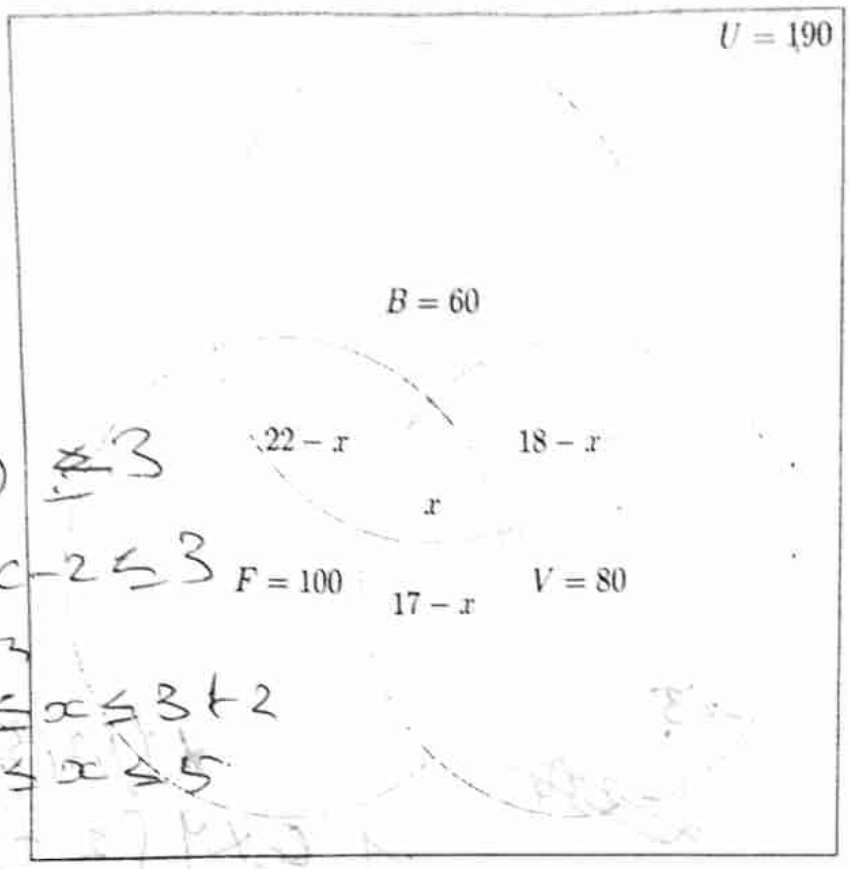
(ii) exactly one of the three games?

(iii) exactly two of the three games?

(iv)) at least one of the three games?

(v) at most two of the three games?

Solution



Let x represent the number of students that play all the three games
From the Venn diagram above,

$$\begin{aligned} F \text{ alone} &= 100 - [22 - x + x + 17 - x] \\ &= 100 - [39 - x] \\ &= 61 + x \end{aligned}$$

$$\begin{aligned} B \text{ alone} &= 60 - [22 - x + x + 18 - x] \\ &= 20 + x \end{aligned}$$

$$\begin{aligned} V \text{ alone} &= 80 - [17 - x + x + 18 - x] \\ &= 45 + x \end{aligned}$$

Hence,

$$\begin{aligned} 61 + x + 22 - x + 20 + x + 17 - x + x + 18 - x + 45 + x + 7 &= 190 \\ \Rightarrow 190 + x &= 190 \\ \Rightarrow x &= 0 \end{aligned}$$

Therefore,
(*i*) no student plays all the three games
(*ii*) For exactly one of the three games, it means playing F alone or B alone or V alone.
That is,

$$\begin{aligned} \textit{exactly one of the three games} &= (61+x)+(20+x)+(45+x) \\ &= 61+20+45 \\ &= 126. \end{aligned}$$

(*iii*) For exactly two of the three games, it means those students playing F and B alone or F and V alone or B and V alone.
That is,
No of students playing exactly two of the three games

$$\begin{aligned} &= F \cap B \textit{ only } + F \cap V \textit{ only } + B \cap V \textit{ only} \\ &= (22-x)+(17-x)+(18-x) \\ &= 22+17+18=57 \end{aligned}$$

(*iv*) For at least one of the three games, it means those students playing exactly one of the games and those playing exactly two and those playing the three games.
That is,

$$\begin{aligned} &= (61+x)+(20+x)+(45+x)+(22-x)+(17-x)+(18-x)+x \\ &= 61+20+45+22+17+18+0 \\ &= 183 \end{aligned}$$

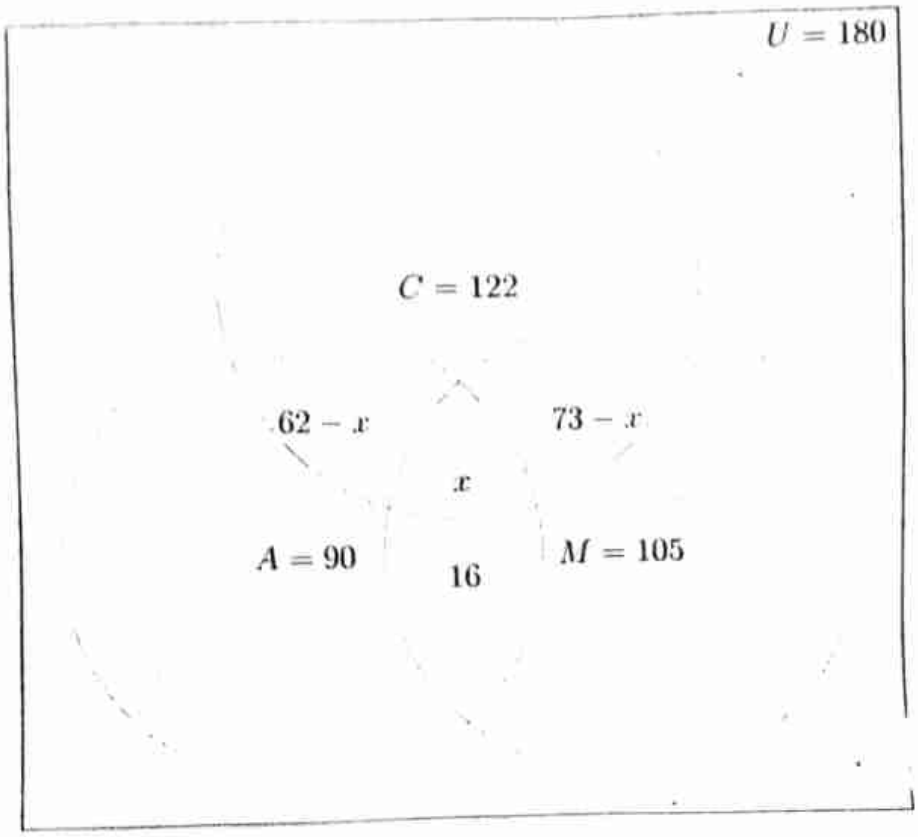
(*v*) For at most two of the three games, it means those students play exactly one of the games and those that play exactly two of the games
That is,

$$\begin{aligned} &= (61+x)+(20+x)+(45+x)+(22-x)+(17-x)+(18-x) \\ &= 61+20+45+22+17+18 \\ &= 183 \end{aligned}$$

Example 4
A sport analyst samples the opinions of 180 sport lovers to know which of these three English Premiership Clubs, Chelsea, Manchester City and Arsenal will emerge as group leader(s) in their UEFA Champions League group draws for the round of 16 in 2014/2015 season. 122 sport lovers tip Chelsea to lead their group, 105 tip Manchester City, while 90 tip Arsenal.

73 tip both Chelsea and Manchester City, 62 tip both Chelsea and Arsenal, while 16 tip both Manchester City and Arsenal but not Chelsea. If the sum of sport lovers that tip all the three clubs to lead their groups and none of these three clubs to lead their groups is 37, how many sport lovers tip at least one of these three clubs.

Solution



Let x represent the number of sport lovers that tip all these three clubs to lead their groups.
From the Venn diagram above,

$$\begin{aligned} C \text{ alone} &= 122 - [73 - x + x + 62 - x] \\ &= 122 - 135 + x \\ &= -13 + x \end{aligned}$$

$$\begin{aligned} M \text{ alone} &= 105 - [73 - x + x + 16] \\ &= 105 - 89 \\ &= 16 \end{aligned}$$

$$\begin{aligned} A \text{ alone} &= 90 - [62 - x + x + 16] \\ &= 90 - 78 \\ &= 12 \end{aligned}$$

Hence,

$$\begin{aligned} x - 13 + 73 - x + 16 + x + 62 - x + 16 + 12 + 37 - x &= 180 \\ \Rightarrow 203 - x &= 180 \\ \Rightarrow x &= 23 \end{aligned}$$

Number of sport lovers that tip at least one of the three clubs is equivalent to those that tip one club only and those that tip two clubs only and those that tip all the three clubs. Therefore,

$$\begin{aligned} \textit{at least one of the three clubs} &= x - 13 + 16 + 12 + 73 - x + 16 + 62 - x + x \\ &= 10 + 16 + 12 + 50 + 16 + 39 + 23 \\ &= 166 \end{aligned}$$

Note that this is equivalent to those sport lovers that did not tip any of the three clubs from all the sampled sport lovers.
That is

$$\begin{aligned} \textit{at least one of the three clubs} &= 180 - 14 \\ &= 166 \end{aligned}$$

1.10 Exercise

1. Three subsets of the whole numbers from 1 *to* 20 are given as:

   $F = \{2, 4, 6, 8, 10, 12, 14, 18\}$, $G = \{1, 2, 3, 4, 5, 7, 9, 11, 16, 18\}$ $H = \{1, 2, 3, 5, 7, 9, 14\}$

   Use a venn digram to show this relationship.
   Find: (*i*) $F \cap G$ (*ii*) $(F \cup G)^c$ (*iii*) $(F \cap H)^c$

2. For any sets X, Y, Z, prove that
   (*a*) $(X \cup Y) \cup Z = X \cup (Y \cup Z)$
   (*b*) $(X \cap Y) \cap Z = X \cap (Y \cap Z)$

3. For any sets X, Y, Z, prove that
   (*a*) $X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$
   (*b*) $X \cap (Y \cup Z) = (X \cap Y) \cup (X \cap Z)$

4. For any sets X, Y, prove that
   (*a*) $(X \cup Y)^c = X^c \cap Y^c$ (*b*) $(X \cap Y)^c = X^c \cup Y^c$

5. Let A and B be two finite sets such that $n(A) = 40$, $n(B) = 38$ and $n(A \cup B) = 60$, find $n(A \cap B)$.

6 If $n(A - B) = 18$, $n(A \cup B) = 70$ and $n(A \cap B) = 25$, find $n(B)$.

7. In a group of 70 students, 37 like Chemistry, 52 like Physics and each student likes at least one of the courses. How many students like both courses?

8. Given that $A = \{a, b, c, d\}$, find $P(A)$, the power set of A.

9. In a survey of 290 newspaper readers, 181 of them read the Daily Times, 142 read Guardian, 117 read punch and each reads at least one of the three papers. If 75 read the Daily Times and the Guardian, 60 read the Daily times and the Punch and 54 read the Guardian and the punch. Draw a Venn diagram to illustrate the information. How many readers read
   (i) all the three papers,
   (ii) exactly two of the papers,
   (iii)exactly one of the papers?

10. In a group of 120 share holders, 72 have shares in First bank, 65 have shares in Eco bank and 53 have shares in GT bank. If 35 of the shareholders have shares in both First bank and Eco bank, 30 have shares in both First bank and GT bank, 21 have shares in both Eco bank and GT bank and each of the share holders has shares in at least one of the three banks. How many shareholders have shares in:
   (i) all the three banks?
   (ii) exactly two of the three banks?

11. A survey of the ratings of 250 subscribers of three leading telecommunication services providers in Nigeria to determine the efficiencies of their customers services in 2014 reveals that 85 subscribers feel provider A offered satisfactory customer services, 75 feel provider B offered satisfactory customers services while 90 subscribers feel C offered satisfactory customer services. 35 subscribers feel providers A and B offered satisfactory customers services, 40 subscribers feel providers B and C offered satisfactory customers services while 45 subscribers feel providers A and C offered satisfactory customers services. If 90 subscribers feel none of these providers offered satisfactory customers services, how many subscribers feel all the three offered satisfactory customers services.

12. It is observed that at the end of 2014/2015 session, 850 stailites of the Faculties of Agriculture, Engineering and Science still have outstanding 100 level first semester courses. Out of these, 119 do not have outstanding in any of CHM101, MTH101 and PHY101. 377 students have outstanding in CHM101, 381 have outstanding in MTH101, and 338 have outstanding in PHY101. 135 students have outstanding in CHM101 and MTH101. 150 have outstanding in MTH101 and PHY101 while 130 have outstanding in CHM101 and PHY101. How many students have outstanding in
   (i) at least one of the three courses CHM101, MTH101 and PHY101.
   (ii) all the three courses.

Assignment

The opinions of 134 lecturers of the a Federal University were sought to decide relevance of the three General Study Courses(GST), A, B. C, to be floated for the students. 59 lecturers feel that course A is relevant. 57 feel that B is relevant while 58 feel that C is relevant.15 lecturers fell that both A and B are relevant, 17 feel that both B and C are relevant while 18 feel that A and C are relevant. If no lecturer feels that all the three courses are not relevant, how many lecturers feel that

(a) all the three GST courses are relevant.

(b) at least two of the GST courses are relevant

(c) A and C only are relevant.

$$n(A \cup B) = n(A) + n(B) - n(A \cap B)$$

$$60 = 40 + 38 - x$$

$$60 = 78 - x$$

$$-78 + 60 = -x$$

$$-18 = -x$$

Chapter 2

REAL NUMBERS AND REAL LINE

2.1 Introduction

Mathematically, the real line or the real number line is the line whose points are the real numbers. The real line is set $\mathbb{R}$ of real numbers which is viewed as a geometric space called the **Euclidean Space of one dimension**. The real line is denoted by the symbol $\mathbb{R}$. The real numbers comprises of all the rational numbers such as the integer(positive and negative natural numbers) and the fraction $\frac{5}{3}$, $\frac{4}{3}$ and all irrational numbers such as $\sqrt{2}$(1.41421356..., the square root of two, an irrational algebraic number) and π(3.14159265..., a transcendental number). Real numbers can be regarded as the points on an infinitely long line called the number line or real line, where the points corresponding to integers are equally spaced.

2.1.1 Properties of real numbers

- All of the numbers that you use in everyday life are real numbers
- Each real numbers corresponds to exactly one point on the number line, and
- Real numbers can be classified as either rational or irrational
- Rational numbers can be expressed as a ratio $\frac{m}{n}$, where m and n are integers and $n \neq 0$.
- The decimal form of a rational numbers is either a terminating or repeating decimal.

Examples:

ratio form	decimal form
$\sqrt{9}$	3.0
$\frac{3}{8}$	0.375
$\frac{3}{7}$	0.428571428571428571 or $\overline{0.428571}$

- A real number that is not rational is irrational.
- The decimal form of an irrational number neither terminates nor repeats
- Examples: $\pi \simeq 3.141592654...$
  $\exp \simeq 2.718281828...$
  $\sqrt{2}$ $\sqrt{3}$ $\sqrt{5}$ $\sqrt{7}$ $\sqrt{11}$ $\sqrt{13}$. They are all prime numbers!

- $\mathbb{Q}$ = Rationals, $\mathbb{I}$=Irrationals, $\mathbb{Z}$=Integers, $\mathbb{N}$=Whole numbers/Natural numbers. $\mathbb{Q} \cup \mathbb{Z} \cup \mathbb{N} \cup \mathbb{I}$.
- The square root of any whole number is either whole or irrational. E.g. $\sqrt{36}$ is a whole number, but $\sqrt{30}$, Since it lies between 5 and 6 must irrational. $\sqrt{30} \simeq 5.477225575...$.

Note: Do not assume that a number is irrational just because it is expressed using the square root symbol. Find its value first. Study and recognize the following numbers: $\sqrt{4}$ $\sqrt{9}$ $\sqrt{16}$ $\sqrt{25}$ $\sqrt{36}$ $\sqrt{49}$ $\sqrt{64}$ $\sqrt{81}$ $\sqrt{100}$ $\sqrt{121}$ $\sqrt{144}$ $\sqrt{169}$

The real number system is an example of a mathematical structure called a **FIELD**. We summarized the properties of a field as follows: For any real numbers a, b, and c

Property	Addition	Multiplication
Commutative	$a+b=b+a$	$a \cdot b = b \cdot a$
Associative	$(a+b)+c=a+(b+c)$	$(a \cdot b) \cdot c$
Identity	$a+0=a=0+a$	$a \cdot 1 = a = 1 \cdot a$
Inverse	$a+(-a)=0=(-a)+a$	If $a \neq 0$ then $a \cdot \frac{1}{a} = 1 = \frac{1}{a} \cdot a$
Distributive	$a(b+c)=ab+ac$	$(b+c)a = ba+ca$

2.2 Indices and Logarithms

2.2.1 Indices

The number given as superscript i.e. a number or variable given as a superscript before a square-root sign showing which root is to be taken is referred to as **an index**. An **index** can also be a subscript or superscript identifying elements a subscripts or superscript numeral that identifies an element or range in a set or sequence.Suppose we have a number b, the product $b \times b \times b \times b \times b \times b \times b \times b = b^8$, and generally, $b \times b \times b \times \ldots \times \ldots \times b = b^m$ is called the mth power of the number b while b is referred to as the base and m is called the **index**.

Some Rules of Indices

Suppose p, q are positive integers.

1. $a^p \times a^q = a^{p+q}$
2. $a^q \div a^p = a^{p-q}$ whenever p, q and $a \neq 0$
3. $(a^p)^q = a^{pq}$
4. $(ab)^q = a^q \times b^q$
5. Whenever b is a non-zero number, then $(\frac{a}{b})^q = \frac{a^q}{b^q}$
6. $a^0 = 1$
7. $a^{-q} = \frac{1}{a^q}$
8. $a^{\frac{1}{q}} = \sqrt[q]{a}$

inverse of of 4
addition
$4+(-4)=0$
$4 \times (1/4) = 1$

So -4 is the additive inverse of 4 while 1/4 is the Multiplicative inverse of 4

$q(b+c) = (qb+qc)$

... in addition is zero while the identity element ...

Example 1: Evaluate $(81)^{-\frac{1}{4}} \times (1024)^{\frac{1}{10}}$
Solution:
$81 = 3^4$ and $1024 = 2^{10}$

$\therefore (81)^{-\frac{1}{4}} \times (1024)^{\frac{1}{10}} = (3^4)^{-\frac{1}{4}} \times (2^{10})^{\frac{1}{10}} = 3^{-1} \times 2 = \frac{2}{3}$

Example 2: Solve $\frac{2^{-a} \cdot 32^{2a-2}}{2^{2a-3} \cdot 64}$
Solution:
$\frac{2^{-a} \cdot 32^{2a-2}}{2^{2a-3} \cdot 64} = \frac{(2^1)^{-a} \cdot (2^5)^{2a-2}}{(2)^{2a} \cdot \frac{1}{2^3} \cdot 2^6} = \frac{2^{-a} \cdot 2^{10a-10}}{2^{2a} \cdot 2^3} = \frac{2^{7a-8}}{2^{2a+3}} = 2^{9a-10-2a-3} = 2^{7a-13}$

Example 3: Solve the equations:
$3^x - 5^{y+1} = 2 \quad 5^y - 3^{x-2} = 2$
Solution:
$3^x - 5^{y+1} = 2$ ——— (i)
$5^y - 3^{x-2} = 2$ ——— (ii)
$3^x - 5^y \cdot 5^1 = 2$
$5^y - 3^x \cdot 3^{-2} = 2$
Let $P = 5^y$, $q = 3^x$
$\therefore q - 5p = 2$ ——— (iii)
$p - \frac{1}{9}q = 2$ ——— (iv)
$q = 2 + 5p$ ——— (v)
substitute q in equation (iv)
$p - \frac{1}{9}(2 + 5p) = 2 \Rightarrow p - \frac{2}{9} - \frac{5}{9}p = 2$
Multiply both sides by 9
$9p - 5p = 18 \Rightarrow 4p = 18 + 2 \Rightarrow 4p = 20 \Rightarrow p = 5$
From (v)
$q = 2 + 5p \Rightarrow q = 2 + 5 \times 5 \Rightarrow = 2 + 25 \Rightarrow q = 27$
when $p = 5$
$5y = 5^1 \Rightarrow y = 1$
From $q = 3^x \quad 27 = 3^x \Rightarrow 3^3 = 3^x \Rightarrow 3 = x \quad \therefore x = 3$
Hence $x = 3$, $y = 1$.

Example 4: Evaluate $\sqrt[3]{9} \times \sqrt[6]{9}$.
Solution:
$\sqrt[3]{9} \times \sqrt[6]{9} = 9^{\frac{1}{3}} \times 9^{\frac{1}{6}} = 9^{\frac{1}{2}} = 3$.

Example 5: Solve the simultaneous equation $3^x = 9^y \qquad 4^{xy} = 2^{x+2}$
Solution:
$3^x = 9^y \qquad (i)$
$4xy = 2x + 2 \qquad (ii)$
from (i) $x = 2y$ and from (ii) , $2xy = x + 2$
$\therefore 2(2y)y = 2y + 2 \Rightarrow 4y^2 = 2y + 2 \Rightarrow 4y^2 - 2y - 2 = 0$

divide through by 2

$$\begin{aligned} 2y^2 - y - 1 = 0 \Rightarrow 2y^2 - 2y - y - 1 &= 0 \\ \Rightarrow 2y(y-1) + 1(y-1) &= 0 \\ \Rightarrow (2y+1)(y-1) &= 0 \\ \Rightarrow 2y + 1 &= 0 \text{ or } y = 1 \\ \Rightarrow 2y &= -1 \text{ or } y = 1 \\ \Rightarrow y &= -\frac{1}{2} \text{ or } y = 1 \end{aligned}$$

when $y = -\frac{1}{2}$,from equation (iii), $x = -2(\frac{1}{2}) \Rightarrow x = -1$
when $y = 1$, from equation (iii), $x = 2 \times 1 \Rightarrow x = 2$
$\therefore$ $x = -1$ or $x = 2$.

Exercise Solve the following exponential equations
(1) $2^{2x} - 6(2^x) + 8 = 0$.
(2) $2^{2x+1} - 5(2^x) + 2 = 0$
(3) $3^2x - 9 = 0$
(4) $7^{2x} - 2 \times 7^x + 1 = 0$
(5) $5^{2x} + 4 \times 5^{x+1} - 125 = 0$.

2.2.2 Logarithms

Some Rules of Logarithms
Suppose p, q are positive integers

1. $\log_a pq = \log_a p + \log_a q$.
2. $\log_a \frac{p}{q} = \log_a p - \log_a q$.
3. $\log_a 1 = 0$ since $a^0 = 1$. In general
$$\log_a b = 0 \Rightarrow a^0 = b.$$
4. $\log_b p^a = a \log_b p$.
5. $\log_a p = \frac{\log_b p}{\log_b a}$.

Example 1: Without using table, show that

$$\frac{\log \frac{3}{2} + \log \frac{8}{10} + \log \frac{1}{30}}{2 \log \frac{\sqrt{2}}{9} + 1\frac{1}{2} \log \frac{4}{25} - 4 \log(\frac{2}{3})} = \frac{2}{3}.$$

Solution:

$$\frac{\log(\frac{3}{2} \times \frac{8}{10} \times \frac{1}{30})}{\log(\frac{\sqrt{2}}{9})^2 + \log(\frac{4}{25})^{\frac{3}{2}} - \log(\frac{2}{3})^4} = \frac{\log(\frac{3}{2} \times \frac{8}{10} \times \frac{1}{30})}{\log(\frac{2}{81} \times \frac{8}{125}) \times \frac{81}{16}}$$

$$= \frac{\log \frac{1}{25}}{\log \frac{1}{125}} = \frac{\log 5^{-2}}{\log 5^{-3}}$$
$$= \frac{-2\log 5}{-3\log 5}$$
$$= \frac{2}{3}.$$

Example 2: Prove from definition of a logarithm that if x, y and z are positive numbers,

$$\log_{xy} yz = \frac{\log_y z + 1}{\log_y x + 1}.$$

Solution:

$$\log_{xy} yz = \frac{\log_y z + 1}{\log_y x + 1}$$

$$\log_{xy} yz = \frac{\log_y yz}{\log_y xy} = \frac{\log_y y + \log_y z}{\log_y x + \log_y y} = \frac{1 + \log_y z}{\log_y x + 1} = \frac{\log_y z + 1}{\log_y x + 1}$$

Example 3: Solve for x if $[\log_3(x^2 + x - 6) - \log_3(x^2 + 7x + 12)] + \frac{2}{3}\log_9 8 = 0$
Solution:

$$[\log_3(x^2 + x - 6) - \log_3(x^2 + 7x + 12)] + \frac{2}{3}\log_9 8 = 0$$
$$\Rightarrow \log_3 \frac{(x^2 + x - 6)}{(x^2 + 7x + 12)} + \frac{2}{3}\left[\frac{3\log_3 2}{2\log_3 3}\right] = 0$$
$$\Rightarrow \log_3 \frac{(x^2 + x - 6)}{(x^2 + 7x + 12)} + \log_3 2 = 0$$
$$\Rightarrow \log_3 \frac{(x^2 + x - 6)}{(x^2 + 7x + 12)} \times 2 = 0$$
$$\Rightarrow \log_3 \frac{(2x^2 + 2x - 12)}{(x^2 + 7x + 12)} = 0$$
$$\Rightarrow (2x^2 + 2x - 12) = (x^2 + 7x + 12) = x^2 - 5x - 24 = 0$$
$$\Rightarrow x^2 - 8x + 3x - 24 = 0$$
$$\Rightarrow x(x - 8) + 3(x - 8) = 0$$
$$\Rightarrow (x - 8)(x + 3) = 0$$

$$\therefore x = -3 \text{ or } x = 8$$

Example 4: Express y in term of x if

$$\log_3(y - 1) = x$$

Solution

$y - 1 = 3^x$

$y = 1 + 3^x$

Example 5 Show that $log_a p^{x-n} + log_a p^{y+n} = (n + y) log_a p$

Solution

From the LHS

$$\begin{aligned} log_a p^{x-n} + log_a p^{y+n} &= log_a p^x \cdot log_a p^{-n} + log_a p^y \times log_a p^n \\ &= log_a \frac{p^x}{p^n} + log_a p^y \times p^n \\ &= log_a (\frac{p^x}{p^n} \times p^y \cdot p^n) \\ &= log_a p^x \times p^y \\ &= (x + y) \log_a p. \end{aligned}$$

2.3 Inequalities and Absolute value

Mathematics is not always about equality. Sometimes we discuss issues of greater than or less than. Inequalities means not "equal". There are two most common inequalities symbol which are $>$ meaning greater than and $<$ meaning less than. We can also have inequalities that include "equals" like $\geq$ greater than or equal to and $\leq$ meaning less than or equal to. There are rules governing the relationships between numbers which are not equal. For any to real numbers a and b. we say that a is greater than $b(a > b)$ if $a - b$ is positive. we say that a is less than b $(a < b)$ if $a - b$ is negative. By definition

$$a > b \; if \; a - b > 0 \; and \; a < b \; if a - b < 0 \tag{2.1}$$

e.g. $6 > -2$ since $6 - (-2) = 8$ is positive, i.e > 0 Also $-4 < -2$ since $-4 - (-2) = -2$ is negative i.e < 0

2.3.1 Law One

We first show that if $a > b$. then

$$a + x > b + x \tag{2.2}$$

where x is any real number
If $a > b$, $a - b = c > 0$ $a + x - (b + x) = a + x - b - x = a - b = c$
$a + x - (b + x) = c > 0$
$a + x > b + x$ by definition
In the same way if $a < b$

$$a + x < b + x \tag{2.3}$$

Thus as with equations we may add the same number to both sides of our inequality and still preserve the inequality. For example:

$7 > -3$
and after adding 5 to both sides
$12 > 2, 5 > 10$ Also,and after adding -4(i.e. subtracting 4) on both sides
$1 > 6$ We cannot however treat inequalities in the same way as equations if we multiply both sides of the inequality by the same number. Rather we have

2.3.2 Law Two

$$If\ a > b = \begin{cases} ax > bx\ if\ x\ is\ positive, \\ ax < bx\ if\ x\ is\ negative \end{cases} \tag{2.4}$$

$$If\ a < b = \begin{cases} ax < bx\ if\ x\ is\ positive, \\ ax > bx\ if\ x\ is\ negative \end{cases} \tag{2.5}$$

We shall prove this for the case $a > b$ If $a - b = c$, where x is positive, then
$ax - bx = cx$ which is positive if x is positive, but negative. If x is negative . Therefore, $ax - bx > 0$ if $x > 0$, $ax - bx < 0$ if $x < 0$ which is the result required. Thus if we multiply both sides of an inequality by a negative number, the inequality sign must be reversed. e.g. $8 > 5$, which becomes $24 > 20$ after multiplication by 3 but $-16 < -10$ after multiplication by -2.

2.3.3 Law Three

$$if\ a > b\ and\ c > d, then\ a + c > b + d \tag{2.6}$$

For example, $8 > 4$, and $-3 > -6$, and $3 > -4$ Note: It does not follow that $a - c > b - d$ e.g, $11 > 10$ and $9 > 2$ but $11 - 9 < 10 - 2$

2.3.4 Law Four

$$if\ a > b\ and\ b > c, then\ a > c \tag{2.7}$$

e.g. $8 > 3$ and $9 > 2$ and $8 > 2$ Note: If $a > b$ and $b < c$, we can say nothing about the relative magnitudes of a and c. For example $9 > 2$ and $2 > 8$ and of course $9 > 8$, but we could equally well have had $9 > 2$ and $2 < 11$ with, of course, $9 < 11$.

2.3.5 Law Five

$$if\ a > b\ and\ c > d, and\ a,\ b,\ c,\ d\ are\ all\ positive\ .\ then\ ac > bd\ and\ \frac{a}{d} > \frac{b}{c} \tag{2.8}$$

For example, $9 > 2$ and $6 > 3$ and of course $9 \times 6 > 2 \times 3$ i.e $54 > 6$.

2.3.6 Law Six

if $a > b$ and a and b are both positive, it follows that $a^2 > b^2$, $a^3 > b^3$, $\frac{1}{a} < \frac{1}{b}$, $\frac{1}{a^2} < \frac{1}{b^2}$. Indeed

$$a^n > b^n \text{ if } n > 0, \qquad a^n < b^n \text{ if } n < 0 \tag{2.9}$$

For example, $3 > 2$ and $3^3 > 2^3$, i.e, $27 > 8$, but $3^2 > 2^2$ i.e. $\frac{1}{9} < \frac{1}{4}$

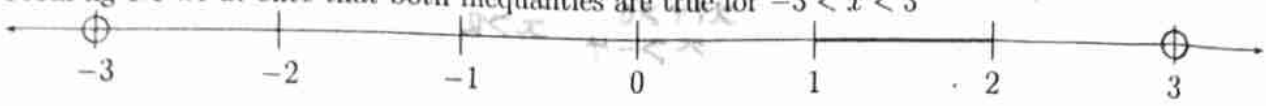
Example 1: For what values of x are both the inequalities $3 + x > 0$ and $3 - x > 0$ true ?

If $3 + x > 0 \Rightarrow x > -3$,

$3 - x > 0$, $-x > -3$, $x < 3$

$-3 < x < 3$.

From fig 1.1 we at once that both inequalities are true for $-3 < x < 3$



Example 2: Find the range of values of x for which

$$\frac{2x+1}{x+2} > \frac{1}{2}.$$

Solution

To preserve the inequality sign, multiply both sides of the inequality by $(x+2)^2$ which is positive. Thus,

$$\begin{aligned}
(2x+1)(x+2) &> \frac{1}{2}(x+2)^2 \\
\Rightarrow 2(2x+1)(x+2) &> (x+2)^2 \\
\Rightarrow (x+2)(4x+2) - (x+2)^2 &> 0 \\
\Rightarrow (x+2)[(4x+2) - (x+2)] &> 0 \\
\Rightarrow (x+2)[4x+2-x-2] &> 0 \\
\Rightarrow (x+2)(4x-x) &> 0 \\
\Rightarrow 3x(x+2) &> 0
\end{aligned}$$

This will be true if $x > 0$ and $x + 2 > 0$, or if $x < 0$ and $x + 2 < 0$ i.e. if both factors are positive or both factors are negative

Equations	$x < -2$	$-2 < x < 0$	$x > 0$
$3x$	-ve	-ve	+ve
$x + 2$	-ve	+ve	+ve
$3x(x+2)$	+ve	-ve	+ve

Thus, the original inequality is true for $x > 0$ or $x < -2$.

Example 3: Determine the range of values of x for which $\frac{x^2+x-2}{x^2+4} > \frac{1}{2}$.

Solution

$x^2 + 4$ being the sum of two squares, is always positive. Thus, we can multiply both sides of the inequality by $x^2 + 4$ and still preserve the sign. Thus

$$\begin{aligned} x^2 + x - 2 &> \frac{1}{2}(x^2 + 4) \\ \Rightarrow 2(x^2 + x - 2) &> (x^2 + 4) \\ \Rightarrow (2x^2 + 2x - 4) &> (x^2 + 4) \\ \Rightarrow 2x^2 - x^2 + 2x - 4 - 4 &> 0 \\ \Rightarrow x^2 + 2x - 8 &> 0 \\ \Rightarrow (x + 4)(x - 2) &> 0 \end{aligned}$$

We draw up our table showing the signs of the individual factors. The individual factors change sign at -4 and 2.

Equations	$x < -4$	$-4 < x < 2$	$x > 2$
$(x + 4)$	-ve	+ve	+ve
$(x - 2)$	-ve	-ve	+ve
$(x + 4)(x - 2)$	+ve	-ve	+ve

Hence, the inequality is true for $x < -4$ or $x > 2$.

Example 4: Solve the inequality

$$\frac{x+3}{x-2} > \frac{x+1}{x-3}$$

Solution

Here, we must multiply by the positive factor $(x-2)^2(x-3)^2$ to obtain

$(x+3)(x-2)^2(x-3)^2 > (x+1)(x-2)^2(x-3)^2$

which yields

$$\begin{aligned} (x-3)(x-2)[(x+3)(x-3) - (x+1)(x-3)] &> 0 \\ (x-3)(x-2)[(x^2-9) - (x^2-x-3)] &> 0 \\ (x-3)(x-2)(x-7) &> 0 \end{aligned}$$

Again, we draw up our table showing the signs of the individual factors which change sign at 2, 3 and 7.

Equations	$x < 2$	$2 < x < 3$	$3 < x < 7$	$x > 7$
$(x - 2)$	-ve	+ve	+ve	+ve
$(x - 3)$	-ve	-ve	+ve	+ve
$(x - 7)$	-ve	-ve	-ve	+ve
$(x - 2)(x - 3)(x - 7)$	-ve	+ve	-ve	+ve

Thus, the original inequality is true if $2 < x < 3$ or $x > 7$

The absolute value or modulus is the measure of the distance from zero(0) to x.The absolute value of x is the positive number having the same magnitude as x. It is written as $| x |$, where

$$| x | = \begin{cases} x \text{ if } x > 0, \\ 0 \text{ if } x = 0, \\ -x \text{ if } x < 0 \end{cases} \tag{2.10}$$

Thus $| 3 | = 3$, $| -6 | = 6$, $| -2 | = 2$ and $| -1 | = 1$. In general if x is positive $| x | = x$, but if x is negative, $| x | = x$. With this notation the range of values of x specified by the inequality $-1 < x < 1$ can be specified more concisely by $| x | < 1$.

Example 5: Find if $| x + 1 | = 5$
Solution:
we have $x + 1 = 5$ or $x + 1 = -5$
$\therefore x = 4$ or $x = -6$.

Example 6: Solve the inequalities $| x - 2 | < | 3 - 2x |$
Solution:
Squaring both sides
$(x - 2)^2 < (3 - 2x)^2$
$x^2 - 4x + 4 < 9 - 12x + 4x^2$
$x^2 - 4x^2 - 4x + 12x + 4 - 9 < 0$
$-3x^2 + 8x - 5 < 0$
$-(3x^2 - 8x + 5) < 0$
$3x^2 - 8x + 5 > 0$
Factorizing the inequalities $3x^2 - 3x - 5x + 5 > 0$
$3x(x - 1) - 5(x - 1) > 0$
$(3x - 5)(x - 1) > 0$
$3x - 5 > 0$ or $x - 1 < 0$
$3x > 5$ or $x > 1$
$x > \frac{5}{3}$ or $x > 1$

Using the table test

Equations	$x < 1$	$1 < x < \frac{5}{3}$	$x > \frac{5}{3}$
$(3x - 5)$	-ve	-ve	+ve
$(x - 1)$	-ve	+ve	+ve
$(3x - 5)(x - 1)$	+ve	-ve	+ve

Hence the solution is $x < 1$, $x > \frac{5}{3}$.

Example 7: Given than a is a positive constant. Solve the inequalities $|x - 3a| > |x - a|$
Solution:
$|x - 3a| > |x - a|$
Squaring both sides
$(x - 3a)^2 > (x - a)^2$

$x^2 - 6ax + 9a^2 > x^2 - 2ax + a^2$
$x^2 - x^2 - 6ax + 2ax + 9a^2 - a^2 > 0$
$-4ax + 8a^2 > 0$
$-4ax > -8a^2$
$x < \frac{8a^2}{4a}$
$\therefore x < 2a$

Example 8: Find the set of values of x satisfy the inequality $|3^x - 8| < 0.5$. Give your answer to 3 significant figures.
Solution:
$|3^x - 8| < 0.5$

$$(3^x - 8)^2 < (0.5)^2 \Rightarrow 3^x(3^x - 8) - 8(3^x - 8) < 0.25 \Rightarrow 3^{2x} - 8(3^x) - 8(3^x) + 64 < 0.25$$
$$3^{2x} - 16(3^x) + 64 - 0.25 < 0 \Rightarrow (3^x)^2 - 16(3^x) + 63.75 < 0$$

Let $P = 3^x$
$P^2 - 16p + 63.75$

$$P = \frac{16 \pm \sqrt{16^2 - 4 \cdot 1 \cdot 63.75}}{2} \Rightarrow P = \frac{16 \pm \sqrt{256 - 255}}{2} \Rightarrow P = \frac{16 \pm \sqrt{1}}{2}$$

$P = \frac{16 \pm 1}{2} \Rightarrow P = \frac{16+1}{2}$ or $P = \frac{16-1}{2} \Rightarrow P = \frac{17}{2}$ or $P = \frac{15}{2}$
$\therefore P = 8.5$or $P = 7.5$
Hence when $P = 8.5$
$3^x = 8.5$ Take the ln of both sides ($\ln 3^x = \ln 8.5$)

$$x \ln 3 = \ln 8.5 \Rightarrow x = \frac{\ln 8.5}{\ln 3} \Rightarrow x = 1.95$$

when $P = 7.5$, $3^x = 7.5$
Take the ln of both sides ($\ln 3^x = \ln 8.5$)

$$\ln 3^x = \ln 7.5 \Rightarrow x \ln 3 = \ln 7.5 \Rightarrow x = \frac{\ln 7.5}{\ln 3} \Rightarrow x = 1.83$$

$\therefore (x - 1.83)(x - 1.95) < 0$
Using table test

Equations	$x < 1.83$	$1.83 < x < 1.95$	$x > 1.95$
$(x - 1.83)$	-ve	+ve	+ve
$(x - 1.95)$	-ve	-ve	+ve
$(x - 1.83)(x - 1.95)$	+ve	-ve	+ve

Hence the solution is $1.83 < x < 1.95$

2.4 Surds

Example 1: Simplify

$$\frac{1}{\sqrt{5}-\sqrt{2}}+\frac{1}{\sqrt{2}+3}-\frac{1}{2-\sqrt{3}}$$

Solution:
We rationalize and add

$$\begin{aligned}\frac{1}{\sqrt{5}-\sqrt{2}}+\frac{1}{\sqrt{2}+3}-\frac{1}{2-\sqrt{3}} &= \frac{1}{\sqrt{5}-\sqrt{2}}\times\frac{\sqrt{5}+\sqrt{2}}{\sqrt{5}+\sqrt{2}}+\frac{1}{\sqrt{2}+3}\times\frac{\sqrt{2}-3}{\sqrt{2}-3}-\frac{1}{2-\sqrt{3}}\times\frac{2+\sqrt{3}}{2+\sqrt{3}}\\ &= \frac{\sqrt{5}+\sqrt{2}}{5+\sqrt{10}-\sqrt{10}-2}+\frac{\sqrt{2}-3}{2-3\sqrt{2}+3\sqrt{2}-9}-\frac{2+\sqrt{3}}{4+2\sqrt{3}-2\sqrt{3}-3}\\ &= \frac{\sqrt{5}+\sqrt{2}}{3}+\frac{\sqrt{2}-3}{-7}-\frac{2+\sqrt{3}}{1}\end{aligned}$$

$$\begin{aligned}&= \frac{7(\sqrt{5}+\sqrt{2})-3(\sqrt{2}-3)-21(2+\sqrt{3})}{21}\\ &= \frac{7\sqrt{5}+7\sqrt{2}-3\sqrt{2}+9-42-21\sqrt{3}}{21}\\ &= \frac{7\sqrt{5}+4\sqrt{2}-21\sqrt{3}-33}{21}\end{aligned}$$

Example 2:
Express

$$\frac{2\sqrt{5}+5\sqrt{2}}{2\sqrt{5}-5\sqrt{2}}$$

in the form $q+r\sqrt{s}$ where q, r and s are rational numbers
Solution:

$$\begin{aligned}\frac{2\sqrt{5}+5\sqrt{2}}{2\sqrt{5}-5\sqrt{2}} &= \frac{2\sqrt{5}+5\sqrt{2}}{2\sqrt{5}-5\sqrt{2}}\times\frac{2\sqrt{5}+5\sqrt{2}}{2\sqrt{5}+5\sqrt{2}}\\ &= \frac{2\sqrt{5}(2\sqrt{5}+5\sqrt{2})+5\sqrt{2}(2\sqrt{5}+5\sqrt{2})}{2\sqrt{5}(2\sqrt{5}-5\sqrt{2})-5\sqrt{2}(2\sqrt{5}+5\sqrt{2})}\\ &= \frac{70+20\sqrt{10}}{70}\\ &= 1+\frac{2}{7}\sqrt{10}\end{aligned}$$

where $q=1, r=\frac{2}{7}$, and $s=10$

Example 3. Find the square root of $7-4\sqrt{3}$
Solution:
$7-4\sqrt{3}=7-\sqrt{16\times 3}=7-\sqrt{48}$

$a = 7, b = 48$
$a^2 - b = 7^2 - 48 = 49 - 48 = 1 = 1^2$
Assume $\sqrt{7 - 4\sqrt{3}} = \sqrt{m} - \sqrt{n}$
Square both sides
$(\sqrt{7 - 4\sqrt{3}}) = \sqrt{m} - \sqrt{n}$
Square both sides
$7 - \sqrt{48} = (\sqrt{m} - \sqrt{n})(\sqrt{m} - \sqrt{n})$
$7 - \sqrt{48} = m + n - \sqrt{4mn}$

$$m + n = 7 \tag{2.11}$$

$$\sqrt{4mn} = \sqrt{48} = 4mn = 48 \tag{2.12}$$

$mn = \frac{48}{4} = 12$
$mn = 12$
From (1.11)

$$m = 7 - n \tag{2.13}$$

Substitute eq. 1.13 in eq. 1.12

$$n(7 - n) = 12 \Rightarrow 7n - n^2 = 12 \Rightarrow 7n - n^2 - 12 = 0 \Rightarrow$$
$$n^2 - 7n + 12 = 0 \Rightarrow (n - 3)(n - 4) = 0 \Rightarrow n = 3 \text{ or } n = 4$$

When $n = 3$, $m = 4$
and when $n = 4$, $m = 3$
Hence $\sqrt{7 - 4\sqrt{3}} = \sqrt{4} - \sqrt{3}$ or $\sqrt{3} - \sqrt{4}$
$= 2 - \sqrt{3}$ or $\sqrt{3} - 2$. **Exercise**

1. Solve the inequality $\frac{x-1}{x-2} > \frac{x-2}{x-3}$
2. Solve the inequality $\frac{2x^2+5x+7}{3x+5} \geq 2$
3. Solve the inequality
$$\frac{x-4}{x-3} < \frac{2x-1}{2}$$
4. If a and b are positive numbers, show that (i) $a + \frac{1}{a} \geq 2$ and (ii) $(a + b)(\frac{1}{a} + \frac{1}{b}) \geq 4$
5. Solve the inequality $|x + 2| < |5 - 2x|$
6. Solve the inequality $|x| > |3x - 2|$
7. Solve the equation $|\frac{x-3}{1+x}| < 1$
8. Rationalize $\frac{1+\sqrt{2}}{1-\sqrt{2}}$
9. Express $\frac{\sqrt{2}+\sqrt{5}}{\sqrt{10}}$ in the form $a\sqrt{5} + b\sqrt{2}$

Chapter 3

MATHEMATICAL INDUCTION

3.1 The Principle of Mathematical Induction

Mathematical Induction is an extremely important tool in Mathematics. It is a technique for providing a statement, theorem or a formula that is asserted about every natural number say. $1, 2, 3, 4, e.t.c.$
For example,

$$1+2+3+...+n=\frac{1}{2}(n+1)$$

This asserts that the sum of consecutive numbers from 1 to n is given by the formula on the right. We want to prove that this will be true for $n = 1$, $n = 2$, $n = 3$, and so on. Now we can test the formula for any given number, say $n = 3$:

$$1+2+3=\frac{1}{2}\cdot 3\cdot 4=6$$

which is true. It is also true for $n = 4$:

$$1+2+3+4=\frac{1}{2}\cdot 4\cdot 5=10$$

But how are we to prove this rule for every value of n?
The method of proof is called the principle of mathematical induction.

The steps involved are as follows:
(1) Verify that the statement is true for $n = 1$, then the statement is true for every natural number n
(2) Assume that the statement is true for $n = k$
(3) Show that $n = k + 1$ is true whenever $n = k$ is true.
The hypothesis of step 2 " *The statement is true for* $n = k$" is called the induction hypothesis. It is what we assume when we prove a theorem by induction.

Example 1

Prove that the sum of the first n natural numbers is given by this formula:

$$1+2+3+...n = \frac{n(n+1)}{2}$$

Solution:
We will do Steps 1 to 3 above.
Step 1: We must show that the formula is true for $n = 1$. We have:

$$1 = \frac{1}{2} \cdot 1 \cdot 2$$

which is true.
Step 2:
We will assume that the formula is true for $n = k$; that is, we will assume:

$$1+2+3+...+k = \frac{k(k+1)}{2} \tag{3.1}$$

This is the induction assumption. Assuming this, we must prove that the formula is true for $n = k+1$. That is, we must show that :

$$1+2+3+...+(k+1) = \frac{(k+1)(k+2)}{2} \tag{3.2}$$

To do this, we will simply add the next term $(k+1)$ to both sides of the induction assumption, equation 1.1

$$\begin{aligned} 1+2+3+...+k+(k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1)+2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2} \end{aligned}$$

We have now fulfilled the three conditions of the principle of mathematical induction.

Example 2

Prove by mathematical induction:

$$1^3+2^3+3^3+...+n^3 = \frac{n^2(n+1)^2}{4}$$

Solution:
For $n = 1$.

$$\begin{aligned} 1^3 &= \frac{1^2 \cdot 2^2}{4} \\ 1 &= \frac{1 \cdot 4}{4} \\ &= 1 \end{aligned}$$

Next, we assume that the formula is true for $n = k$
For convenience, we will denote the sum up to n by $S(n)$. That is, that

$$S(k) = \frac{k^2(k+1)^2}{4}$$

We must show that the formula is also true for $n = k + 1$;

$$S(k+1) = \frac{(k+1)^2(k+2)^2}{4}$$

To do that, add the next cube to $S(k)$:

$$\begin{aligned} S(k+1) &= S(k) + (k+1)^3 \\ &= \frac{k^2(k+1)^2}{4} + (k+1)^3 \\ &= \frac{k^2(k+1)^2 + 4(k+1)^3}{4} \\ &= \frac{(k+1)^2[k^2 + 4(k+1)]}{4} \\ &\quad \textit{on taking } (k+1)^2 \textit{ as a factor.} \\ &= \frac{(k+1)^2(k^2 + 4k + 4)}{4} \\ &= \frac{(k+1)^2(k+2)^2}{4} \end{aligned}$$

Example 3

Prove by induction:

$$1 \cdot 3 + 2 \cdot 3^3 + \ldots + n \cdot 3^n = \frac{(2n-1)3^{n+1} + 3}{4}$$

Solution:
For $n = 1$, $n \cdot 3^n = 1 \cdot 3^1 = 3$

$$\begin{aligned} &= \frac{(2n-1)3^{n+1} + 3}{4} \\ &= \frac{(2-1)3^{1+1} + 3}{4} \\ &= 3 \end{aligned}$$

For $n = 1$ it is true.

Now, for $n = k$

$$1 \cdot 3 + 2 \cdot 3^2 + \ldots + k \cdot 3^k = \frac{(2k-1)3^{k+1} + 3}{4}$$

We assume that it is true for $n = k$.

For $n = k + 1$:

$$\begin{aligned}
1 \cdot 3 + 2 \cdot 3^2 + \ldots + k \cdot 3^k + (k+1) \cdot 3^{k+1} &= \frac{(2k+1)3^{k+2} + 3}{4} \\
&= \frac{(2k-1)3^{k+1} + 3}{4} + \frac{(k+1) \cdot 3^{k+1}}{1} \\
&= \frac{(2k-1)3^{k+1} + 3}{4} + \frac{[4(k+1)3^{k+1}]}{1} \\
&= \frac{(2k-1)3^{k+1} + 4(k+1)3^{k+1} + 3}{4} \\
&= \frac{2k \cdot 3^{k+1} - 3^{k+1} + 4k \cdot 3^{k+1} + 4 \cdot 3^{k+1} + 3}{4} \\
&= \frac{2k \cdot 3^{k+1} + 4k \cdot 3^{k+1} + 4 \cdot 3^{k+1} - 3^{k+1} + 3}{4} \\
&= \frac{6k \cdot 3^{k+1} + 3 \cdot 3^{k+1} + 3}{4} \\
&= \frac{(6k+3)3^{k+1} + 3}{4} \\
&= \frac{3(2k+1)3^{k+1} + 3}{4} \\
&= \frac{(2k+1)3^{k+1} \cdot 3 + 3}{4} \\
&= \frac{(2k+1)3^{k+1+1} + 3}{4} \\
&= \frac{(2k+1)3^{k+2} + 3}{4}.
\end{aligned}$$

Hence we say that the statement is true for $n = k \Rightarrow$ the statement is true for $n = k + 1$.

Example 4

Prove by mathematical induction:

$$\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \ldots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}.$$

for $n = 1$

$$\frac{1}{1\cdot 3} = \frac{1}{2\cdot 1+1} = \frac{1}{3}$$

For $n = k$, we denote the sum by $S(n)$

$$S(k) = \frac{k}{2k+1}.$$

Next we show that the formula is true for $n = k+1$; that is,

$$S(k+1) = \frac{k+1}{2k+3}.$$

Thus,

$$\begin{aligned} S(k+1) &= S(k) + \textit{The next term , whose denominator is the product of the next odd num} \\ &= \frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)} \\ &= \frac{k(2k+3)+1}{(2k+2)(2k+3)} \\ &= \frac{2k^2+3k+1}{(2k+1)(2k+3)} \\ &= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)} \qquad 2k^2+2k+k+1 \\ &= \frac{k+1}{2k+3} \end{aligned}$$

Example 5

Prove by mathematical induction that $\sum_{r=1}^{n} r(r+1) = \frac{n(n+1)(n+2)}{3}$

Solution

For $n = 1$

LHS: $n(n+1) = 1(1+1) = 2$

RHS: $\frac{(1)(2)(3)}{3} = \frac{6}{3}$

$\therefore$ it is true for $n = 1$

For $n = k$

$$\begin{aligned} k(k+1) &= \frac{k(k+1)(k+2)}{3} \\ \therefore \sum_{r=1}^{n} r(r+1) &= \frac{k(k+1)(k+2)}{3} \end{aligned}$$

We assume it is true for $n = k$

For $n = k + 1$

$$\sum_{r=1}^{n} r(r+1) + (k+1)(k+2) = \frac{k(k+1)(k+2)(k+3)}{3}$$

For LHS:

$$\begin{aligned}
&= \frac{k(k+1)(k+2) + (k+)(k+2}{3} \\
&= \frac{k(k+1)(k+2) + 3(k+1)(k+2)}{3} \\
&= \frac{(k+1)[k(k+2) + 3(k+2)]}{3} \\
&= \frac{(k+1)(k+2)[(k+3)]}{3} \\
&= \frac{(k+1)(k+2)(k+3)}{3}
\end{aligned}$$

Since LHS = RHS, we say that the statement is true for $n = k \Rightarrow$ the statement is true for $n = k + 1$

Exercises

1. For all positive inters n, $1^2 + 2^2 + \ldots + n^2 = \frac{(n)(n+1)(2n+1)}{6}$
2. Prove that this rule of exponents is true for every natural number n:

$$(ab)^n = a^n b^n$$

3. Prove that the sum of the first n odd numbers is equal to the nth square

$$1 + 3 + 5 + 7 + \ldots + (2n-1) = n^2$$

4. Prove by induction that

$$\sum_{r=1}^{n} r^2(r+1) = \frac{n}{12}(n+1)(n+2)(3n+1)$$

5. Prove by mathematical induction that

$$1 + 3 + 3^2 + \ldots + 3^{n-1} = \frac{3^n - 1}{2}$$

6. Prove this statement by method of induction:

$$\sum_{r=1}^{n} ap^{r-1} = \frac{a(p^n - 1)}{p - 1}$$

7. Prove that $8^n - 7n + 6$ is divisible by 7 for all positive integer values of n

8. Prove that $x^n - y^n$ has $x - y$ as a factor for all positive integers n

Chapter 4

SEQUENCES AND SERIES

4.1 Sequence

A Sequence is a set of numbers. Each number in the sequence is called a term and the general term can be denoted a_n. Examples are:

1. 2, 6, 10, 14, ...
2. 6, 18, 54, 162, ...
3. -1, -2, -3, -4, ...

The general term a_n for the above sequences are $a_n = 4n - 2$, $a_n = 6 \times 3^{n-1}$, $a_n = -1(n)$ respectively.

4.2 Arithmetic Progression

An Arithmetic progression (AP) is a sequence of numbers generated by adding or subtracting a fixed number called a common difference to a preceding term to get a term. The nth term of a linear sequence is given by

$$a_n = a - (n - 1)d$$

where a is the first term and d is the common difference. For example

1. The sequence 2, 5, 8, 11, 14, ... is an arithmetic sequence with the first term a and a common difference $d = 3$
2. The sequence 8, 4, 2, −2, ... is an arithmetic progression with the first term $a = 8$ and the common difference $d = 4$.

Example 1 Find the nth term of the arithmetic sequence 7, 2, −3, −8, ...

$$\begin{aligned} a_n &= a + (n-1)d \\ &= 7 + (n-1)5 \\ &= 7 + 5n - 5 \\ &= 2 + 5n. \end{aligned}$$

Example 2 Find the common difference between successive terms of the arithmetic sequence for which the first term is 5 and the $32nd$ term is -119

Solution We are to find d given that $a = 5$ $n = 32$ and $a_{32} = -119$

$$\begin{aligned} -119 &= 5 + (32-1)d \\ -124 &= 31d \\ -4 &= d \\ d &= -4. \end{aligned}$$

$a_n = a + (n-1)d$

$-119 = 5(32-1)d$

$-119 = 5(31)d$

$-119 = 31d$

4.2.1 Sum of n terms

Sum of terms of an arithmetic sequence is called an arithmetic series or progression. The general arithmetic sequence is often denoted by $a, a+d, a+2d, a+3d$ e.t.c
To sum the series of the first n terms of the sequence,

$$S_n = a + (a+d) + (a+2d) + \ldots + (a+(n-1)d).$$

The order can be reversed to give

$$S_n = (a+(n-1)d) + (a+(n-2)d) + \ldots + a.$$

Adding the two expressions for S_n gives

$$\begin{aligned} 2S_n &= (2a+(n-1)d + (2a+(n-1)d + \ldots + (2a+(n-1) \\ &= n[2a+(n-1)d] \\ S_n &= \frac{n}{2}(2a+(n-1)d). \end{aligned}$$

$S_n = \frac{n}{2}(a+l)$

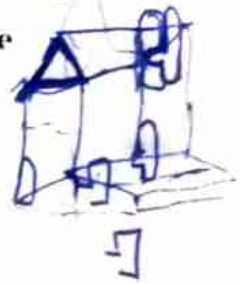
$l = a + (n-1)d$

An alternative form of S_n is given in terms of its first and last term a and l where

$$l = a + (n-1)d$$

and thus

$$S_n = \frac{n}{2}(a+l).$$



Example 1 Sum the Series $5 + 9 + 13 + \ldots$ to 20 terms
Solution:
$a = 1.$ $d = 7.$ $S_n = 396$

$$\Rightarrow \frac{n}{2}[2 + 7(n-1)] = 396$$

$$\Rightarrow n(7n-5) = 792$$

$$\Rightarrow 7n^2 - 5n - 792 = 0$$

$$\Rightarrow (7n+72)(n-11) = 0$$

$$\Rightarrow n = 11$$

There are 11 terms

$S_n = \frac{n}{2}[2a + (n-1)d]$

$396 = \frac{n}{2}[10 + (n-1)4]$

$792 = n[10 + 4n - 4]$

$792 = 10n + 4n^2 - 4n$

$796 = 6n + 4n^2$

$6n + 4n^2 - 796 = 0$

$a_n = a + (n-1)d$

$= -1 + (n-1)-1$

$= 1 + n + 1$

4.3 Geometric Sequences

The series of numbers 4, 8, 16, 32, 64, ... is an example of a Geometric progression (GP). Each term in the progression is found by multiplying the previous number by 2.
Thus, the general form of a geometric sequence with n terms is $a, ar, ar^2, ..., ar^{n-1}$ where a is the first term, the ratio of consecutive terms r, is known as the common ratio and the nth term is ar^{n-1}.

Example 1
What is the first term, common ration and nth term of the given geometric sequence $1, -2, 4, -8, 16$
Solution
The first term $a = 1$, the common ratio $r = -2$.
The nth term is given by

$$a_n = ar^{n-1} = 1 \times (-2)^{n-1} = -2^{n-1}$$

Example 2
Find the eighth term of the geometric sequence $8, 4, 2, ...$
Solution
$a = 8, \quad r = \frac{1}{2}$

$$a_8 = 8(\frac{1}{2})^{8-1} = \frac{8}{2^7} = \frac{1}{16}$$

4.3.1 Sum of n terms

The formular for the sum of n terms of a G.P. is found as follows

$$S_n = a + ar + ar^2 + ... + ar^{n-1} \qquad eq(1)$$

Multiply both sides by r

$$rS_n = ar + ar^2 + ar^3 + ... + ar^{n-1} + ar^n \qquad eq(2)$$

Subtracting eq(2) from (1) gives

$$S_n - rS_n = a - ar^n$$
$$\Rightarrow S_n(1 - r) = a(1 - r^n)$$
$$\Rightarrow S_n = \frac{a(1 - r^n)}{1 - r}$$

$(r \neq 1)$]
When $r > 1$, the formular could be written as

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

Example 1 Find the sum of the first 10 terms of $4, 6, 9, ...$
Solution

First term $a = 4$ common ratio $r = 1.5$ number of terms $n = 10$

$$S_{11} = \frac{4(1 - 5^{10} \quad 1)}{1\,5 - 1} = 453.4$$

Example 2

Find the sum of the first 8 terms of $1, -\frac{1}{2}, \frac{1}{4}, -\frac{1}{8}, \frac{1}{16}, \ldots$

Solution

First term $a = 1$, common ratio $r = -\frac{1}{2}$ numbers of terms $n = 8$

$$S_8 = \frac{1(1 - (-\frac{1}{2})^8)}{1 - (-\frac{1}{2})} = \frac{1(1 + \frac{1}{2})}{1 + \frac{1}{2}} = 1$$

4.3.2 Infinite Geometric Progression

The sum of the terms of an infinite geometric series is given by

$$S_\infty = \frac{a}{1 - r} \quad |r| < 1$$

where a is the first term and r is the common ratio

Example

Find the sum of the infinite geometric series

Solution

$$-\frac{1}{2} + \frac{1}{16} - \frac{1}{128} + \ldots$$

$$a_1 = 4, \quad r = -\frac{1}{8}$$

$$S_\infty = \frac{4}{1 - (-\frac{1}{8})} = \frac{4}{1 + \frac{1}{8}} = \frac{32}{9}$$

Exercise

1. Find the nth term and sum of the first n terms of the arithmetic progressions
   (a) 12, 21, 30, 39, ...
   (b) $1, 2\frac{1}{2}, 4, 5\frac{1}{2}, \ldots$
   (c) $1\frac{1}{2}, 5, 8\frac{1}{2}, 12, \ldots$

2. The first term of an arithmetic series is 16 and the last term is 60. The sum of the arithmetic series is 342. Find the common difference

3. The sum of the 3rd term and the 7th term of a sequence equals 6 and their product equals 8. Find the sum of the first 16 terms of the sequence

4. Find the sum of all the integers between 100 and 400 that are divisible by 7

5. The first three terms of an A.P. are $x, 2x + 1 and 5x + 1$. Find x and the sum of the first 10 terms.

6. Write down the formulae for the *nth* term of these sequences
   (a) 3, 6, 12, 24, ...
   (b) $\frac{1}{4}, \frac{1}{12}, \frac{1}{36}, \frac{1}{100}$
   (c) 10, 100, 1000, 10000, ...

7. Find the sum of
   (a) 6 terms of 5, 10, 21, ...
   (b) 8 terms of 64, 16, 4, ...
   (c) 10 terms of $\frac{1}{2}, \frac{3}{4}, \frac{9}{8}$...

8. The *3rd* and *6th* terms of a G.P. are 108 and -32 respectively. Find the sum of the first 7 terms.

9. The sum of the first two terms of a G.P. is $2\frac{1}{2}$, and the sum of the first four terms is $3\frac{11}{18}$. Find the G.P. if $r > 0$.

10. Find the sum to infinity of the following
    (a) $5 + 1 + 0.2 + 0.04 + ...$
    (b) $6 - 4 + \frac{8}{3}$....

11. Find the sum to infinity of the geometric progression whose first term is 9 and the second term is -3.

[1] Bronson, P.C. (2003): Differential Equations. Schaum's Outline of Theory and Problems of Differential Equations, Second Edition.

Chapter 5

POLYNOMIALS

A polynomial is the sum of separate terms of positive powers of a variable with a constant term. Thus, expressions like $3x^4-2x^3+2.5x-8, 4x^3-2x^2+6x-5$ and $-4x^6-7x^5+\frac{1}{3}x^2-7x+1$ are examples of polynomials. In general, a polynomial in x is an expression of the form:

$$a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + ... + a_2x^2 + a_1x^1 + a_0$$

where n is a positive integer and $a_0, a_1, a_2, ..., a_n$ are the coefficients of the polynomial, with $a_n \neq 0$. The integer n is called the degree of the polynomial.

5.1 Methods of Solutions of Quadratic Equations

Given that a, b *and* c are constants, the polynomial of the form $p(x) = ax^2 + bx + c$ is called a quadratic polynomial(quadratic function) and when $p(x) = 0$ then $ax^2 + bx + c = 0$ is called a quadratic equation in x.

5.1.1 By Factorization

Example 1: Solve the equation $2x^2 + 7x - 4 = 0$
Solution

$$\begin{aligned} 2x^2 + 7x - 4 = 0 \Rightarrow (2x-1)(x+4) &= 0 \\ \Rightarrow 2x - 1 &= 0 \text{ or } x + 4 = 0 \\ \Rightarrow x &= \frac{1}{2} \text{ or } x = -4 \end{aligned}$$

Example 2: Solve by factoring:

don't get

$$\frac{1}{x} + 3 = \frac{2}{x+2}$$

$3x+1=$

$2x^2 + 8x - x - 4$

$2x^2 + 7x - 4$

Solution

$$\begin{aligned} x + 2 + 3x^2 + 6x &= 2x \\ \Rightarrow 3x^2 + 5x + 2 &= 0 \\ \Rightarrow (3x + 2)(x + 1) &= 0 \\ \Rightarrow 3x + 2 &= 0 \text{ or } (x + 1) = 0 \\ \Rightarrow x &= \frac{-2}{3} \text{ or } x = -1. \end{aligned}$$

5.1.2 Completing the Square Method

Example 1: Solve the following quadratic equation by completing the square method

$$x^2 - 6x - 2 = 0$$

a b c

Solution:

$$x^2 - 6x = 2$$

Add the square of one-half of the coefficient of x to both sides

$$\begin{aligned} x^2 - 6x + 9 &= 2 + 9 \\ (x - 3)^2 &= 11 \\ \text{hence } x - 3 &= \pm\sqrt{11} \end{aligned}$$

$$i.e\ x - 3 = \sqrt{17} \text{ or } x - 3 = -\sqrt{17} \Rightarrow x = 3 + \sqrt{17} \text{ or } x = 3 - \sqrt{17}$$

are the two roots of the equation.

Example 2: Solve $2x^2 + 16x - 9 = 0$ by completing the square
Solution:
$2x^2 + 16x - 9 = 0$
Make the coefficient of x^2 term equal to 1

$$\begin{aligned} x^2 + 8x - \frac{9}{2} &= 0 \Rightarrow \\ x^2 + 8x &= \frac{9}{2} \Rightarrow \\ x^2 + 8x + 16 &= \frac{9}{2} + 16 \\ (x + 4)^2 &= \frac{41}{2} \Rightarrow \\ x + 4 &= \pm\sqrt{\frac{41}{2}} \Rightarrow \\ x &= -4 \pm \sqrt{\frac{41}{2}} \end{aligned}$$

$\therefore$ the roots are $-4+\sqrt{\frac{41}{2}}$ or $-4-\sqrt{\frac{41}{2}}$

5.1.3 The Quadratic Formula

Consider the general quadratic equation:

$$ax^2+bx+c=0$$

When we divide through by a, we obtain

$$x^2+\frac{b}{a}x+\frac{c}{a}=0$$

Subtracting $\frac{c}{a}$ from each side, we have

$$x^2+\frac{b}{a}x=-\frac{c}{a}$$

Half of $\frac{b}{a}$ is $\frac{b}{2a}$, which when squared is $\frac{b^2}{4a^2}$. Adding $\frac{b^2}{4a^2}$ to both sides gives

$$x^2+\frac{b}{a}x+\frac{b^2}{4a^2}=-\frac{c}{a}+\frac{b^2}{4a^2}$$

Writing the left side as a perfect square, and combing fractions on the right side, we have

$$\left(x+\frac{b}{2a}\right)^2=\frac{b^2-4ac}{4a^2}$$

$$x+\frac{b}{2a}=\frac{\pm\sqrt{b^2-4ac}}{2a}$$

which gives

$$x=-\frac{b}{2a}\pm\frac{\sqrt{b^2-4ac}}{2a}$$

i.e.

$$x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

To use this formula, we must use the appropriate values of a,b and c.

Example 1: Solve $2x^2-7x-5=0$

Solution:

$a=2\ b=-7\ c=-5$

$$x=\frac{-(-7)\pm\sqrt{(-7)^2-4(2)(-5)}}{2(2)}$$

$$=\frac{7\pm\sqrt{49+40}}{4}$$

$$=\frac{7\pm\sqrt{89}}{4}$$

Example 2: Find the roots of the equation $\frac{x+4}{x+2} = \frac{x+5}{2}$

$$\begin{aligned} 2(x+4) &= (x+5)(x+2) \\ 2x+8 &= x^2+7x+10 \Rightarrow \\ x^2+5x+2 &= 0 \end{aligned}$$

$$a = 1, b = 5, c = 2$$

$$\begin{aligned} x &= \frac{-5 \pm \sqrt{25-4(1)(2)}}{2} \\ &= \frac{-5 \pm \sqrt{17}}{2} \\ &= \frac{-5 \pm 4.123}{2} \end{aligned}$$

$$\therefore \textit{ the roots are } -5 + \frac{4.123}{2} = \frac{-0.877}{2} = -0.44$$

$$\textit{or } -5 - \frac{4.123}{2} = \frac{-9.123}{2} = -4.56$$

5.1.4 Graphical Solution

To solve the equation $ax^2 + bx + c = 0$ graphically, we let $y = ax^2 + bx + c$ and graph the function. The roots of the equation are the $x-$ coordinates of the points for which $y = 0$ ($x-$ *intercepts*). If the graph does not intercept the x - axis, the roots are imaginary. The graph is usually called a parabola.

5.2 Minimum values of a quadratic function

We know that a quadratic function is a function of the form

$$f(x) = ax^2 + bx + c$$

If we let

$$y = ax^2 + bx + c$$

when $a > 0$, the parabola has a minimum value and the curve opens downward

Example 1: Find the minimum value of $y = 3x^2 - 2x + 1$ and for what value of x does it occur?

Solution:
$a = 3 > 0$ hence it is a minimum value and occurs when

$$x = \frac{-b}{2a} = \frac{2}{2(3)} = \frac{2}{6} = \frac{1}{3}$$

$$\text{minimum value} = y = \frac{4ac - b^2}{4a} = \frac{4(3)(1) - (-2)^2}{4(3)}.$$

Example 2: Find the maximum value of $6x - 9 - x^2$.
Solution:
Rewrite as $y = -x^2 + 6x - 9$
$-1 < 0$, thus, maximum value occurs when

$$x = \frac{-b}{2a} = \frac{-6}{2(-1)} = 3$$

maximum value

$$y = \frac{4ac - b^2}{4a} = \frac{4(-1)(-9) - 36}{4(-1)} \frac{36 - 36}{-4} = 0$$

5.3 Roots of Quadratic Equations

5.3.1 Nature of Roots of a Quadratic Equation

The position of the curve relative to the x - axis depends on the nature of the roots of the quadratic equation $ax^2 + bx + c = 0$. Thus, the nature of the roots is determined by the discriminant $b^2 - 4ac$.
Assuming a, b, c are real numbers, then

1. If $b^2 - 4ac < 0$ the roots are imaginary
2. If $b^2 - 4ac > 0$ the roots are real and unequal
3. If $b^2 - 4ac = 0$ the roots are real and equal

Example 1: If the equation $x^2 - 3x + 1 = p(x - 3)$ has real and equal roots, find the values of p.
Solution:

$$\begin{aligned} x^2 - 3x - p(x - 3) + 1 &= 0 \Rightarrow \\ x^2 - 3x - px + 3p + 1 &= 0 \Rightarrow \\ x^2 - x(3 + p) + (1 + 3p) &= 0 \end{aligned}$$

for real roots, $b^2 - 4ac = 0$. Here, $a = 1, b = -(3 + p)$ and $c = (1 + 3p)$. Hence,

$$[-(3 + p)]^2 - 4(1)(1 + 3p) = 0 \Rightarrow 9 + 6p + p^2 - 4 - 12p = 0$$

$$p^2 - 6p + 5 = 0 \Rightarrow (p - 5)(p - 1) = 0 \Rightarrow p = 5 \text{ or } 1$$

5.3.2 Sum and Product of roots

If the equation $ax^2 + bx + c = 0, a \neq 0$, has roots α and β, then, given the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\alpha = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } \beta = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Adding α and β gives

$$\alpha + \beta = \frac{-b + \sqrt{b^2 - 4ac} - b - \sqrt{b^2 - 4ac}}{2a} = \frac{-2b}{2a} = \frac{-b}{a}$$

Product of α and β gives

$$\alpha\beta = (\frac{-b + \sqrt{b^2 - 4ac}}{2a})(\frac{-b - \sqrt{b^2 - 4ac}}{2a}) = \frac{b^2 - (b^2 - 4ac)}{4a^2} = \frac{4ac}{4a^2} = \frac{c}{a}$$

Example 1: If α and β are the roots of the equation $3x^2 - x - 5 = 0$, find $\alpha + \beta$, and $\alpha\beta$
Solution:

$$\alpha + \beta = \frac{-b}{a} = \frac{1}{3}$$

$$\alpha\beta = \frac{c}{a} = \frac{-5}{3}$$

Example 2: If α, β are the roots of $2x^2 - x - 2 = 0$, find the values of $(a)\alpha^2 + \beta^2$ $(b)\alpha - \beta$ $(c)\alpha^2 - \beta^2$ $(d)\frac{1}{\alpha} + \frac{1}{\beta}$

Solution:
$\alpha + \beta = \frac{1}{2}, \alpha\beta = -1$

$(a)\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$

$(b)\alpha - \beta$

$$(\alpha - \beta)^2 = (\alpha + \beta)^2 - 4\alpha\beta = \frac{1}{4} - 4(-1)$$

$$= \frac{1}{4} + 4 = \frac{17}{4}$$

$$\alpha - \beta = \sqrt{\frac{17}{4}} = \frac{\sqrt{17}}{2}.$$

5.3.3 Formulation of equations when the roots are known

Example 1: If $\frac{1}{2}$ and $\frac{3}{2}$ are the roots of an equation form the equation.
Solution:
Let the equation be $x^2 - (\alpha + \beta) + \alpha\beta = 0$

$$\alpha + \beta = \frac{1}{2} + \frac{3}{2} = 2$$

$$\alpha\beta = (\frac{1}{2})(\frac{3}{2}) = \frac{3}{4}$$

The required equation is

$$x^2 - 2x + \frac{3}{4} = 0$$

Multiply through by 4

$$4x^2 - 8x + 3 = 0$$

Example 2 If α and β are the roots of the equation $ax^2 + bx + c = 0$, form an equation whose roots are $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$

Solution:

If the roots are $\frac{\alpha^2}{\beta}$ and $\frac{\beta^2}{\alpha}$

Sum of new roots

$$\frac{\alpha^2}{\beta} + \frac{\beta^2}{\alpha} = \frac{\alpha^3 + \beta^3}{\alpha\beta} = \frac{(\alpha+\beta)^3 - 3\alpha\beta(\alpha+\beta)}{\alpha\beta} = \frac{(-\frac{b}{a})^3 - 3\frac{c}{a}(-\frac{b}{a})}{\frac{c}{a}}$$

$$= \frac{-\frac{b^3}{a^3} + \frac{3bc}{a^2}}{\frac{c}{a}} = \frac{-\frac{b^3+3abc}{a^3}}{\frac{c}{a}} = \frac{3abc - b^3}{a^3} \times \frac{a}{c} = \frac{3abc - b^3}{a^2c}$$

Product of new roots

$$(\frac{\alpha^2}{\beta})(\frac{\beta^2}{\alpha}) = \alpha\beta = \frac{c}{a}$$

Required equation is

$$x^2 - (\frac{3abc - b^3}{a^2c})x + \frac{c}{a} = 0$$
$$a^2cx^2 - (3abc - b^3)x + ac^2 = 0$$

5.3.4 Roots of cubic Equations

Considering the general cubic equations:

$$ax^3 + bx^2 + cx + d = 0 \ldots\ldots(*)$$

(with $a \neq 0$) dividing through by a we obtain.

$$x^3 + \frac{bx^2}{a} + \frac{cx}{a} + \frac{d}{a} = 0$$

Let the roots of the cubic equation in equation (*) be α, β, γ.

$$x^3 + \frac{bx^2}{a} + \frac{cx}{a} + \frac{d}{a} = (x-\alpha)(x-\beta)(x-\gamma)$$

By expanding the right-hand side, we have:

$$(x-\alpha)(x-\beta)(x-\gamma) = x^3 - (\alpha+\beta+\gamma)x^2 + (\alpha\beta + \alpha\gamma + \beta\gamma)x - \alpha\beta\gamma$$

Thus,

$$x^3 + \frac{bx^2}{a} + \frac{cx}{a} + \frac{d}{a} = x^3 - (\alpha+\beta+\gamma)x^2 + (\alpha\beta + \alpha\gamma + \beta\gamma)x - \alpha\beta\gamma$$

By comparing the coefficients, we have:

1. The sum of roots is given by

$$\alpha + \beta + \gamma = \frac{b}{a}$$

2. The sum of products of roots, taking two roots at a time, is given by

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

3. The product of the roots is given by:

$$\alpha\beta\gamma = \frac{-d}{a}$$

Example 1: The roots of $x^3 + bx + c = 0$ are 1, 2, and p. Find b, C, and p
Solution:
Given: $x^3 + bx + c = 0$ The roots are 1, 2 and P.

1. Sum of roots: $1 + 2 + p = 0$ $3 + p = 0$ $p = -3$
2. Products of roots, taking two at a time: $1(2) + 1(p) + 2(p) = b$ $2 + p + 2p = b$ $2 + 3p = b$ $b = 2 + 3p = 2 + 3(-3)$ $b = -7$
3. Products of roots: $(1)(2)(p) = -c$ $2p = -c$, $c = -2p = -2(-3)$ $c = 6$

Exercise

1. The roots of the equation $x^2 + px + q = 0$ are α and β. Given that the roots differ by $2\sqrt{3}$ and that the sum of the reciprocal of the roots is 4, find the possible values of p and q

2. If α and β are the roots of the equation $3x^2 - x - 5$, form the equations whose roots are (a) $5\alpha, 5\beta$ (b) α^3, β^3

3. Find the value of p for which the equation $(x - 2)(x - 3) = p$ has roots which differ by 2

4. If α, β are the roots of $2x^2 - 2x - 1 = 0$, form the equation whose roots are $\frac{\alpha}{\beta}, \frac{\beta}{\alpha}$

5. One root of the equation $27x^2 + bx + 8 = 0$ is known to be the square of the other. Find b

6. Find a if the equation $(5a + 1)x^2 - 8ax + 3a = 0$ has equal roots

7. Find the maximum or minimum values (as appropriate) of the following functions and the value of x where they occur

   (a) $2x^2 + x - 5$

   (b) $(2 - x)(x + 3)$

(c) $1 - 2x - x^2$

8. The roots of a cubic equation α, β, γ are such that

$$\alpha + \beta + \gamma = \frac{3}{2}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{-1}{2}$$

$$\alpha\beta\gamma = \frac{-7}{2}$$

Obtain the cubic equation where roots are $\alpha^2, \beta^2, \gamma^2$

9. The cubic equation $2x^3 + 5x^2 - x - 1 = 0$ has roots p, q, s. Find the equation whose roots are p^3, q^3, s^3.

10. If the roots of $8x^3 + 143x^2 - 10x + 1 = 0$ are a, b, c, obtain new cubic equations whose roots are:

(a) a^2, b^2, c^2

(b) a^3, b^3, c^3

(c) $\frac{1}{a^3}, \frac{1}{b^3}, \frac{1}{c^3}$

(d) $\frac{1}{a^3}, \frac{1}{b^3}, \frac{1}{c^3}$

5.4 Algebra of Polynomials

A polynomial is the sum of separate terms of positive powers of a variable with a constant term. In general, a polynomial in x is an expression of the form:

$$a_n x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + ... + a_2x^2 + a_1x^1 + a_0$$

where n is a positive integer and $a_0, a_1, a_2, ..., a_n$ are the coefficients of the polynomial, with $a_n \neq 0$. The integer n is called the degree of the polynomial.
Expressions like $3x^4 - 2x^3 + 2.5x - 8, 4x^3 - 2x^2 + 6x - 5$ and $-4x^6 - 7x^5 + \frac{1}{3}x^2 - 7x + 1$ are examples of polynomials.

Example 1: If $P(x) = 3x^3 + 2x^2 - 10x - 12$ and $Q(x) = a(x-1)^3 + b(x-1)^2 + c(x-1) + d$, with $P(x) = Q(x)$. Find the values of a, b, c and d

Solution
Given: $P(x) = 3x^3 + 2x^2 - 10x - 12$ and $Q(x) = a(x-1)^3 + b(x-1)^2 + c(x-1) + d$

$$= a(x^3 - 3x^2 + 3x - 1) + b(x^2 - 2x + 1) + cx - c + d$$

$$= ax^3 - 3ax^2 + 3x - a + bx^2 - 2bx + b + cx - c + d$$

$$= ax^3 + (-3a + b)x^2 + (3a - 2b + c)x - (a - b + c - d)$$

since $P(x) = Q(x)$ Thus $3x^3 + 2x^2 - 10x - 12 = ax^3 + (-3a + b)x^2 + (3a - 2b + c)x - (a - b + c - d)$ on collecting coefficients of the x in each power yield

$$\Rightarrow a = 3 \qquad (for\ x^3)$$

$$\Rightarrow -3a + b = 2 \qquad (for\ x^2)$$

$$b = 2 + 3a = 2 + 3(3) = 11$$

$$\Rightarrow 3a - 2b + c = -10 \qquad (for\ x)$$

$$c = -10 - 3a + 2b = -10 - 3(3) + 2(11) = 3$$

$$\Rightarrow a - b + c - d = 12 \qquad (for\ constant)$$

$d = -12 + a - b + c = -12 + 3 - 11 + 3 = -17$ Therefore, $a = 3, b = 11, c = 3$ and $d = -17$.

5.4.1 Addition and Subtraction of polynomials

If $f(x)$ and $g(x)$ are two polynomials, their sum $f(x) + g(x)$ and their difference $f(x) - g(x)$ are also polynomials.

Example 2 Two polynomials are $f(x) = 2x^5 - 3x^4 + 5x^3 - 7x^2 + 1$ and $g(x) = -5x^5 + 3x^4 + 3x^3 - 5x^2 + 6x - 7$. Find (i) $f(x) + g(x)$ (ii) $2f(x) - 3g(x)$

Solution

Given $f(x) = 2x^5 - 3x^4 + 5x^3 - 7x^2 + 1$ $g(x) = -5x^5 + 3x^4 + 3x^3 - 5x^2 + 6x - 7$

1 $f(x) + g(x) = (2x^5 - 3x^4 + 5x^3 - 7x^2 + 1) + (-5x^5 + 3x^4 + 3x^3 - 5x^2 + 6x - 7)$

$$= 2x^5 - 3x^4 + 5x^3 - 7x^2 + 1 - 5x^5 + 3x^4 + 3x^3 - 5x^2 + 6x - 7$$

$$= 2x^5 - 5x^5 - 3x^4 + 3x^4 + 5x^3 + 3x^3 - 7x^2 - 5x^2 + 6x + 1 - 7$$

$$= (2 - 5)x^5 - (3 - 3)x^4 + (5 + 3)x^3 - (7 + 5)x^2 + 6x - 6$$

$$= -3x^5 + 8x^3 - 12x^2 + 6x - 6$$

2 $2f(x) - 3g(x) = 2(2x^5 - 3x^4 + 5x^3 - 7x^2 + 1) - 3(-5x^5 + 3x^4 + 3x^3 - 5x^2 + 6x - 7)$

$$= 4x^5 - 6x^4 + 10x^3 - 14x^2 + 2 + 15x^5 - 9x^4 - 9x^3 5x^2 - 18x + 21$$

$$= 19x^5 - 15x^4 + x^3 - 23x^2 - 18x + 23$$

$= 19x^2 - 15x^4 + x^3 + x^2 - 18x + 23$

5.4.2 Multiplication of Polynomials

Product of two or more polynomials can also be found and yielding another polynomial.

Example 3:

If $g(x) = 2x^2 - x - 4$ and $h(x) = x^2 + x - 1$, find the product of $g(x) and h(x)$. Hence, find the coefficients of $x^3 and x^4$ in the product

Solution

Given $g(x) = 2x^2 - x - 4$ and $h(x) = x^2 + x - 1$ $g(x)\,h(x) = (2x^2 - x - 4)(x^2 + x - 1)$

$$= 2x^2(x^2 + x - 1) - x(x^2 + x - 1) - 4(x^2 + x - 1)$$

$$= 2x^4 + 2x^3 - 2x^2 - x^3 - x^2 + x - 4x^2 - 4x + 4$$

$$= 2x^4 + x^3 - 7x^2 - 3x + 4$$

The coefficient of x^3 is 1, and the coefficient of x^4 is 2.
Note that in solving problems on algebra of polynomials, there are needs to repeatedly apply axioms such as the commutative and associative axioms for addition and multiplication, and the distributive axiom:

$$a(b + c) = ab + ac = (b + c)a$$

5.4.3 Division of Polynomials

A polynomial $f(x)$ can be divided by another polynomial $g(x)$, (mostly degree of polynomial of $g(x)$ is less than the degree of polynomial of $f(x)$), and thus obtain a quotient $q(x)$ as well as a remainder, $r(x)$. It must be noted that $g(x) \neq 0$. Both $q(x)$ and $r(x)$ are also polynomials. Thus,

$$Polynomial \;\; f(x) = ((Divisor\; g(x)\; X\; Quotient q(x)) + Remainder(x)$$

Example 4: Divide $2x^4 + 3x^3 + 9x^2 - 25x - 6$ by $x^2 + x - 6$
Solution Division of polynomial is usually done by long division (similar to long division of numbers), using $x^n - term$ of the divisor.
$f(x) = g(x).q(x) + r(x)$

$$\Rightarrow \frac{f(x)}{g(x)} = q(x) + \frac{r(x)}{g(x)}$$

Note that the degree of r(x) is less than the degree of the divisor $g(x)$. So, the quotient is $(2x^2 + 5x + 2)$ and the remainder is $(-57x - 18)$. Thus,

$$(2x^4 + 3x^3 + 9x^2 - 25x - 6) = [(x^2 + x - 6).(2x^2 + 5x + 2)] + (-57x - 18)$$

5.5 Remainder Theorem

If $f(x) = x^2 - 5x + 1$ is divided by $g(x) = x - 2$, the division yields:
Thus, it is observed that the divisor above is $(x-2)$. On $x = 2$ into $f(x) = (2)^2 - 5(2) + 1 = -5$

which is exactly the remainder. Hence, it can be concluded that if a polynomial $P(x)$ is divided by $(x - k)$, then the remainder is $P(x)$ or $P(x = k)$. This is known as Remainder Theorem. This can be proved as:

$$Polynomial f(x) = (divisor g(x) X quotient q(x)) + remainder R$$

$$i.e.\ f(x) = [g(x).q(x)] + R, \qquad ,let g(x) = x - k$$
$$f(x) = (x - k)q(x) + R \qquad ,let x = k$$
$$f(k) = (k - k)q(x) + R$$
$$f(k) = 0 + R$$
$$f(k) = R.$$

Also, if $f(x)$ is divided by $ax + b$, then $f(x) = (ax + b)q(x) + R$. This is also remainder theorem. This can also be proved as:

$$f(x) = (ax + b)q(x) + R \quad 'let\ x = \frac{-b}{a}$$
$$f(\frac{-b}{a}) = (a.\frac{-b}{a} + b)q(\frac{-b}{a}) + R$$
$$f(\frac{-b}{a}) = 0.q(\frac{-b}{a}) + R$$
$$\Rightarrow f(\frac{-b}{a}) = R$$

Example 5: If $f(x) = 2x^3 - 3x^2 - 5x + 1$. What are the remainders when $f(x)$ is divided by (a) $x - 3$ (b) $x + 3$ (c) $2x - 5$?

Solution Given: $f(x) = 2x^3 - 3x^2 - 5x + 1$
(a) Divisor is $x - 3$. Let $x = 3$.,
Then, the remainder $R = f(3) = 2(3)^3 - 3(3)^2 - 5(3) + 1 = 13$
(b) Divisor is $x + 3$. Let $x = -3$.
Then, the remainder $R = f(-3) = 2(-3)^3 - 3(-3)^2 - 5(-3) + 1 = -65$
(c) Divisor is $2x - 5$. Let $x = \frac{5}{2}$.
Then, the remainder $R = f(\frac{5}{2}) = 2(\frac{5}{2})^3 - 3(\frac{5}{2})^2 - 5(\frac{5}{2}) + 1 = 1$

Note: If a polynomial of degree $n \geq 2$ is divided by $(x - a)(x - b)$, the remainder will be $Ax + B$ where both A and B are constants. Thus,

$$f(x) = (x - a)(x - b) + (Ax + B)$$

Remember that $R = Ax + B$.

Example 6: Find the remainder when $x^5 - x - 1$ is divided by $(x - 1)(x - 2)$
Solution
Given: $f(x) = x^5 - x - 1$

$$(x - 1)(x - 2)q(x) + (Ax + B)$$
$$\Rightarrow x^5 - x - 1 = (x - 1)(x - 2)q(x) + (Ax + B) \quad \ldots\ldots(1)$$

Putting $x = 1$ into equation (1) yields $-1 = A + B$ $\ldots\ldots(2)$

Putting $x = 2$ into equation (1) yields $29 = 2A + B$ (3)

Solving equations (2) and (3) simultaneously for A and B, we have

$$A = 30 \text{ and } B = -3$$

Hence, the remainder is $30x - 31$

5.6 Factor Theorem

As earlier stated, if a polynomial $f(x)$ is divided by another polynomial $g(x)$, the result yields the quotient with a remainder $r(x) = R$

$$i.e. f(x) = g(x).q(x) + R, \quad if g(x) = x - k$$

$$f(x) = (x - k)q(x) + R$$

If the remainder equals zero, then:

$$f(x) = (x - k)(q(x)$$

This implies that $x - k$, which is the divisor, is a factor of $f(x)$.

Factor Theorem
Let $f(x)$ be a polynomial and k is a real constant. Then $(x - k)$ is a factor of $f(x)$, if and only if $f(k) = 0$

Example 1:Prove that $(x - 2)$ is a factor of $x^3 - 5x^2 + 4x + 4$
Proof
Given: $f(x) = x^3 - 5x^2 + 4x + 4$ If $x = 2$, $f(2) = (2)^3 - 5(2)^2 + 4(2) + 4$ Since $f(2) = 0$ which means the remainder equals zero,thus by factor theorem $(x-2)$ is a factor of $x^3 - 5x^2 + 4x + 4$

Example 2: Given that:$x^2 - 3x + 2$ is a factor of $x^4 + ax^2 + bx + 8$, find the values of a and b and completely factorize the polynomial.
Solution
Given: $f(x) = x^4 + ax^2 + bx + 8$ since $x^2 - 3x + 2 = (x - 1)(x - 2)$, then both $(x - 1)$ and $(x - 2)$ are both factors of $f(x) = x^4 + ax^2 + bx + 8$ By factor theorem,

$$when \quad x = 1, \quad f(1) = a + b + 9 = 0 \ldots\ldots\ldots (1)$$

$$when \quad x = 2, \quad f(2) = 4a + 2b + 24 = 0 \ldots\ldots\ldots (2)$$

On solving equations (1) and (2) simultaneously, we get

$$a = -3 \text{ and } b = -6$$

Thus, $f(x) = x^4 - 3x^2 - 6x + 8$

Exercise

1. If $h(x) = 3x^6 - 7x^5 - 11x^4 - 4x^3 + x^2 + 13x - 7$, find (i) h(0) (ii) h(-1) (iii) h(1) (iv) h(2) (v) h(3) (vi) h(11)

2. In each of the following equal polynomials, find the values of the unknown constants
   (i) $ax^4 + bx^3 + cx^2 + d = (x + 1)(2x + 3)(x - 4)(x - 5)$
   (ii) $A(x - 1)^3 + B(x - 1)^2 + c(x - 1) - D = 7x^3 + 6x^2 + 2x - 7$
   (iii) $(x - 1)(Ax^2 + Bx + C) + D = x^3 + 2x^2 + x + 5$
   (iv) $(x - k)^4 = x^4 + ax^3 + bx^2 + cx + 16$

3. In each of the pair of polynomials in number (i) to (iv), find (a) $f(x)+g(x)$ (b) $f(x)-g(x)$ (c)$3f(x) + 4g(x)$ (d) $2f(x) - g(x)$ (e) $f(x).g(x)$
   (i) $f(x)6x^5 + 7x^4 - 5x^2 + 1$; $g(x) = 3x^4 + 7x^3 - 8x^2 + 5x - 2$
   (ii) $f(x) = 4x^7 + 7x^6 - 5x^5 + 1$; $g(x)3x^4 - 7x^2 + 8$
   (iii) $f(x) = x^3 - 5x^2 + 4x + 8$; $g(x)x + 1$
   (iv) $f(x) = x^3 - 9x^2 + 7x + 6$; $g(x)x^3 - x^2 - x + 1$

4. In each of the following polynomials, find both the quotient and remainder when $g(x)$ is divided by $h(x)$:
   (i) $g(x) = 3x^3 + 8x^2 - 4x + 25$: $h(x) = x + 5$ (ii) $g(x) = 2x^4 - 3x^2 + 4x + 1$: $h(x)2x - 5$
   (iii) $(g(x) = x^4 - 1; h(x) = x - 1$ (iv) $g(x) = 4x^3 - 3x^2 - 2x - 1$. $h(x)x^2 + 2x + 1$

5. Find the remainder when:
   (i) $f(x) = 3x^3 + 2x^2 + x + 1$ *is divided by* $g(x) = x - 3$
   (ii) $f(x) = 4x^4 - 8x^3 + 1$ *is divided by* $g(x) = (x - 1)(x + 2)$
   (iii) $f(x)2x^3 - 5x^2 + x - 2$ *is divided by* $g(x) = 2x - 1$

5.6.1 Partial Fractions

A rational function is a function obtained by dividing one polynomial by another.

$$R(x) = \frac{Q(x)}{P(x)}.$$

(Note that $P(x) \neq 0$). The following simple rational fractions are known as Partial fractions:

$$\frac{A}{(x-k)} \qquad \frac{Bx+c}{(ax^2+bx+c)} \qquad \frac{d}{(x-k)^2}$$

Here A, B, C and D are constants.
Any rational function $\frac{Q(x)}{P(x)}$ can be written as the sum of a polynomial and partial fractions. The polynomial is the quotient when $Q(x)$ is divided by $P(x)$. The partial fraction is found as follows:

- For each linear factor $(x - k)$, include $\frac{A}{x-k}$.
- For each quadratic factor $(ax^2 + bx + c)$ of $P(x)$, include $\frac{Bx+C}{(ax^2+bx+c)}$

- For each repeated linear factor $(x-k)^2$, include $\frac{B}{x-k}+\frac{E}{(x-k)^2}$

The constants A, B, C, D and E are found by putting in values for x, or by equating coefficients of powers of x.

Example 1: Resolve $\frac{3}{(x+2)(2x-1)}$ into partial fractions

Solution

Given: $\frac{3}{(x+2)(2x-1)}=\frac{A}{x+2}+\frac{B}{2x-1}$ $\frac{3}{(x+2)(2x-1)}=\frac{A(2x-1)+B(x+2)}{(x+2)(2x-1)}$

Multiply both sides by $(x+2)(2x-1)$

$$\Rightarrow 3=A(2x-1)+B(x+2).....(*)\ .$$

Putting $x=-2$ into equation(*) gives:

$$3=-5A$$

$$A=\frac{-3}{5}$$

And putting $x=\frac{1}{2}$ into equation (*) gives:

$$3=\frac{5}{2}B$$

$$B=\frac{6}{5}$$

Thus,

$$\frac{3}{(x+2)(2x-1)}=\frac{A}{x+2}+\frac{B}{2X-1}=\frac{\frac{-3}{5}}{x+2}+\frac{\frac{6}{5}}{2x-1}=\frac{6}{5(2x+1)}-\frac{3}{5(x+2)}$$

Example 2: Resolve $\frac{3x}{(x+1)(x-2)}$ into partial fractions.

Solution

Let

$$\frac{3x}{(x+1)(x-2)}=\frac{A}{x+1}+\frac{B}{x-2}$$

Thus ,

$$3x=A(x-2)+B(x+1).....(*)$$

Putting $x=-1$, into equation (*) gives:

$$-3=-3A$$

$$A=1$$

Also Putting $x=2$ into equation (*) gives:

$$6=3B$$

$$\frac{3x}{(x+1)(x-2)} = \frac{A}{(x+1)} + Bx - 2 = \frac{1}{x+1} + \frac{2}{3(x-2)}$$

Example 3: Resolve $\frac{2x+3}{(x+1)(2x+1)(x-3)}$ into partial fractions

Solution

$$\frac{2x+3}{(x+1)(2x+1)(x-3)} = \frac{A}{x+1} + \frac{B}{2x+1} + \frac{C}{x-3}$$

$$\Rightarrow 2x+3 = A(2x+1)(x-3) + B(x+1)(x-3) + C(x+1)(2x+1) \quad (*)$$

Putting $x = -1$ into equation (*) gives:

$$1 = A(-1)(-4)$$

$$1 = 4A$$

$$A = \frac{1}{4}$$

Putting $x = \frac{-1}{2}$ into equation (*) gives:

$$2 = B(\frac{1}{2})(\frac{-7}{2})$$

$$2 = \frac{-7}{4}B$$

$$B = \frac{-8}{7}$$

And putting $x = -3$ into equation (*) gives:

$$9 = C(4)(7)$$

$$9 = 28C$$

$$C = \frac{9}{28}$$

Hence,

$$\frac{2x+3}{(x+1)(2x+1)(x-3)} = \frac{A}{x+1} + \frac{B}{2x+1} + \frac{C}{x-3} = \frac{1}{4(x+1)} - \frac{8}{9(2x+1)} + \frac{9}{28(x-3)}$$

Example 4: Resolve the following into partial fractions

$$\frac{x+1}{(x-1)(x^2+1)}$$

Solution

Let

$$\frac{x+1}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$$

$$x+1 = A(x^2+1) + (Bx+C)(x-1)$$

Where $x = 1$

$$\Rightarrow 2 = 2A$$

$$A = 1$$

So.

$$\frac{x+1}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1} = \frac{1}{x-1} + \frac{Bx+C}{x^2+1}$$

$$\frac{x+1}{(x-1)(x^2+1)} - \frac{1}{x-1} = \frac{Bx+C}{x^2+1}$$

$$\frac{x+1-x^2-1}{(x-1)(x^2+1)} = \frac{Bx+C}{x^2+1}$$

$$\frac{x-x^3}{(x-1)(x^2+1)} = \frac{Bx+C}{x^2+1}$$

$$\frac{-x(x-1)}{(x-1)(x^2+1)} = \frac{Bx+C}{x^2+1}$$

$$\frac{-x}{x^2+1} = \frac{Bx+C}{x^2+1}$$

Thus.

$$-x = Bx + C$$

Then

$$B = -1, C = 0$$

Hence.

$$\frac{x+1}{(x-1)(x^2+1)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+1}$$

$$\frac{1}{x-1} - \frac{x}{x^2+1}$$

Example 5: Resolve $\frac{3x+1}{(x+1)(x-1)^2}$ into partial fractions.

Solution

Let

$$\frac{3x+1}{(x+1)(x-1)^2} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2}$$

$$3x+1 = A(x-1)^2 + B(x+1)(x-1) + C(x+1) \quad (*)$$

Put $x = -1$ into equation (*) to give:

$$-2 = 4A$$

$$A = \frac{-1}{2}$$

Putting $x = 1$ into equation (*) gives:

$$4 = 2C$$

$$C = 2$$

Equate the coefficients of x^2 in equation(*) gives:

$$B = -A = \frac{1}{2}$$

$$\frac{3x+1}{(x+1)(x-1)^2} = \frac{A}{x+1} + \frac{B}{x-1} + \frac{C}{(x-1)^2} = \frac{-1}{2(x+1)} + \frac{1}{2(x-1)} + \frac{2}{(x-1)^2}$$

Exercise

Express the following in partial fraction:

1. $\frac{2}{x(x+1)}$
2. $\frac{1}{(x-1)(x+3)}$
3. $\frac{4x}{(x+2)(x+3)}$
4. $\frac{x-5}{x^2-9}$
5. $\frac{4x-3}{2x^2-5x-3}$
6. $\frac{6}{(x+1)(x-2)(x+3)}$
7. $\frac{6}{(x+2)(x-1)^2}$
8. $\frac{2x-3}{x^2(x-1)}$
9. $\frac{2x^2-5x+6}{(2x-1)^2(x+2)}$
10. $\frac{5x^2-2x+11}{(x+1)(x^2-3x+5)}$

① the each linear factor $(x+1)$ includ $\frac{A}{x^2}$

② for each

Chapter 6

BINOMIAL THEOREM

6.1 Permutation and Combination

6.1.1 Factorials

Mathematically, it is established that $1 \times 2 \times 3 \times 4 \times 5 \times 6 \times 7 = 7!$ This implies that $n! = n \times (n-1) \times (n-2) \times \cdots \times 3 \times 2 \times 1$ Or alternatively, $n! = n(n-1)(n-2) \cdots 3\ 2\ 1$ $= n(n-1)(n-2)!$ $= n(n-1)!$ It should be noted that $1! = 1$ *and* $0! = 1$

Permutation: Permutation is the number of ways n-number of objects can be arranged by taking r-number of the objects or the whole n-number of objects. Mathematically, it is denoted as ${}^nP_r$ (Which reads, n permutation r).

$$ {}^nP_r = \frac{n!}{(n-r)!} $$

Example: In how many ways can 8 objects be arranged from 10 objects.

Solution

Number of ways

$$ = {}^{10}P_8 = \frac{10!}{(10-8)!} = \frac{10!}{2!} = \frac{10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 1,814,400\ ways $$

Example: Find the number of ways of arranging 4 different conical flasks from 6 conical flasks.

Solution

Number of ways

$$ = {}^6P_4 = \frac{6!}{(6-4)!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1} = 360 ways $$

In addition, the permutation or arrangements of n-numbers of objects being taken together when p-number of objects are alike,q-number of the objects are alike and r-number of objects are alike is

$$ \frac{n!}{p!q!r!} \qquad where (p+q+r) = n $$

Example: Find the number of arrangements which can be made by using the letters of the

word 'NIGERIAN'.

Solution

There are 8 letters in the word "NIGERIAN" , out of which N appears twice, I appears twice and G,E.R,A appears once each.

So, number of ways

$$= \frac{8!}{2!2!1!1!1!1!} = \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{2 \times 1 \times 2 \times 1} = 10,080$$

Combination

As earlier stated, permutation takes into consideration the number of r-objects taken from n-objects. For instance, if two letters are to be arranged from 5 letters : a, b, c, d, e. The arrangement will be:

$ab \quad ac \quad ad \quad ae \quad ba \quad bc \quad bd \quad be \quad ca \quad cb \quad cd \quad ce \quad da \quad db \quad dc \quad de \quad ea \quad eb \quad ec \quad ed$

Thus the number of arrangement is 20.

i.e.

$$^5P_2 = \frac{5!}{(5-2)!} = \frac{5!}{3!} = \frac{5 \times 4 \times 3 \times 2 \times 1}{3 \times 2 \times 1} = 20$$

However, suppose we are not interested in the actual order of the letters but only in the number of selection of the letters . Then the selection will be;

$ab \quad ac \quad ad \quad ae \quad bc \quad bd \quad be \quad cd \quad ce \quad de$

(Here ab is the same as ba, and so on)

Thus, the number of selection is 10

Hence, combination (otherwise called selection) is the set of r-objects chosen from n-objects where attention is not given to the order of the objects.

Mathematically, ^nC_r (which reads n combination r) is denoted by:

$$^nC_r = \frac{^nP_r}{r!} = \frac{n!}{r!(n-r)!}$$

Example: In how many ways can four boys be chosen from 6 boys?

Solution

Number of ways

$$= ^6C_4 = \frac{6!}{4!(6-4)!} = \frac{6!}{4!2!} = \frac{6 \times 5 \times 4 \times 3 \times 2 \times 1}{4 \times 3 \times 2 \times 1 \times 2 \times 1} = 15$$

Example: In how ways can a team of 11 players/boys be selected from 14 boys?

Solution

Number of ways

$$= ^{14}C_{11} = \frac{14!}{11!(14-11)!} = \frac{14 \times 13 \times 12 \times 11!}{11! \times 3 \times 2 \times 1} = 364 ways.$$

Note

1. $n! = n(n-1)(n-2)\cdots 3\cdot 2\cdot 1 = n(n-1)!$
2. ${}^nP_r = \frac{n!}{(n-r)!} = n(n-1)(n-2)\cdots(n-r+1)$
3. ${}^nP_n = \frac{n!}{(n-n)!} = \frac{n!}{0!} = n! = n(n-1)(n-2)\cdots 3\cdot 2\cdot 1$
4. ${}^nC_r = \frac{{}^nP_r}{r!} = \frac{n!}{r!(n-r)!}$
5. ${}^{n+1}C_r = {}^nC_r + {}^nC_{r-1}$
6. ${}^nC_r = {}^nC_{n-r}$

Exercise 6.1

1. Find the values of the following: (i) $\frac{17!}{8!10!}$ (ii) $\frac{10!}{8!}$ (iii) ${}^8P_3$ (iv) ${}^{17}P_{15}$ (v) ${}^3C_2$ (vi) ${}^{100}C_{86}$
2. In how many ways can the following words be arranged: (i) MATHEMATICS (ii) STATIONARY (iii) LAWAL (iv)MEDIAN
3. Six people sit at a round table. In how many ways can two of them sit next to each other?
4. Seven men volunteer to dig a well in a community. Find the number of ways five of them can be chosen for the job.

6.1.2 Binomial Expansion

The binomial expansion is the algebraic expressions with only two terms which are raised to a certain power.

For instance:

$(a+b)^0 = 1$

$(a+b)^1 = a + b$

$(a+b)^2 = (a+b)(a+b) = a^2 + 2ab + b^2$

$(a+b)^3 = (a+b)(a+b)(a+b) = a^3 + 3a^2b + 3ab^2 + b^3$

$(a+b)^4 = (a+b)(a+b)(a+b)(a+b) = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + a^4$

$(a+b)^5 = (a+b)(a+b)(a+b)(a+b)(a+b) = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$

Thus,

$(a+b)^0 = 1$

$(a+b)^1 = a + b$

$(a-b)^2 = a^2 + 2ab + b^2$

$(a+b)^3 = a^3 + 3a^2b + 3ab^2 - b^3$

$(a+b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + a^4$

$(a+b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$

The following points are noted from the above expansion;

1. Each expansion is homogeneous (that is, the sum of the powers of a and b in each term is constant and equal to the power of the binomial)

2. While the powers of a are decreasing, the powers of b are ascending.

3. With the powers arranged in this way, the coefficients of the form the terms shown below:

$(a+b)^0$ 1
$(a+b)^1$ 1 1
$(a+b)^2$ 1 2 1
$(a+b)^3$ 1 3 3 1
$(a+b)^4$ 1 4 6 4 1
$(a+b)^5$ 1 5 10 10 5 1

This pattern of coefficients is known as Pascal's Triangle.

Example: Expand (i) $(a+b)^5$ (ii) $(x+1)^5$ (iii) $(3x-2)^5$

Solution

(i) $(a+b)^5 = (a+b)(a+b)(a+b)(a+b)(a+b) = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$

(ii) $(x+1)^5$, here let $a = x$ and $b = 1$.

Thus
$$\begin{aligned}(x+1)^5 &= x^5 + 5x^4(1) + 10x^3(1)^2 + 10x^2(1)^3 + 5x(1)^4 + (1)^5 \\ &= x^5 + 5x^4 + 10x^3 + 10x^2 + 5x + 1.\end{aligned}$$

(iii) For $(3x-2)^5$, let $a = 3x$ and $b = -2$.

$$\begin{aligned}(3x-2)^5 &= (3x+(-2))^5 \\ &= (3x)^5 + 5(3x)^4(-2) + 10(3x)^3(-2)^2 + 10(3x)^2(-2)^3 + 5(3x)(-2)^4 + (-2)^5 \\ &= 243x^5 - 810x^4 1080x^3 - 720x^2 + 240x - 32.\end{aligned}$$

Example: Find the exact value of $(1.01)^5$ using the binomial expansion.

Solution
Since $(a+b)^5 = a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + b^5$ Then, $(1.05)^5 = (1+0.01)^5$ (where $a = 1$ and $b = 0.01$)

$$= (1)^5 + 5(1)^4(0.01) + 10(1)^3(0.01)^2 + 10(1)^2(0.01)^3 + 5(1)(0.01)^4 + (0.01)^5$$

$$1 + 0.05 + 0.001 + 0.00001 + 0.00000005 + 0.0000000001$$

$$= 1.0510100501$$

We see from binomial expansion that

$$(a+b)^0 = 1$$

$$(a+b)^1 = a+b$$

$$(a+b)^2 = a^2+2ab+b^2$$

$$= ({}^2C_0)a^2b^0 + ({}^2C_1)a^1b^1 + ({}^2C_2)a^0b^2$$

$$(a+b)^3 = a^3+3a^2b+3ab^2+b^3$$

$$= ({}^3C_0)a^3b^0 + ({}^3C_1)a^2b^1 + ({}^3C_2)a^1b^2 + ({}^3C_3)a^0b^3$$

$$(a+b)^4 = a^4+4a^3b+6a^2b^2+4ab^3+a^4$$

$$= ({}^4C_0)a^4b^0 + ({}^4C_1)a^3b^1 + ({}^4C_2)a^2b^2 + ({}^4C_3)a^1b^3 + ({}^4C_4)a^0b^4$$

$$(a+b)^5 = a^5+5a^4b+10a^3b^2+10a^2b^3+5ab^4+b^5$$

$$= ({}^5C_0)a^5b^0 + ({}^5C_1)a^4b^1 + ({}^5C_2)a^3b^2 + ({}^5C_3)a^2b^3 + ({}^5C_4)a^1b^4 + ({}^5C_5)a^0b^5$$

Hence,

$$(a+b)^n = ({}^nC_0)a^nb^0 + ({}^nC_1)a^{n-1}b^1 + ({}^nC_2)a^{n-2}b^2 + \cdots + ({}^nC_{n-1})a^1b^{n-1} + ({}^nC_n)a^0b^n$$

This results in binomial theorem.

Binomial theorem gives the general algebraic expansion of $(a+b)^n$ as

$$({}^nC_0)a^nb^0 + ({}^nC_1)a^{n-1}b^1 + ({}^nC_2)a^{n-2}b^2 + \cdots + ({}^nC_{n-1})a^1b^{n-1} + ({}^nC_n)a^0b^n$$

Notes

1. There are $(n+1)$ terms in the expansion $(a+b)^n$
2. Each term is homogeneous of degree n.
3. The coefficients are symmetrical (that is, the first coefficient is ${}^nC_0$, the second coefficient is ${}^nC_1$, ... the last coefficient is ${}^nC_n$).
4. The coefficient of b^r is ${}^nC_r a^{n-r}$ and the coefficient of a^{n-r} is ${}^nC_r b^r$

Example Find the first four terms of $(x+1)^{11}$ in descending powers of x

Solution. For $(x+1)^{11}$, here $n=11$. Thus,

$$(x+1)^{11} = {}^{11}C_0x^{11}1^0 + {}^{11}C_1x^{11-1}1^1 + {}^{11}C_2x^{11-2}1^2 + {}^{11}C_3x^{11-3}1^3 +$$

$$= (1)x^{11} + (11)x^{10} + (55)x^9 + (165)x^8 +$$

$$= x^{11} + 11x^{10} + 55x^9 + 165x^8 +$$

Exercise 7.2

1. Expand the following binomial expansion and simplify as far as poss
   (i) $(b+6)^6$ (ii) $(2x+3y)^7$ (iii) $(1-2x)^6$ (iv) $(3a-2b)^8$ (v) $\left(\frac{2}{x}+\frac{x}{7}\right)^4$

2. Expand the following up to the terms indicated:
   (i) $(1-x)^7$, x^2 term
   (ii) $(1-x^2)^7$, x^6 term
   (iii) $(2a-3b)^6$, a^2b^4 term
   (iv) $(x^3+\frac{1}{x})^7$, x^8 term

3. Using binomial theorem, expand;
   (i) $(1.001)^8$
   (ii) $(0.99)^5$
   (iii) $(1.01)^9$

Chapter 7

MATRICES AND DETERMINANTS

7.1 Matrices

Definition of a matrix: A *matrix* is a rectangular array of numbers or variables enclosed in a bracket. The dimension of a given matrix is determined by its number of rows and number of columns. The row number always precedes the column number. Eg. If a matrix is an $m \times n$ matrix, it means that the matrix has m number of rows and n number of columns.

In general, a matrix $\mathbf{A}$ of order $m \times n$ can be written as

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} = [a_{ij}]_{m \times n}.$$

The quantities a_{ij}, $1 \leq i \leq m$, $1 \leq j \leq n$ are known as the *entries* or *components* or *elements* of the matrix. a_{ij} denotes the element in the ith row and the jth column.

Examples of matrices are

(i) $\mathbf{B} = \begin{bmatrix} a & b \\ c & e \end{bmatrix}$, 2×2 matrix.

(ii) $\mathbf{Y} = \begin{bmatrix} 1 & 4 & 3 \\ 2 & 1 & 2 \end{bmatrix}$, 2×3 matrix.

(iii) $\mathbf{Z} = \begin{bmatrix} 0 & 2 \\ 1 & 5 \\ 3 & 1 \end{bmatrix}$, 3×2 matrix.

(iv) $\mathbf{X} = \begin{bmatrix} x_{11} \\ x_{21} \\ \vdots \\ x_{n1} \end{bmatrix}$, $n \times 1$ matrix.

7.2 Types of Matrices

The major types of matrices include the following:

(i) **Square matrix**: A matrix **A** is called a *square matrix* if it has equal number of rows and columns. For example,

$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ is 2×2 (square) matrix and $\mathbf{B} = \begin{bmatrix} 2 & 1 & 3 \\ 1 & 1 & 4 \\ 5 & 2 & 1 \end{bmatrix}$ is 3×3 (square) matrix.

(ii) **Null or Zero matrix**: A matrix **A** is called a *null matrix* if all its components are zeros. It is denoted as **0**. Examples are:

$[0 \quad 0 \quad 0]$ is a null matrix of order 1×3 and $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is 3×3 null matrix.

(iii) **Identity or Unit matrix**: A square matrix whose entries on the main diagonal are ones and other entries are zeros is called an *Identity* or *unit matrix*. Examples are:

$$\mathbf{I}_{2\times2} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \mathbf{I}_{3\times3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{I}_{4\times4} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

(iv) **Diagonal matrix**: A square matrix whose entries are zeros except on the main (or leading or principal) diagonal is called a *diagonal matrix*. Examples are

$$\mathbf{D}_{2\times2} = \begin{bmatrix} a & 0 \\ 0 & e \end{bmatrix} = \text{diag}\begin{bmatrix} a & e \end{bmatrix}, \quad \mathbf{D}_{3\times3} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{3}{7} & 0 \\ 0 & 0 & 6 \end{bmatrix} = \text{diag}\begin{bmatrix} 1 & \frac{3}{7} & 6 \end{bmatrix}$$

$$\mathbf{D}_{4\times4} = \begin{bmatrix} 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix} = \text{diag}\begin{bmatrix} 3 & 0 & 7 & 2 \end{bmatrix}.$$

(v) **Upper and Lower diagonal matrices**: An *Upper diagonal matrix*, **U**, is a square matrix whose entries underneath the leading diagonal are zeros. A *Lower diagonal matrix*, **L**, is a square matrix whose **all** entries above the leading diagonal are zeros. Examples are:

$$\mathbf{U} = \begin{bmatrix} 2 & 1 & 3 \\ 0 & 4 & 2 \\ 0 & 0 & 3 \end{bmatrix}, \quad \mathbf{L} = \begin{bmatrix} 2 & 0 & 0 \\ 1 & 4 & 0 \\ 5 & 3 & 5 \end{bmatrix}.$$

(vi) **Symmetric matrix**: A square matrix in which entries $a_{ij} = a_{ji}$ is called a *symmetric matrix*. That is, $a_{12} = a_{21}$, $a_{13} = a_{31}$ etc. Examples are:

$$\mathbf{A} = \begin{bmatrix} a & b & c \\ b & d & e \\ c & e & f \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 2 & 7 & 3 \\ 7 & 1 & 5 \\ 3 & 5 & 9 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

(vii) **Skew symmetric matrix**: A square matrix with all entries on the leading diagonal equal to zero while the off-diagonal entries are given as $a_{ij} = -a_{ji}$. Examples are:

$$\mathbf{A} = \begin{bmatrix} 0 & 7 & -1 \\ -7 & 0 & -5 \\ 1 & 5 & 0 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

(vii) **Row matrix**: A *row matrix* is a matrix with one row. Eg. $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} 2 & 1 & 4 \end{bmatrix}$.

(viii) **Column matrix**: A *column matrix* is a matrix with a single column, such as

$$\mathbf{A} = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 1 \\ 4 \\ 1 \\ 3 \end{bmatrix}.$$

Equality of two matrices

Two $m \times n$ matrices, $\mathbf{A}$ and $\mathbf{B}$, are said to be *equal* if and only if all their corresponding entries are equal. That is, for all $1 \leq i \leq m$ and $1 \leq j \leq n$, the entries $a_{ij} = b_{ij}$. Two equal matrices $\mathbf{A}$ and $\mathbf{B}$ can be written as $\mathbf{A} = \mathbf{B}$. Two equal matrices have the same order.

7.3 Algebra of Matrices

Addition/Subtraction of Matrices

Let $\mathbf{A} = [a_{ij}]_{m \times p}$ and $\mathbf{B} = [b_{ij}]_{m \times p}$ be two matrices of the same order, the *sum of the two matrices*, denoted as $\mathbf{A} + \mathbf{B}$, is defined as the matrix

$$\mathbf{A} + \mathbf{B} := [a_{ij} + b_{ij}].$$

Let $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$,

$$\mathbf{A} + \mathbf{B} = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} & a_{13} + b_{13} \\ a_{21} + b_{21} & a_{22} + b_{22} & a_{23} + b_{23} \\ a_{31} + b_{31} & a_{32} + b_{32} & a_{33} + b_{33} \end{bmatrix}$$

Let $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ be matrices of the same order. The matrix addition satisfies the following properties:

- *Commutativity*: $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$.
- *Associativity*: $(\mathbf{A} + \mathbf{B}) + \mathbf{C} = \mathbf{A} + (\mathbf{B} + \mathbf{C})$.
- $\mathbf{A} + \mathbf{0} = \mathbf{A}$.

Remark: The sum of two or more matrices is not defined if they are not of the same order.

Subtraction of matrices Let $\mathbf{A} = [a_{ij}]_{m\times p}$ and $\mathbf{B} = [b_{ij}]_{m\times p}$ be two matrices of the same order.

$$\mathbf{A} - \mathbf{B} = [a_{ij} - b_{ij}]$$

Given that $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$.

$$\mathbf{A} - \mathbf{B} = \begin{bmatrix} a_{11} - b_{11} & a_{12} - b_{12} & a_{13} - b_{13} \\ a_{21} - b_{21} & a_{22} - b_{22} & a_{23} - b_{23} \\ a_{31} - b_{31} & a_{32} - b_{32} & a_{33} - b_{33} \end{bmatrix}$$

Remark $\mathbf{A} - \mathbf{B} \neq \mathbf{B} - \mathbf{A}$.

Examples

1. Let $\mathbf{A} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} e & b \\ f & g \end{bmatrix}$. $\mathbf{A} + \mathbf{B} = \begin{bmatrix} a+e & b+b \\ c+f & d+g \end{bmatrix}$.

2. Let $\mathbf{C} = \begin{bmatrix} 5 & 1 & 3 \\ 2 & 1 & 6 \end{bmatrix}$, $\mathbf{E} = \begin{bmatrix} 1 & 2 & 1 \\ 4 & 3 & 2 \end{bmatrix}$, $\mathbf{F} = \begin{bmatrix} 3 & 2 & 4 \\ 1 & 3 & 2 \end{bmatrix}$

$$\mathbf{C} + \mathbf{E} = \begin{bmatrix} 6 & 3 & 4 \\ 6 & 4 & 8 \end{bmatrix}$$

$$\mathbf{E} + \mathbf{F} = \begin{bmatrix} 4 & 4 & 5 \\ 5 & 6 & 4 \end{bmatrix}$$

$$(\mathbf{C} + \mathbf{E}) + \mathbf{F} = \begin{bmatrix} 6+3 & 3+2 & 4+4 \\ 6+1 & 4+3 & 8+2 \end{bmatrix} = \begin{bmatrix} 9 & 5 & 8 \\ 7 & 7 & 10 \end{bmatrix}$$

$$\mathbf{C} + (\mathbf{E} + \mathbf{F}) = \begin{bmatrix} 9 & 5 & 8 \\ 7 & 7 & 10 \end{bmatrix}.$$

3. $\begin{bmatrix} x & 1 \\ 4 & 2 \\ 2 & 3 \end{bmatrix} + \begin{bmatrix} 1 & -x \\ 3 & -3x \\ 1 & 4 \end{bmatrix} = \begin{bmatrix} x+1 & 1-x \\ 7 & 2-3x \\ 3 & 7 \end{bmatrix}$.

4. $\mathbf{M} = \begin{bmatrix} 2 & 3 \\ 1 & 2 \\ 5 & 4 \end{bmatrix}$ and $\mathbf{N} = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$ The sum $\mathbf{M} + \mathbf{N}$ is not defined since $\mathbf{M}$ and $\mathbf{N}$ are not of the same order.

5. Let $\mathbf{A} = \begin{bmatrix} e & f \\ d & e \\ f & j \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} a & b \\ c & d \\ b & e \end{bmatrix}$. $\mathbf{A} - \mathbf{B} = \begin{bmatrix} e-a & f-b \\ d-c & e-d \\ f-b & j-e \end{bmatrix}$.

6. Let $\mathbf{C} = \begin{bmatrix} 2 & 3 & 11 \\ 4 & 3 & 2 \\ 13 & 8 & 7 \end{bmatrix}$ and $\mathbf{E} = \begin{bmatrix} 4 & 2 & 1 \\ 1 & 2 & 1 \\ 4 & 2 & 8 \end{bmatrix}$. $\mathbf{C} - \mathbf{E} = \begin{bmatrix} -2 & 1 & 10 \\ 3 & 1 & 1 \\ 9 & 6 & -1 \end{bmatrix}$

$$\mathbf{E} - \mathbf{C} = \begin{bmatrix} 4-2 & 2-3 & 1-11 \\ 1-4 & 2-3 & 1-2 \\ 4-13 & 2-8 & 8-7 \end{bmatrix} = \begin{bmatrix} 2 & -1 & -10 \\ -3 & -1 & -1 \\ -9 & -6 & 1 \end{bmatrix}.$$

Multiplication of Matrices

Scalar multiplication of matrices: The multiplication of a matrix $\mathbf{A}$ by a scalar α is defined as $\alpha\mathbf{A} := [\alpha a_{ij}]$. If $\alpha = 0$, $\alpha\mathbf{A}$ will give a null matrix.
Let α, β be real numbers. Let $\mathbf{A}$ and $\mathbf{B}$ be matrices. The following properties hold:

(i) $\alpha(\mathbf{A} + \mathbf{B}) = \alpha\mathbf{A} + \alpha\mathbf{B}$ (distributive property)

(ii) $(\alpha + \beta)\mathbf{A} = \alpha\mathbf{A} + \beta\mathbf{B}$

(iii) $(\alpha\beta)\mathbf{A} = \alpha(\beta\mathbf{A})$

(iv) $(\alpha\mathbf{A})^T = \alpha\mathbf{A}^T$

(v) $0 \cdot \mathbf{A} = 0$

(vi) $\mathbf{IA} = \mathbf{A}$.

Examples

1. Let $\mathbf{A} = \begin{bmatrix} 3 & 1 & 4 \\ 2 & 2 & 5 \\ 1 & -2 & 1 \end{bmatrix}$, $5\mathbf{A} = \begin{bmatrix} 15 & 5 & 20 \\ 10 & 10 & 25 \\ 5 & -10 & 5 \end{bmatrix}$.

2. Let $\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$, $-1\mathbf{B} = \begin{bmatrix} -b_{11} & -b_{12} \\ -b_{21} & -b_{22} \end{bmatrix}$.

3. Given that $\mathbf{M} = \begin{bmatrix} 1 & 5 & 7 \\ 3 & -2 & 1 \\ 1 & 4 & 2 \end{bmatrix}$ and $\mathbf{N} = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 3 & 1 \\ 1 & -2 & -3 \end{bmatrix}$, evaluate $k(\mathbf{M} + \mathbf{N})$, where k is a scalar.

   Solution

$$k(\mathbf{M} + \mathbf{N}) = k \begin{bmatrix} 1+3 & 5+1 & 7+2 \\ 3+1 & -2+3 & 1+1 \\ 1+1 & 4+(-2) & 2+(-3) \end{bmatrix} = k \begin{bmatrix} 4 & 6 & 9 \\ 4 & 1 & 2 \\ 2 & 2 & -1 \end{bmatrix} = \begin{bmatrix} 4k & 6k & 9k \\ 4k & 1k & 2k \\ 2k & 2k & -1k \end{bmatrix}$$

Matrix multiplication

Let $\mathbf{A} = [a_{ij}]$ be a matrix of order $m \times n$, and $\mathbf{B} = [b_{jk}]$ be a matrix of order $n \times p$. Then

$$\mathbf{AB} = \sum_{k=1}^{p} a_{ik}b_{kj} = [c_{ik}] = \mathbf{C}$$

is a matrix of order $m \times p$.

The following general rules are necessary for matrix multiplication:

(i) To evaluate $\mathbf{C} = \mathbf{AB}$, the number of columns of $\mathbf{A}$ must equal number of rows of $\mathbf{B}$. In other words, the matrices $\mathbf{A}$ and $\mathbf{B}$ must be compatible.

(ii) The number of rows in $\mathbf{AB}$ equals the number of rows in the matrix $\mathbf{A}$. The number of columns in $\mathbf{AB}$ equals the number of columns in the matrix $\mathbf{B}$.

(iii) Multiplication of matrix is not commutative, $\mathbf{AB} \neq \mathbf{BA}$.

(iv) Multiplication of matrix is associative, $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$

(v) $\mathbf{IA} = \mathbf{A}$.

(vi) For the product of two matrices, $\mathbf{A}$ and $\mathbf{B}$ to be defined, the number of columns of $\mathbf{A}$ must be the same as the number of rows of $\mathbf{B}$.

(v) Distributive property holds: $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$ and $(\mathbf{B} + \mathbf{C})\mathbf{A} = \mathbf{BA} + \mathbf{CA}$.

(vi) Equality, $\mathbf{AB} = \mathbf{BA}$ holds if $\mathbf{A}$ and $\mathbf{B}$ are diagonal.

Examples

1. Let $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ and $\mathbf{B} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \\ b_{31} & b_{32} & b_{33} \end{bmatrix}$.

$$\mathbf{AB} = \begin{bmatrix} a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} & a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} & a_{11}b_{13} + a_{12}b_{23} + a_{13}b_{33} \\ a_{21}b_{11} + a_{22}b_{21} + a_{23}b_{31} & a_{21}b_{12} + a_{22}b_{22} + a_{23}b_{32} & a_{21}b_{13} + a_{22}b_{23} + a_{23}b_{33} \\ a_{31}b_{11} + a_{32}b_{21} + a_{33}b_{31} & a_{31}b_{12} + a_{32}b_{22} + a_{33}b_{32} & a_{31}b_{13} + a_{32}b_{23} + a_{33}b_{34} \end{bmatrix}$$

$$= \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix} = \mathbf{C}.$$

2. Let $\mathbf{M} = \begin{bmatrix} 1 & 8 \\ 9 & -2 \end{bmatrix}$, $\mathbf{N} = \begin{bmatrix} 2 & 0 \\ 4 & -1 \end{bmatrix}$

$$\mathbf{MN} = \begin{bmatrix} (1 \times 2) + (8 \times 4) & (1 \times 0) + (8 \times (-1)) \\ (9 \times 2) + (-2 \times 4) & (9 \times 0) + (-2 \times -1) \end{bmatrix} = \begin{bmatrix} 34 & -8 \\ 10 & 2 \end{bmatrix}$$

$$\mathbf{NM} = \begin{bmatrix} 2 & 0 \\ 4 & -1 \end{bmatrix} \begin{bmatrix} 1 & 8 \\ 9 & -2 \end{bmatrix} = \begin{bmatrix} 2 & 16 \\ -5 & 34 \end{bmatrix}.$$

Note that $\mathbf{MN} \neq \mathbf{NM}$.

3 $\mathbf{G}_{2\times 3} = \begin{bmatrix} 2 & 6 & 3 \\ 1 & 0 & -1 \end{bmatrix}$ and $\mathbf{H}_{3\times 3} = \begin{bmatrix} 2 & -1 & 4 \\ 3 & 1 & 2 \\ 4 & 0 & 1 \end{bmatrix}$.

$$\mathbf{GH}_{2\times 3} = \begin{bmatrix} 4 + 18 + 12 & -2 + 6 + 0 & 8 + 12 + 3 \\ 2 + 0 - 4 & -1 + 0 + 0 & 4 + 0 - 1 \end{bmatrix}$$

$$= \begin{bmatrix} 34 & 4 & 23 \\ -2 & -1 & 3 \end{bmatrix}$$

4. Given that $\mathbf{E} = \begin{bmatrix} 8 & 2 \\ 6 & 1 \\ 2 & 3 \end{bmatrix}$, $\mathbf{F} = \begin{bmatrix} -4 & 2 & -1 \\ 3 & 5 & 1 \end{bmatrix}$, evaluate $\mathbf{EF}$.

$\mathbf{E}$ is 3×2 matrix, $\mathbf{F}$ is 2×3 matrix, therefore $\mathbf{EF}$ will give a 3×3 matrix.

$$\mathbf{EF} = \begin{bmatrix} 8 & 2 \\ 6 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} -4 & 2 & -1 \\ 3 & 5 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} (8 \times -4) + (2 \times 3) & (8 \times 2) + (2 \times 5) & (8 \times -1) + (2 \times 1) \\ (6 \times -4) + (1 \times 3) & (6 \times 2) + (1 \times 5) & (6 \times -1) + (1 \times 1) \\ (2 \times -4) + (3 \times 3) & (2 \times 2) + (3 \times 5) & (2 \times -1) + (3 \times 1) \end{bmatrix}.$$

$$= \begin{bmatrix} -26 & 26 & -6 \\ -21 & 17 & -5 \\ 1 & 19 & 1 \end{bmatrix}$$

5. $\mathbf{P} = \begin{bmatrix} 2 & 3 \\ 1 & 9 \end{bmatrix}$, find $\mathbf{IP}$.

$$\mathbf{IP} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 3 \\ 1 & 9 \end{bmatrix} = \begin{bmatrix} 2+0 & 3+0 \\ 0+1 & 0+9 \end{bmatrix} = \begin{bmatrix} 2 & 3 \\ 1 & 9 \end{bmatrix} = \mathbf{P}.$$

Transposed Matrix

Let $\mathbf{A}$ be an $m \times n$ matrix, the transpose of $\mathbf{A}$, denoted as $\mathbf{A}^T$ or $T(\mathbf{A})$ is an $n \times m$ matrix whose ijth entry is the jith entry of $\mathbf{A}$. This shows that $a_{ij}^T = a_{ji}$.

Theorem: Let $\mathbf{A}$ be an $m \times n$ matrix, then

(i) $\mathbf{A}$ is symmetric implies that $\mathbf{A}^T = \mathbf{A}$.

(ii) $\mathbf{A}$ is anti-symmetric implies that $\mathbf{A}^T = -\mathbf{A}$.

Given any matrix $\mathbf{A}$, its transpose is obtained by changing all the rows of $\mathbf{A}$ into the columns of $\mathbf{A}^T$ while maintaining the order.

Remark

Let $\mathbf{A}$ and $\mathbf{B}$ be two matrices

(i) $(\mathbf{A}^t)^T = \mathbf{A}$

(ii) $\mathbf{A}\mathbf{A}^T$ and $\mathbf{A}^T\mathbf{A}$ are symmetric.

(iii) Let α be a scalar, $(\alpha\mathbf{A})^T = \alpha\mathbf{A}^T$

$(\mathbf{A} + \mathbf{B})^T = \mathbf{A}^T + \mathbf{B}^T$

$\mathbf{A}$ is an $m \times n$ matrix, its transpose is an $n \times m$ matrix.

Examples

(i) $\mathbf{A} = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$, $\mathbf{A}^T = \begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix}$

(ii) $\mathbf{B} = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 7 & 9 \\ 8 & 6 & 5 \end{bmatrix}$, $\mathbf{B}^T = \begin{bmatrix} 1 & 2 & 8 \\ 3 & 7 & 6 \\ 4 & 9 & 5 \end{bmatrix}$

(iii) $\mathbf{C} = \begin{bmatrix} 1 & 4 \\ 3 & 2 \\ 6 & 1 \end{bmatrix}$, $\mathbf{C}^T = \begin{bmatrix} 1 & 3 & 6 \\ 4 & 2 & 1 \end{bmatrix}$

(iv) $\mathbf{E} = \begin{bmatrix} 2 & 8 \\ -7 & 3 \end{bmatrix}$, $\mathbf{E}^T = \begin{bmatrix} 2 & -7 \\ 8 & 3 \end{bmatrix}$

(iv) $\mathbf{F} = \begin{bmatrix} 4 & -1 \\ -1 & 7 \end{bmatrix}$, $\mathbf{F}^T = \begin{bmatrix} 4 & -1 \\ -1 & 7 \end{bmatrix}$ $\mathbf{F}$ is symmetric since $\mathbf{F} = \mathbf{F}^T$.

(iv) If $\mathbf{G} = \begin{bmatrix} 1 & 3 & 1 & 5 \\ -2 & 4 & -1 & -3 \\ 3 & 7 & -5 & 1 \\ 2 & 6 & 1 & -2 \end{bmatrix}$, then $\mathbf{G}^T = \begin{bmatrix} 1 & -2 & 3 & 2 \\ 3 & 4 & 7 & 6 \\ 1 & -1 & -5 & 1 \\ 5 & -3 & 1 & -2 \end{bmatrix}$.

7.4 Determinant

A determinant of order n can be regarded as a scalar associated with an $n \times n$ square matrix $\mathbf{A} = [a_{jk}]$. It is denoted as $\det(\mathbf{A})$ or $|\mathbf{A}|$.

To find the determinant of a 3×3 matrix.

$$|\mathbf{A}| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}.$$

we expand the determinant of the first row. This is obtained by multiplying the elements of the first row by the determinant obtained by omitting the row and column of that element, with an alternating sign.

$$\begin{aligned} |\mathbf{A}| &= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} \\ &= a_{11}(a_{22}a_{33} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{31}a_{22}) \\ &= a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}. \end{aligned}$$

If $\mathbf{A} = \begin{bmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{bmatrix}$

$$\det(\mathbf{A}) = \begin{vmatrix} a & b & c & d \\ e & f & g & h \\ i & j & k & l \\ m & n & o & p \end{vmatrix}$$

$$= a\begin{vmatrix} f & g & h \\ j & k & l \\ n & o & p \end{vmatrix} - b\begin{vmatrix} e & g & h \\ i & k & l \\ m & o & p \end{vmatrix} + c\begin{vmatrix} e & f & h \\ i & j & l \\ m & n & p \end{vmatrix} - d\begin{vmatrix} e & f & g \\ i & j & k \\ m & n & o \end{vmatrix}.$$

In general, let $\mathbf{A}$ be an n-dimensional square matrix.

$$|\mathbf{A}| = \begin{vmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{vmatrix}$$

$$= \sum_{k=1}^{n} (-1)^{j+k} a_{jk} M_{jk} \quad (j = 1, 2, \ldots, \text{or } n)$$

$$= \sum_{j=1}^{n} (-1)^{j+k} a_{jk} M_{jk} \quad (k = 1, 2, \ldots, \text{or } n).$$

If $n = 1$, $\det(\mathbf{A}) = \det(\mathbf{A}) = a_{11}$.
If $n = 2$, $\det(\mathbf{A}) = \det(\mathbf{A}) = a_{11}a_{22} - a_{21}a_{12}$.
From the above M_{jk} represents the determinant of the submatrix of $\mathbf{A}$ obtained from $\mathbf{A}$ by omitting the row and column of the entry a_{jk}.

Examples

Find the determinant of the following matrices

(a) $\mathbf{A} = \begin{bmatrix} 3 & 7 & 1 \\ 2 & 4 & 10 \\ 6 & -3 & 1 \end{bmatrix}$

(b) $\mathbf{B} = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 4 & 1 \\ -1 & 3 & -2 \end{bmatrix}$

(c) $\mathbf{BA}$ given that $\mathbf{A} = \begin{bmatrix} 1 & 2 \\ 5 & 3 \end{bmatrix}$. $\mathbf{B} = \begin{bmatrix} -4 & 1 \\ 2 & 5 \end{bmatrix}$

Solution

(a) $\det(\mathbf{A}) = \begin{vmatrix} 3 & 7 & 1 \\ 2 & 4 & 10 \\ 6 & -3 & 1 \end{vmatrix} = 3\begin{vmatrix} 4 & 10 \\ -3 & 1 \end{vmatrix} - 7\begin{vmatrix} 2 & 10 \\ 6 & 1 \end{vmatrix} + 1\begin{vmatrix} 2 & 4 \\ 6 & -3 \end{vmatrix}$

$= 3(4 + 30) - 7(2 - 60) + 1(-6 - 24) = 478.$

(b) $\det(\mathbf{B}) = \begin{vmatrix} 1 & 0 & 3 \\ 2 & 4 & 1 \\ -1 & 3 & -2 \end{vmatrix} = 1\begin{vmatrix} 4 & 1 \\ 3 & -2 \end{vmatrix} - 0\begin{vmatrix} 2 & 1 \\ -1 & -2 \end{vmatrix} + 3\begin{vmatrix} 2 & 4 \\ -1 & 3 \end{vmatrix}$

$= 1(-8 - 3) - 0(-4 + 1) + 3(6 + 4) = 19.$

(c) $\det(\mathbf{BA}) = \begin{vmatrix} -4+5 & -8+3 \\ 2+25 & 4+15 \end{vmatrix} = \begin{vmatrix} 1 & -5 \\ 27 & 19 \end{vmatrix} = 154.$

Properties of a Determinant

(i) When two rows or columns of a given matrix are interchanged. the determinant changes sign. Eg. Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 4 & 2 \\ 1 & 4 & 1 \end{bmatrix}$

$$|\mathbf{A}| = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 4 & 2 \\ 1 & 4 & 1 \end{vmatrix} = 1\begin{vmatrix} 4 & 2 \\ 4 & 1 \end{vmatrix} - 2\begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} + 1\begin{vmatrix} 3 & 4 \\ 1 & 4 \end{vmatrix}$$

$$= 1(4 - 8) - 2(3 - 2) + 1(12 - 4) = 2.$$

Interchanging two rows gives

$$|\mathbf{A}_1| = \begin{vmatrix} 1 & 2 & 1 \\ 1 & 4 & 1 \\ 3 & 4 & 2 \end{vmatrix} = 1\begin{vmatrix} 4 & 1 \\ 4 & 2 \end{vmatrix} - 2\begin{vmatrix} 1 & 1 \\ 3 & 2 \end{vmatrix} + 1\begin{vmatrix} 1 & 4 \\ 3 & 4 \end{vmatrix}$$

$$= 1(8 - 4) - 2(2 - 3) + 1(4 - 12) = -2.$$

Interchanging two columns gives

$$|\mathbf{A}_2| = \begin{vmatrix} 1 & 1 & 2 \\ 3 & 2 & 4 \\ 1 & 1 & 4 \end{vmatrix} = 1\begin{vmatrix} 2 & 4 \\ 1 & 4 \end{vmatrix} - 1\begin{vmatrix} 3 & 4 \\ 1 & 4 \end{vmatrix} + 2\begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix}$$

$$= 1(8 - 4) - 1(12 - 4) + 2(3 - 2) = -2.$$

(ii) The determinant of a given matrix whose two rows or two columns are equal is zero

$$|\mathbf{A}| = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 5 & 1 \\ 3 & 5 & 1 \end{vmatrix} = 1\begin{vmatrix} 5 & 1 \\ 5 & 1 \end{vmatrix} - 2\begin{vmatrix} 3 & 1 \\ 3 & 1 \end{vmatrix} + 4\begin{vmatrix} 3 & 5 \\ 3 & 5 \end{vmatrix}$$

$$= 1(5 - 5) - 2(3 - 3) + 4(15 - 15) = 0$$

(iii) The determinant of a given square matrix is equal to the determinant of its transpose.

$$|\mathbf{A}| = \begin{vmatrix} 1 & 2 & 1 \\ 3 & 4 & 2 \\ 1 & 4 & 1 \end{vmatrix} = 2$$

$$|\mathbf{A}^t| = \begin{vmatrix} 1 & 3 & 1 \\ 2 & 4 & 4 \\ 1 & 2 & 1 \end{vmatrix} = 1\begin{vmatrix} 4 & 4 \\ 2 & 1 \end{vmatrix} - 3\begin{vmatrix} 2 & 4 \\ 1 & 1 \end{vmatrix} + 1\begin{vmatrix} 2 & 4 \\ 1 & 2 \end{vmatrix}$$

$$= 1(4-8) - 3(2-4) + 1(4-4) = 2.$$

(iv) The product of the determinant of two square matrices has the same value as the product of the determinants of the factors. That is, If $\mathbf{A}$ and $\mathbf{B}$ are $n \times n$ matrices, $\det(\mathbf{A})\det(\mathbf{B}) = \det(\mathbf{AB})$.

(v) When a factor is common to a row or a column, it can be factored out.

(vi) The determinant of a matrix with zero row or zero column is zero.

(vii) The determinant of a diagonal matrix is equal to the product of its diagonal elements.

7.5 The Inverse of a Matrix

Definition: Let $\mathbf{A}$ be a square matrix. The *adjoint* of $\mathbf{A}$ is defined as the transpose of the matrix in which cofactors of $\mathbf{A}$ is used to replace the corresponding elements of $\mathbf{A}$. It is denoted as adj($\mathbf{A}$).

$$\text{Let } \mathbf{A}_{3\times 3} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

Let M denotes the minors and c denotes the cofactor. The difference between the numbers called *cofactors* and the numbers called *minors* is that some minors have a negative sign according to the following pattern:

$$\begin{bmatrix} + & - & + & + & \cdots \\ - & + & - & - & \cdots \\ + & - & + & + & \cdots \\ - & + & - & - & \cdots \\ \cdots & \cdots & \cdots & \cdots & \cdots \end{bmatrix}$$

The cofactors of $\mathbf{A}$ are obtained as shown below:

$$c_{11} = +\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} = +(a_{22}a_{33} - a_{32}a_{23}) = +M_{11}$$

$$c_{12} = -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} = -(a_{21}a_{33} - a_{31}a_{23}) = -M_{12}$$

$$c_{13} = +\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} = +(a_{21}a_{32} - a_{31}a_{22}) = +M_{13}$$

$$c_{21} = -\begin{vmatrix} a_{12} & a_{13} \\ a_{32} & a_{33} \end{vmatrix} = -(a_{12}a_{33} - a_{32}a_{13}) = -M_{21}$$

$$c_{22} = +\begin{vmatrix} a_{11} & a_{13} \\ a_{31} & a_{33} \end{vmatrix} = +(a_{11}a_{33} - a_{31}a_{13}) = +M_{22}$$

$$c_{23} = -\begin{vmatrix} a_{11} & a_{12} \\ a_{31} & a_{32} \end{vmatrix} = -(a_{11}a_{32} - a_{31}a_{12}) = -M_{23}$$

$$c_{31} = +\begin{vmatrix} a_{12} & a_{13} \\ a_{22} & a_{23} \end{vmatrix} = +(a_{12}a_{23} - a_{22}a_{13}) = +M_{31}$$

$$c_{32} = -\begin{vmatrix} a_{11} & a_{13} \\ a_{21} & a_{23} \end{vmatrix} = -(a_{11}a_{23} - a_{21}a_{13}) = -M_{32}$$

$$c_{33} = +\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = +(a_{11}a_{22} - a_{21}a_{12}) = +M_{33}$$

$$\text{adj}(\mathbf{A}) = \begin{bmatrix} c_{11} & c_{12} & c_{13} \\ c_{21} & c_{22} & c_{23} \\ c_{31} & c_{32} & c_{33} \end{bmatrix}^T = \begin{bmatrix} c_{11} & c_{21} & c_{31} \\ c_{12} & c_{22} & c_{32} \\ c_{13} & c_{23} & c_{33} \end{bmatrix}.$$

Definition: Let $\mathbf{A}$ be an $n \times n$ matrix, the *inverse of* $\boldsymbol{A}$, denoted as $\mathbf{A}^{-1}$ is an $n \times n$ matrix such that $\mathbf{A}\mathbf{A}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$, where $\mathbf{I}$ denotes the $n \times n$ unit matrix. It is obtained by using the equation:

$$\mathbf{A}^{-1} = \frac{\text{adj}\mathbf{A}}{|\mathbf{A}|}.$$

Properties of an Inverse of a Matrix

(i) If there exists an inverse of a given matrix, the inverse is unique. In other words, if a matrix has both a left inverse and a right inverse then the two are equal.

(ii) If the inverse of a matrix $\mathbf{A}$ exists, then $\mathbf{A}$ is said to be a *non-singular* matrix. If $\mathbf{A}$ does not have an inverse, then $\mathbf{A}$ is said to be a *singular* matrix.

(iii) A matrix $\mathbf{A}$ is a left inverse matrix of the matrix $\mathbf{B}$ if $\mathbf{AB}$ is the identity matrix. It is a right inverse if $\mathbf{BA}$ is the identity. A matrix $\mathbf{B}$ with a two-sided inverse is an invertible matrix. That two-sided inverse is denoted $\mathbf{B}^{-1}$.

(iv) The product of invertible matrices is invertible.

(v) If $\mathbf{A}$ is an $n \times n$ matrix, then $\mathbf{A}^{-1}$ must be an $n \times n$ matrix.

(vi) $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$.

Examples

(i) Find the inverse of the matrix $\mathbf{A} = \begin{bmatrix} 1 & 3 \\ 5 & 2 \end{bmatrix}$.

Solution

$$\text{adj}\mathbf{A} = \begin{bmatrix} 2 & -5 \\ -3 & 1 \end{bmatrix}^T = \begin{bmatrix} 2 & -3 \\ -5 & 1 \end{bmatrix}$$

$$|\mathbf{A}| = -13$$

$$\therefore \mathbf{A}^{-1} = \frac{\text{adj}\mathbf{A}}{|\mathbf{A}|} = \frac{-1}{13}\begin{bmatrix} 2 & -3 \\ -5 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{-2}{13} & \frac{3}{13} \\ \frac{5}{13} & \frac{-1}{13} \end{bmatrix}.$$

Note that $\mathbf{A}\mathbf{A}^{-1} = \begin{bmatrix} 1 & 3 \\ 5 & 2 \end{bmatrix}\begin{bmatrix} \frac{-2}{13} & \frac{3}{13} \\ \frac{5}{13} & \frac{-1}{13} \end{bmatrix} = \mathbf{I}$.

(ii) Evaluate the inverse of the diagonal matrix $\mathbf{D} = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

Solution

The cofactors are:

$$c_{11} = \begin{vmatrix} -1 & 0 \\ 0 & 2 \end{vmatrix} = -2, \quad c_{12} = -\begin{vmatrix} 0 & 0 \\ 0 & 2 \end{vmatrix} = 0, \quad c_{13} = \begin{vmatrix} 0 & -1 \\ 0 & 0 \end{vmatrix} = 0$$

$$c_{21} = -\begin{vmatrix} 0 & 0 \\ 0 & 2 \end{vmatrix} = 0, \quad c_{22} = \begin{vmatrix} 3 & 0 \\ 0 & 2 \end{vmatrix}, \quad c_{23} = \begin{vmatrix} 3 & 0 \\ 0 & 0 \end{vmatrix} = 0$$

$$c_{31} = \begin{vmatrix} 0 & 0 \\ -1 & 0 \end{vmatrix} = 0, \quad c_{32} = -\begin{vmatrix} 3 & 0 \\ 0 & 0 \end{vmatrix} = 0, \quad c_{33} = \begin{vmatrix} 3 & 0 \\ 0 & -1 \end{vmatrix}.$$

$$\det \mathbf{D} = 3\begin{vmatrix} -1 & 0 \\ 0 & 2 \end{vmatrix} - 0\begin{vmatrix} 0 & 0 \\ 0 & 2 \end{vmatrix} + 0\begin{vmatrix} 0 & -1 \\ 0 & 0 \end{vmatrix} = -6$$

$$\text{adj}(\mathbf{D}) = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & -3 \end{bmatrix}^T = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & -3 \end{bmatrix}$$

$$\mathbf{D}^{-1} = \begin{bmatrix} \frac{1}{3} & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}.$$

(iii) Let $\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 5 \\ 3 & -1 & 2 \end{bmatrix}$.

Evaluate (a) $\text{adj}\mathbf{A}$ (b) $\mathbf{A} \cdot \text{adj}\mathbf{A}$ (c) $|\mathbf{A}|$ (d) $\mathbf{A}^{-1}$.

Solution

(a) The adjoint of the matrix

$$\text{adj}\mathbf{A} = \begin{bmatrix} +\begin{vmatrix} 1 & 5 \\ -1 & 2 \end{vmatrix} & -\begin{vmatrix} 3 & 5 \\ 3 & 2 \end{vmatrix} & +\begin{vmatrix} 3 & 1 \\ 3 & -1 \end{vmatrix} \\ -\begin{vmatrix} 2 & -1 \\ -1 & 2 \end{vmatrix} & +\begin{vmatrix} 1 & -1 \\ 3 & 2 \end{vmatrix} & -\begin{vmatrix} 1 & 2 \\ 3 & -1 \end{vmatrix} \\ +\begin{vmatrix} 2 & -1 \\ 1 & 5 \end{vmatrix} & -\begin{vmatrix} 1 & -1 \\ 3 & 5 \end{vmatrix} & +\begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} \end{bmatrix}$$

$$= \begin{bmatrix} 7 & 9 & -6 \\ -3 & 5 & 7 \\ 11 & -8 & -5 \end{bmatrix}^T = \begin{bmatrix} 7 & -3 & 11 \\ 9 & 5 & -8 \\ -6 & 7 & -5 \end{bmatrix}$$

(b) $\mathbf{A} \cdot \text{adj}\mathbf{A} = \begin{bmatrix} 1 & 2 & -1 \\ 3 & 1 & 5 \\ 3 & -1 & 2 \end{bmatrix} \begin{bmatrix} 7 & -3 & 11 \\ 9 & 5 & -8 \\ -6 & 7 & -5 \end{bmatrix}$

$$= \begin{bmatrix} 7+18+6 & -3+10-7 & 11-16+5 \\ 21+9-30 & -9+5+35 & 33-8-25 \\ 21-9-12 & -9-5+14 & 33+8-10 \end{bmatrix} \begin{bmatrix} 31 & 0 & 0 \\ 0 & 31 & 0 \\ 0 & 0 & 31 \end{bmatrix} = 31 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = |\mathbf{A}|\mathbf{I}$$

(c) $\det(\mathbf{A}) = 1\begin{vmatrix} 1 & 5 \\ -1 & 2 \end{vmatrix} - 2\begin{vmatrix} 3 & 5 \\ 3 & 2 \end{vmatrix} - 1\begin{vmatrix} 3 & 1 \\ 3 & -1 \end{vmatrix}$

$$= 1(7) - 2(-9) - 1(-6) = 31.$$

(d) $\mathbf{A}^{-1} = \dfrac{\text{adj}\mathbf{A}}{|\mathbf{A}|} = \dfrac{1}{13}\begin{bmatrix} 7 & -3 & 11 \\ 9 & 5 & -8 \\ -6 & 7 & -5 \end{bmatrix}$.

7.6 Application of Matrices

Matrix has a lot of applications in all fields especially Engineering (e.g. Ohm's law: Voltage = Current × Resistance; Kirchhoff's law: first law states that the sum of all potential drops around a closed loop is balanced by the sum of all potential sources around the same loop, and the second law states that the currents entering each node must equal the algebraic sum of the currents exiting it; Simple Harmonic oscillators), Sciences and Social Sciences

Polynomial of Matrices

Suppose that $\mathbf{A}$ is a square matrix and α is a scalar, let f and g be polynomials. The following properties hold:

(i) $(f+g)(\mathbf{A}) = f(\mathbf{A}) + f(\mathbf{B})$

(ii) $(\alpha f)(\mathbf{A}) = \alpha f(\mathbf{A})$

(iii) $(fg)(\mathbf{A}) = f(\mathbf{A})g(\mathbf{A})$

(iv) $f(\mathbf{A})g(\mathbf{A}) = g(\mathbf{A})f(\mathbf{A})$.

Example

Let $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix}$. $\mathbf{A}^2 = \begin{bmatrix} 7 & 1 \\ 3 & 4 \end{bmatrix}$ and $f(t) = 3t^2 - t + 2$. Find $f(\mathbf{A})$.

Solution

$$f(\mathbf{A}) = 3\mathbf{A}^2 - \mathbf{A} + 2\mathbf{I}$$

$$= \begin{bmatrix} 21 & 3 \\ 9 & 16 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 3 & -1 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 21 & 2 \\ 6 & 19 \end{bmatrix}.$$

Characteristic Polynomial

Let $\mathbf{A}$ be an $n \times n$ matrix. The matrix $\mathbf{M} = \mathbf{A} - t\mathbf{I}_n$ where $\mathbf{I}_n$ denotes the $n \times n$ identity matrix and t is an indeterminate, may be obtained by subtracting t down the diagonal of A. The negative of $\mathbf{M}$ is the matrix $t\mathbf{I}_n - \mathbf{A}$, and its determinant

$$\Delta(t) = \det(t\mathbf{I}_n - \mathbf{A}) = (-1)^n \det(\mathbf{A} - t\mathbf{I}_n)$$

which is a polynomial in t of degree n is called the *characteristic polynomial* of $\mathbf{A}$.

Cayley-Hamilton Theorem
Every matrix is a root of its characteristic polynomial.

Example

Let $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix}$.
Its characteristic polynomial is

$$\det(t\mathbf{I} - \mathbf{A}) = |t\mathbf{I}_n - \mathbf{A}| = \left| \begin{bmatrix} t & 0 \\ 0 & t \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 3 & 4 \end{bmatrix} \right| = \begin{vmatrix} t-2 & -1 \\ -3 & t-4 \end{vmatrix}$$

$$= (t-2)(t-4) - 3 = t^2 - 6t + 5 = \Delta(t)$$

$$\Delta(\mathbf{A}) = \mathbf{A}^2 - 6\mathbf{A} + 5\mathbf{I}$$

$$= \begin{bmatrix} 7 & 6 \\ 18 & 19 \end{bmatrix} - \begin{bmatrix} 12 & 6 \\ 18 & 24 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

near Equations

linear equation in n unknowns $x_1, x_2, \ldots, x_n$ is an equation which can be written in the
ndard form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

here b is a constant and $a_1, a_2, \ldots, a_n$ are coefficients of the unknowns $x_1, x_2, \ldots, x_n$ respectively.
A solution of the linear system is a list of values for the unknowns $x_1 = c_1, x_2 = c_2, \ldots, x_n =$
such that

$$a_1c_1 + a_2c_2 + \cdots + a_nc_n = b.$$

pplication of Inverse in Solving Matrices

olving simultaneous equation

1. Solve the simultaneous equation

$$\begin{aligned} x + y &= 7 \\ 5x - 3y &= -5. \end{aligned}$$

Solution
In the matrix form, we have

$$\begin{pmatrix} 1 & 1 \\ 5 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 7 \\ -5 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 5 & -3 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} x \\ y \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} 7 \\ -5 \end{pmatrix}.$$

$\mathbf{AX} = \mathbf{B}$ where $\mathbf{A}$ is the matrix of coefficients and $\mathbf{B}$ is the column vector of constants.
If there exists an inverse. multiply both sides of the system by $\mathbf{A}^{-1}$

$$\mathbf{A}^{-1}\mathbf{AX} = \mathbf{A}^{-1}\mathbf{B}.$$

Recall $\mathbf{A}^{-1}\mathbf{A} = \mathbf{I}$ and $\mathbf{IX} = \mathbf{X}$. We obtain

$$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B}.$$

$$\mathbf{A}^{-1} = \frac{\text{adj}\mathbf{A}}{|\mathbf{A}|} = \frac{\begin{pmatrix} -3 & -1 \\ -5 & 1 \end{pmatrix}}{-8}$$

$$|\mathbf{A}| = \begin{vmatrix} 1 & 1 \\ 5 & -3 \end{vmatrix} = -3 - 5 = -8$$

$$\mathbf{A}^{-1}\mathbf{B} = \frac{1}{-8}\begin{pmatrix} -3 & -1 \\ -5 & 1 \end{pmatrix}\begin{pmatrix} 7 \\ -5 \end{pmatrix}$$

$$= \frac{1}{-8}\begin{pmatrix} -16 \\ -40 \end{pmatrix} = \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} x \\ y \end{pmatrix}$$

Therefore, $x = 2$ and $y = 5$

2. Solve the simultaneous equation

$$\begin{aligned} x_1 + x_2 &= -3 \\ -x_1 - 2x_2 &= 10. \end{aligned}$$

Solution
In the matrix form, we have

$$\begin{pmatrix} 1 & 1 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} -3 \\ 10 \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ -1 & -2 \end{pmatrix}, \quad \mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \text{ and } \mathbf{B} = \begin{pmatrix} -3 \\ 10 \end{pmatrix}$$

$\mathbf{AX} = \mathbf{B}$.

$$|\mathbf{A}| = \begin{vmatrix} 1 & 1 \\ -1 & -2 \end{vmatrix} = -1$$

$$\text{adj}\mathbf{A} = \begin{pmatrix} -2 & 1 \\ -1 & 1 \end{pmatrix}^T = \begin{pmatrix} -2 & -1 \\ 1 & 1 \end{pmatrix}$$

$$\mathbf{A}^{-1} = \frac{\text{adj}\mathbf{A}}{|\mathbf{A}|} = \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix}.$$

$\mathbf{X} = \mathbf{A}^{-1}\mathbf{B}$.

$$\mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ -1 & -1 \end{pmatrix} \begin{pmatrix} -3 \\ 10 \end{pmatrix} = \begin{pmatrix} 4 \\ -7 \end{pmatrix}.$$

This gives $x_1 = 4$ and $x_2 = -7$.

3. Solve the system

$$\begin{aligned} x + y - 2z &= 5 \\ 2x - 2y + z &= 7 \\ x + 3y - z &= 10. \end{aligned}$$

Solution
In matrix form, we have

$$\begin{pmatrix} 1 & 1 & -2 \\ 2 & -2 & 1 \\ 1 & 3 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \\ 10 \end{pmatrix}.$$

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & -2 \\ 2 & -2 & 1 \\ 1 & 3 & -1 \end{pmatrix},$$

$$\text{adj}\mathbf{A} = \begin{pmatrix} -1 & 3 & 8 \\ -5 & 1 & -2 \\ -3 & -5 & -4 \end{pmatrix}^T = \begin{pmatrix} -1 & -5 & -3 \\ 3 & 1 & -5 \\ 8 & -2 & 1 \end{pmatrix}.$$

$$|\mathbf{A}| = 1(-1) - 1(-3) - 2(8) = -14$$

$$\mathbf{A}^{-1} = -\frac{1}{14}\begin{pmatrix} -1 & -5 & -3 \\ 3 & 1 & -5 \\ 8 & -2 & -4 \end{pmatrix}$$

$$\mathbf{A}^{-1}\mathbf{B} = -\frac{1}{14}\begin{pmatrix} -1 & -5 & -3 \\ 3 & 1 & -5 \\ 8 & -2 & -4 \end{pmatrix}\begin{pmatrix} 5 \\ 7 \\ 10 \end{pmatrix} = -\frac{1}{14}\begin{pmatrix} -5-35-30 \\ 15+7-50 \\ 40-14-40 \end{pmatrix}$$

$$= -\frac{1}{14}\begin{pmatrix} -70 \\ -28 \\ -14 \end{pmatrix} = \begin{pmatrix} 5 \\ 2 \\ 1 \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Therefore, the solution of the system is $x = 5$, $y = 2$ and $z = 1$.

4. Solve the system

$$\begin{aligned} 2x - 5y - z &= 4 \\ x + 3y - 2z &= -8 \\ 3x - 2y + z &= 8. \end{aligned}$$

Solution

In matrix form,

$$\mathbf{AX} = \begin{pmatrix} 2 & -5 & -1 \\ 1 & 3 & -2 \\ 3 & -2 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ -8 \\ 8 \end{pmatrix} = \mathbf{B}.$$

$$|\mathbf{A}| = \begin{vmatrix} 2 & -5 & -1 \\ 1 & 3 & -2 \\ 3 & -2 & 1 \end{vmatrix} = 2(-1) + 5(7) - 1(-11) = 44.$$

$$\text{adj}\mathbf{A} = \begin{pmatrix} + \begin{vmatrix} 3 & -2 \\ -2 & 1 \end{vmatrix} & - \begin{vmatrix} 1 & -2 \\ 3 & 1 \end{vmatrix} & + \begin{vmatrix} 1 & 3 \\ 3 & -2 \end{vmatrix} \\ - \begin{vmatrix} -5 & -1 \\ -2 & 1 \end{vmatrix} & + \begin{vmatrix} 2 & -1 \\ 3 & 1 \end{vmatrix} & - \begin{vmatrix} 2 & -5 \\ 3 & -2 \end{vmatrix} \\ + \begin{vmatrix} -5 & -1 \\ 3 & -2 \end{vmatrix} & - \begin{vmatrix} 2 & -1 \\ 1 & -2 \end{vmatrix} & + \begin{vmatrix} 2 & -5 \\ 1 & 3 \end{vmatrix} \end{pmatrix}^T$$

$$= \begin{pmatrix} -1 & -7 & -11 \\ 7 & 5 & -11 \\ 13 & 3 & 11 \end{pmatrix}^T = \begin{pmatrix} -1 & 7 & 13 \\ -7 & 5 & 3 \\ -11 & -11 & 11 \end{pmatrix}$$

$$\mathbf{A}^{-1}\mathbf{B} = \frac{1}{44}\begin{pmatrix} -1 & 7 & 13 \\ -7 & 5 & 3 \\ -11 & -11 & 11 \end{pmatrix}\begin{pmatrix} 4 \\ -8 \\ 8 \end{pmatrix} = \frac{1}{44}\begin{pmatrix} -4-56+104 \\ -28-40+24 \\ -44+88+88 \end{pmatrix} = \frac{1}{44}\begin{pmatrix} 44 \\ -44 \\ 132 \end{pmatrix}$$

$$= \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

$x = 1$, $y = -1$, $z = 3$.

5. The following system of equations were obtained for current flow in a d.c circuit using Kirchhoff's law:

$$\begin{aligned} 5I_1 - 3I_2 + 4I_3 &= 79 \\ 7I_1 + 4I_2 - 5I_3 &= 105 \\ 4I_1 - 5I_2 + 5I_3 &= 40 \end{aligned}$$

Use the method of inverse to find the values of I_1, I_2 and I_3 in Amperes.

Solution
Let

$$V = \begin{bmatrix} 5 & -3 & 4 \\ 7 & 4 & -5 \\ 4 & -5 & 5 \end{bmatrix}; B = \begin{bmatrix} 79 \\ 105 \\ 40 \end{bmatrix}; X = \begin{pmatrix} I_1 \\ I_2 \\ I_3 \end{pmatrix};$$

$$X = V^{-1}B; V^{-1} = \frac{\text{Adjoint of } V}{\det(V)}$$

$$\det(V) = 5(20-25) - -3(35+20) + 4(-35-16) = -64.$$

To obtain the adjoint of A:

$c_{11} = +(20-25) = -5;$ $c_{12} = -(35--20) = -55;$ $c_{13} = +(-35-16) = -51$

$c_{21} = -(-15+20) = -5;$ $c_{22} = +(25-16) = +9;$ $c_{23} = -(-25--12) = +13$

$c_{31} = +(15-16) = -1;$ $c_{32} = -(-25-28) = +53;$ $c_{33} = +(20+21) = 41$

$$\text{Adjoint of } V = \begin{pmatrix} -5 & -55 & -51 \\ -5 & 9 & 13 \\ -1 & 53 & 41 \end{pmatrix}^T = \begin{pmatrix} -5 & -5 & -1 \\ -55 & 9 & 53 \\ -51 & 13 & 41 \end{pmatrix};$$

$$V^{-1} = \begin{pmatrix} \frac{5}{64} & \frac{5}{64} & \frac{1}{64} \\ \frac{55}{64} & -\frac{9}{64} & -\frac{53}{64} \\ -\frac{51}{64} & \frac{13}{64} & \frac{41}{64} \end{pmatrix}$$

$$X = V^{-1}B = \begin{pmatrix} \frac{5}{64} & \frac{5}{64} & \frac{1}{64} \\ \frac{55}{64} & -\frac{9}{64} & -\frac{53}{64} \\ -\frac{51}{64} & \frac{13}{64} & \frac{41}{64} \end{pmatrix} \begin{pmatrix} 79 \\ 105 \\ 40 \end{pmatrix} = \begin{pmatrix} \frac{960}{64} \\ \frac{1280}{64} \\ \frac{1024}{64} \end{pmatrix} = \begin{pmatrix} 15 \\ 20 \\ 16 \end{pmatrix}$$

$\therefore I_1 = 15\text{Amps}, I_2 = 20\text{Amps}, I_3 = 16\text{Amps}.$

Crammer's Rule

Suppose that a system which has n linear equations in n unknown is given by

$$\mathbf{AX} = \mathbf{B}$$

$\mathbf{A} = [a_{ij}]$ is the square matrix of coefficients of $\mathbf{X}$ Let $\mathbf{A}_i, 1 \leq i \leq n$ represent the matrix obtained from $\mathbf{A}$ by exchanging the i column of $\mathbf{A}$ by the column vector of $\mathbf{B}$ Let $\Delta = \det(\mathbf{A})$

$N_1 = \det(\mathbf{A}_1)$, $N_2 = \det(\mathbf{A}_2)$,...,$N_n = \det(\mathbf{A}_n)$. There exists a solution for the system if and only if $\Delta \neq 0$. The solution is

$$x_1 = \frac{N_1}{\Delta}, \quad x_2 = \frac{N_2}{\Delta}, \ldots, x_n = \frac{N_n}{\Delta}.$$

This is known as the *Crammer's rule* for solving systems of linear equations

Example

1. Solve the system

$$\begin{aligned} x + y - 2z &= 5 \\ 2x - 2y + z &= 7 \\ x + 3y - z &= 10. \end{aligned}$$

Solution
First, calculate the determinant Δ,

$$\Delta = \begin{vmatrix} 1 & 1 & -2 \\ 2 & -2 & 1 \\ 1 & 3 & -1 \end{vmatrix} = -14.$$

Since $\Delta \neq 0$, the system has a unique solution.

$$N_x = \begin{vmatrix} 5 & 1 & -2 \\ 7 & -2 & 1 \\ 10 & 3 & -1 \end{vmatrix} = 5(-1) - 1(-17) - 2(41) = -70$$

$$N_y = \begin{vmatrix} 1 & 5 & -2 \\ 2 & 7 & 1 \\ 1 & 10 & -1 \end{vmatrix} = 1(-17) - 5(-3) - 2(13) = -28$$

$$N_z = \begin{vmatrix} 1 & 1 & 5 \\ 2 & -2 & 7 \\ 1 & 3 & 10 \end{vmatrix} = 1(-41) - 1(13) + 5(8) = -14.$$

The unique solution of the system is

$$x = \frac{N_x}{\Delta} = 5, \quad y = \frac{N_y}{\Delta} = 2, \quad z = \frac{N_z}{\Delta} = 1.$$

2. Solve the system

$$\begin{aligned} 2x - y + 2z &= 9 \\ x - 2y - z &= 4 \\ 3x + 4y + z &= 6. \end{aligned}$$

Solution
In matrix form

$$\begin{pmatrix} 2 & -1 & 2 \\ 1 & -2 & -1 \\ 3 & 4 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 9 \\ 4 \\ 6 \end{pmatrix}.$$

$$\Delta = \begin{vmatrix} 2 & -1 & 2 \\ 1 & -2 & -1 \\ 3 & 4 & 1 \end{vmatrix} = 2(2) + 1(4) + 2(10) = 28$$

The system has a unique solution since $\Delta \neq 0$.

$$N_x = \begin{vmatrix} 9 & -1 & 2 \\ 4 & -2 & -1 \\ 6 & 4 & 1 \end{vmatrix} = 9(2) + 1(10) + 2(28) = 84$$

$$N_y = \begin{vmatrix} 2 & 9 & 2 \\ 1 & 4 & -1 \\ 3 & 6 & 1 \end{vmatrix} = 2(10) - 9(4) + 2(-6) = -28$$

$$N_z = \begin{vmatrix} 2 & -1 & 9 \\ 1 & -2 & 4 \\ 3 & 4 & 6 \end{vmatrix} = 2(-28) + 1(-6) + 9(10) = 28.$$

The unique solution of the system is

$$x = \frac{N_x}{\Delta} = \frac{84}{28} = 3, \quad y = \frac{N_y}{\Delta} = \frac{-28}{28} = -1, \quad z = \frac{N_z}{\Delta} = \frac{28}{28} = 1.$$

3. Solve the system

$$\begin{aligned} 50x + 30y + 20z &= 190 \\ 15x + 25y + 15z &= 100 \\ 30x + 15y + 5z &= 90. \end{aligned}$$

Solution

In matrix form, we have

$$\begin{pmatrix} 50 & 30 & 20 \\ 15 & 25 & 15 \\ 30 & 15 & 5 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 190 \\ 100 \\ 90 \end{pmatrix}.$$

Since each of the rows has common factor, the system can be written as

$$\begin{pmatrix} 5 & 3 & 2 \\ 3 & 5 & 3 \\ 6 & 3 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 19 \\ 20 \\ 18 \end{pmatrix}.$$

$$\Delta = \begin{vmatrix} 5 & 3 & 2 \\ 3 & 5 & 3 \\ 6 & 3 & 1 \end{vmatrix} = 5\begin{vmatrix} 5 & 3 \\ 3 & 1 \end{vmatrix} - 3\begin{vmatrix} 3 & 3 \\ 6 & 1 \end{vmatrix} + 2\begin{vmatrix} 3 & 5 \\ 6 & 3 \end{vmatrix}$$

$$= 5(5-9) - 3(3-18) + 2(9-30) = -17.$$

$\therefore \Delta \neq 0$, the system has a unique solution.

$$N_x = \begin{vmatrix} 19 & 3 & 2 \\ 20 & 5 & 3 \\ 18 & 3 & 1 \end{vmatrix} = 19\begin{vmatrix} 5 & 3 \\ 3 & 1 \end{vmatrix} - 3\begin{vmatrix} 20 & 3 \\ 18 & 1 \end{vmatrix} + 2\begin{vmatrix} 20 & 5 \\ 18 & 3 \end{vmatrix}$$

$$= 19(5-9) - 3(20-54) + 2(60-90) = -76 + 102 - 60$$

$$= -34$$

$$x = \frac{N_x}{\Delta} = \frac{-34}{-17} = 2$$

$$N_y = \begin{vmatrix} 5 & 19 & 2 \\ 3 & 20 & 3 \\ 6 & 18 & 1 \end{vmatrix} = 5\begin{vmatrix} 20 & 3 \\ 18 & 1 \end{vmatrix} - 19\begin{vmatrix} 3 & 3 \\ 6 & 1 \end{vmatrix} + 2\begin{vmatrix} 3 & 20 \\ 6 & 18 \end{vmatrix}$$

$$= 5(20-54) - 19(3-18) + 2(54-120) = -170 + 285 - 132 = -17$$

$$y = \frac{N_y}{\Delta} = \frac{-17}{-17} = 1.$$

$$N_z = \begin{vmatrix} 5 & 3 & 19 \\ 3 & 5 & 20 \\ 6 & 3 & 18 \end{vmatrix} = 5\begin{vmatrix} 5 & 20 \\ 3 & 18 \end{vmatrix} - 3\begin{vmatrix} 3 & 20 \\ 6 & 18 \end{vmatrix} + 19\begin{vmatrix} 3 & 5 \\ 6 & 3 \end{vmatrix}$$

$$= 5(90-60) - 3(54-120) + 19(9-30) = 150 + 198 - 399 = -51$$

$$z = \frac{N_z}{\Delta} = \frac{-51}{-17} = 3.$$

Gauss Elimination method of solving Systems of Linear Equations

A linear system can be transform into another equivalent system by (i) interchanging two equations of the system (ii) using a nonzero number to multiply an equation and (iii) addition of a multiple of some other equation to one equation. Equivalent linear systems have the same solution.

Example

1. Solve the system

$$\begin{aligned} 3x_2 + 4x_3 &= 6 \\ 2x_1 - x_2 + x_3 &= 15 \\ x_1 + x_2 + 2x_3 &= 9. \end{aligned}$$

Solution

Rearrange equation 1 and equation 2 of the system

$$\begin{aligned} x_1 + x_2 + 2x_3 &= 9 \\ 2x_1 - x_2 + x_3 &= 15 \\ 3x_2 + 4x_3 &= 6. \end{aligned}$$

In matrix form, we have

$$\begin{pmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 0 & 3 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 9 \\ 15 \\ 6 \end{pmatrix}.$$

From the rearranged system, multiply equation 1 by -2 and add to equation 2:

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & -3 & -3 \\ 0 & 3 & 4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 9 \\ -3 \\ 6 \end{pmatrix}$$

which gives

$$\begin{aligned} x_1 + x_2 + 2x_3 &= 9 \\ -3x_2 - 3x_3 &= -3 \\ 3x_2 + 4x_3 &= 6 \end{aligned}$$

Multiply equation 2 of the last system by 1 and add to equation 3

$$\begin{pmatrix} 1 & 1 & 2 \\ 0 & -3 & -3 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 9 \\ -3 \\ 3 \end{pmatrix}.$$

We obtain

$$\begin{aligned} x_1 + x_2 + 2x_3 &= 9 \\ -3x_2 - 3x_3 &= -3 \\ x_3 &= 3. \end{aligned}$$

The solution is $x_1 = 5$, $x_2 = -2$ and $x_3 = 3$.
The four systems have the same solution since they are equivalent. The Gaussian elimination method is also called the *echelon method* or *triangularization*.

2. Solve the system

$$\begin{aligned} 2x - y + z &= 2 \\ x + 3y - z &= 6 \\ y - z &= 2. \end{aligned}$$

Solution
In matrix form

$$\begin{pmatrix} 2 & -1 & 1 \\ 1 & 3 & -1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 2 \\ 6 \\ 2 \end{pmatrix}.$$

Rearrange equation 1 and equation 2 of the system

$$\begin{aligned} x + 3y - z &= 6 \\ 2x - y + z &= 2 \\ y - z &= 2 \end{aligned} \qquad \begin{pmatrix} 1 & 3 & -1 \\ 2 & -1 & 1 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ 2 \\ 2 \end{pmatrix}.$$

Multiply equation 1 by -2 and add to equation 2,

$$\begin{aligned} x + 3y - z &= 6 \\ -7y + 3z &= -10 \\ y - z &= 2 \end{aligned} \qquad \begin{pmatrix} 1 & 3 & -1 \\ 0 & -7 & 3 \\ 0 & 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ -10 \\ 2 \end{pmatrix}.$$

Multiply equation 2 by $1/3$ and add to equation 3,

$$\begin{aligned} x + 3y - z &= 6 \\ -7y + 3z &= -10 \\ -4/3y &= -4/3 \end{aligned} \qquad \begin{pmatrix} 1 & 3 & -1 \\ 0 & -7 & 3 \\ 0 & -4/3 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 6 \\ -10 \\ -4/3 \end{pmatrix}.$$

$\therefore\ y = 1, z = -1, x = 2.$

(i) $\begin{aligned} 3x - y &= 1 \\ 4x - 3y &= -2 \end{aligned}$

(ii) $\begin{aligned} x + y &= \frac{3}{4} \\ x - y &= \frac{1}{4}. \end{aligned}$

(iii) $\begin{aligned} x + y &= 6 \\ 2x - 6y &= -8. \end{aligned}$

9. Solve the system using (i) inverse method (ii) Cramer's rule.

(a) $\begin{aligned} x + y + z &= 2 \\ 5x - 2y - 2z &= 3 \\ 3x - 5y + z &= -8. \end{aligned}$

(b) $\begin{aligned} 2x - y + z &= 2 \\ x + 3y - z &= 6 \\ x - y - 2z &= 3. \end{aligned}$

(c) $\begin{aligned} 2x - y + 2z &= -1 \\ x + 3y - z &= 5 \\ 3x + y - 3z &= 7. \end{aligned}$

(d) $\begin{aligned} 4x - y + z &= 10 \\ x - 3y - z &= 4 \\ 3x - 7y - 5z &= 8. \end{aligned}$

(e) $\begin{aligned} x + y - z &= -6 \\ 2x + 2y - 3z &= -17 \\ 5x + 3y + z &= -4. \end{aligned}$

(f) $\begin{aligned} 2x + y - z &= 1 \\ 3x + 2y + z &= 2 \\ x - y - 2z &= 5. \end{aligned}$

(g) $\begin{aligned} -x + 2y - 2z &= -1 \\ 3x - 2y + z &= 10 \\ x - 5y + 3z &= -7. \end{aligned}$

(h) $\begin{aligned} 2x - y + 2z &= -5 \\ 3x + 2y + z &= 6 \\ x + 3y - z &= 11. \end{aligned}$

(i) $\begin{aligned} x - 2y + z &= -3 \\ 2x - 3y + z &= -1 \\ 4x - 5y - z &= 9. \end{aligned}$

(j) $\begin{aligned} 3x + y - z &= 1 \\ x + 3y + 4z &= -9 \\ 2x - 4y - 3z &= -4. \end{aligned}$

(k) $\begin{aligned} x + y + z &= 1.5 \\ 2x - y + 2z &= 0.75 \\ x + 4y - 2z &= 3. \end{aligned}$

(l)
$$\begin{aligned} 6x - 12y + 3z &= 8 \\ x + y - z &= -2/3 \\ 2x + y + z &= 1 \end{aligned}$$

(m)
$$\begin{aligned} 3x - y + 3z &= 450 \\ 5x + 2y - 2z &= 800 \\ 2x + y - 2z &= 200 \end{aligned}$$

(n)
$$\begin{aligned} x + y - z &= 150 \\ 4x - 3y + z &= 700 \\ 3x - 4y + 2z &= 550. \end{aligned}$$

10 Use Gaussian elimination method to solve the system

(i)
$$\begin{aligned} 3x_2 + 2x_3 &= 8 \\ 2x_1 - x_2 - x_3 &= 1 \\ x_1 + 2x_2 + x_3 &= 6. \end{aligned}$$

(ii)
$$\begin{aligned} x - y + z &= 100 \\ 2x - 3y + 4z &= 200 \\ x + 2y - z &= 300. \end{aligned}$$

11 A car accelerating at uniform acceleration between two points has a velocity given by $v = u + at$, where u is the initial velocity, v is the final velocity, and t is the time taken to pass between the two points. At time $t = 1.5s$, $v = 55m/s$ and at time $t = 2.5s$, $v = 75m/s$. With the use of determinant, obtain the values of u and a.

12 The acceleration $\ddot{x}$, velocity $\dot{x}$ and distance x in a mechanical system are related by the simultaneous equations:

$$\begin{aligned} 4\ddot{x} - 10\dot{x} + 10x &= -6 \\ 1.5\ddot{x} + 0.5\dot{x} - 2x &= 8.25 \\ 0.5\ddot{x} - \dot{x} + 5x &= 6 \end{aligned}$$

(i.) Represent the information in the matrix form $AX = B$ and find the inverse of A.
(ii.) Find the values of $\ddot{x}$, $\dot{x}$ and x.

13. Three different bakeries A, B, and C require flour (F), margarine (M) and sugar (S) to bake certain snack each for an event. Bakery A requires 2.5kg of flour, 1.5kg of margarine and 0.5kg of sugar. Bakery B requires 3.5kg of flour, 2.5kg of margarine and 1.5kg of sugar. Bakery C requires 4.5kg of flour, 1.5kg of margarine and 2.5kg of sugar. If the total costs of the ingredients for each of the bakeries are \$160, \$240 and \$280 respectively, find the cost of each of the ingredient.

14. A newspaper vendor supplies Daily Sun (S), Vanguard (V) and Daily Times (T) to three different organisations owned by Mr. Andy, Mr. Greg and Mr. Fred respectively. In each week, Mr. Andy buys 3 Daily Sun, 4 Vanguard and 2 Daily Times, Mr. Greg buys 4 Daily Sun, 2 Vanguard and 4 Daily Times, Mr. Fred buys 2 Daily Sun, 4 Vanguard and 4 Daily Times. If they pay N1200, N1400, and N1340 respectively for each week.
(i) represent the information in matrix form $AX = B$.
(ii) find the determinant and the inverse of the matrix A.
(iii) how much does each newspaper cost?

15. Mrs Ayo owns three different shops for the sale of the following brands of detergents Omo (M), Klin (K) and Zip (Z). Last week, she makes the income of $142, $126 and $145 respectively in the three shops. If the sales are 30 packs of Omo, 40 packs of Klin and 20 packs of Zip for the first shop, 25 packs of Omo, 30 packs of Klin and 15 packs of Zip for the second shop and 40 packs of Omo, 10 packs of Klin and 25 packs of Zip for the third shop.
(i) Interpret the information in matrix form.
(ii) Obtain the price of each brand of detergent.

16. The following sectors: Palm oil production (P), Soap production (S) and Cream production (C) require certain units of items from themselves in order to ensure smooth running of their business. Palm oil sector requires 0.5 of its own item, 1.2 from Soap sector and 0.2 from Cream sector. Soap sector requires 1.5 of its own item, 3.5 from Palm oil sector and 0.5 from Cream sector. Cream sector requires 0.5 of its own item, 4.0 from Palm oil sector and 1.2 of Cream. If the total demand for each is given respectively as 69, 200 and 193.
(i) Obtain the matrix interpretation of the information.
(ii) Find the demand for each of the sector.

17. The initial velocity u, final velocity v and distance S of a system is given by the simultaneous equation

$$\begin{aligned} 0.8u - 1.2v + 2.4S &= 195.84 \\ 0.5u - 0.75v + 0.1S &= -25.44 \\ 3.5u - 0.5v + 10S &= 1108 \end{aligned}$$

(i.) Represent the information in the matrix form $AX = b$.
(ii.) Find the inverse of A.
(ii.) Use the inversion method to find the values of u, v and S.

18. The systems of forces acting on a body in equilibrium led to the systems of linear equations:

$$\begin{aligned} F_1 - F_2 &= 7 \\ F_1 + 2F_2 - F_3 &= 23 \\ F_2 - F_3 &= 0 \end{aligned}$$

Use Gaussian elimination method to find the three forces F1, F2, F3 (Newtons).

19. A d.c. circuit is composed of three closed loops in which Kirchhoff's laws gives the equations for current I in the closed loops as

$$\begin{aligned} 100I_1 + 5I_2 - 10I_3 &= 931 \\ 250I_1 - 15I_2 - 20I_3 &= 262 \\ 300I_1 - 50I_2 - 10I_3 &= 745 \end{aligned}$$

If the currents are measured in milliamperes, use inversion method to find I_1, I_2 and I_3.

20. A bricklayer intends to mould blocks and has planned for certain bags of cement for each heap of sand. If he buys 2 heaps of sand and 6 bags of cement, he pays ₦15,200.00. If he decides to buy 3 heaps of sand and 9 bags of cement, he pays ₦22,800.00. Use matrix inversion to determine the cost of each heap of sand and the cost of each bag of cement.

Chapter 8

COMPLEX NUMBERS

8.1 Algebra of Complex Numbers

Addition and Subtraction of Complex Number
If $z^1 = a + ib$ and $z^2 = c + id$ $z^1 + z^2 = (a + ib) + (c + id)$ $= (a + c) + i(b + d)$
Similarly,
$z_1 - z_2 = (a + ib) - (c + id) = (a - c) + i(b - d)$

Example 1: Simplify: (i) $(6 + 3i) + (7 - i)$ (ii) $(7 + 2i) + (3 - 4i) - (5 + i)$
Solution
(i) $(6 + 3i) + (7 - i) = (6 + 7) + i(3 - 1) = 13 + 2i$
(ii) $(7 + 2i) + (3 - 4i) - (5 + i) = (10 - 2i) - (5 + i) = 5 - 3i$

Multiplication of Complex Numbers
If $z_1 = a + ib$ and $z_2 = c + id$, then $z_1.z_2 = (a + ib)(c + id) = ac + iad + ibc - bd$
$= (ac - bd) + i(ad + bc)$

Example 2: Express the following in the form $a + ib$.

(i)$(3 + i)(4 - 21)$ (ii) $2i(1 - 3i) + 3(4 - i)$

Solution: (i) $(3 + i)(4 - 2i) = 12 - 6i + 4i + 2 = 14 - 2i$

(ii) $2i(1 - 3i) + 3(4 - i) = 2i + 6 + 12 - 3i = 18 - i$

Division of Complex Numbers
Let $z_1 = (a + ib)$, $z_2 = (c + id)$

$$\frac{z_1}{z_2} = \frac{z_1 \times z_2}{z_1 \times z_2} = \frac{(a+ib)(c-id)}{(c+id)(c-id)} = \frac{(ac+bd)+(bc-ad)i}{(c^2+d^2)}$$

Example 3: Evaluate the following (i) $\frac{2+i}{3-i}$ (ii) $\frac{3-4i}{5+2i}$
Solution
(i) $\frac{2+i}{3-i} = \frac{(2+i)(3+i)}{(3-i)(3+i)} = \frac{6+2i+3i-1}{9+1} = \frac{5+5i}{10} = \frac{1}{2} + \frac{1}{2}i$

(ii) $\frac{3-4i}{5+2i} = \frac{(3-4i)(5-2i)}{(5+2i)(5-2i)} = \frac{15-8+i(-20-6)}{25+4} = \frac{7}{29} - \frac{26i}{29}$

Modulus of Complex Numbers
The modulus of a complex number $a + ib$ denoted by $|a + ib|$ is defined as

$$|a + ib| = a^2 + b^2$$

Example 4: $|3 - 4i| = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$

Example 5: Simplify $\frac{5+5i}{3-4i} + \frac{20}{4i+3i}$
Solution

$$\begin{aligned}
\frac{(5+5i)(3+4i)}{(3-4i)(3+4i)} + \frac{20(4-3i)}{(4i+3i)(4i-3i)} &= \frac{15+20i+15i-20}{9+16} + \frac{80-60i}{16+9} \\
&= \frac{-5+35i}{25} + \frac{80-60i}{16+9} \\
&= \frac{-5+35i}{25} + \frac{80-60i}{25} \\
&= \frac{-5+80+i(35-60)}{25} \\
&= \frac{75-25i}{25} \\
&= 3-i
\end{aligned}$$

8.2 Argard Diagram

This is the method of representing a complex number graphically.
Example 1: Find the modulus and the arguments of the complex number $\frac{(1+i)(2+i)}{3-i}$

Solution

$$\begin{aligned}
\frac{(1+i)(2+i)}{3-i} &= \frac{2+i+2i-1}{3-i} \\
&= \frac{1+3i}{3-i} \\
&= \frac{(1+3i)(3+i)}{(3-i)(3+i)} \\
&= \frac{2+i+2i-1}{3-i} = \frac{3i+1}{3-i} \\
&= \frac{(3i+1)(3+i)}{(3-i)(3+i)} = \frac{2+i+2i-1}{3-i} \\
&= \frac{10i}{10} = i
\end{aligned}$$

$$r = \sqrt{0^2 + 1^2} = 1$$

$$\theta = tan^{-1}\frac{b}{a} = tan^{-1}\frac{1}{0} = tan^{-1}\infty$$
$$= \frac{\pi}{2} \text{ or } 90^o$$

$\therefore$ the modulus is 1 and the argument is $\frac{\pi}{2}$.

Example 2: Express $2 + 2\sqrt{3}i$ in polar form

$$R = \sqrt{2^2 + (2\sqrt{3})^2} = \sqrt{16} = 4$$
$$\theta = tan^{-1}\frac{2\sqrt{3}}{2} = tan^{-1}\sqrt{3} = 60^o = \frac{\pi}{3}rad$$
$$z = r(cos\theta + sin\theta)$$
$$Z = 4(cos\frac{\pi}{3} + sin\frac{\pi}{3}) = 4(cos60^o + isin60^o)$$

De-Moivre's Theorem

$$e^{i\theta} = cos\theta + isin\theta$$
$$re^{i\theta} = r(cos\theta + isin\theta)$$

The above is known as Euler's formular
Let

$$Z_1 = a_1 + ib_1 = r_1(cos\theta_1 + isin\theta_1)$$
$$Z_2 = a_2 + ib_2 = r_2(cos\theta_2 + isin\theta_2)$$

Then

$$Z_1Z_2 = (a_1 + ib_1)(a_2 + ib_2) = r_1r_2(cos(\theta_1 + \theta_2) + i(sin\theta_1 + \theta_2) = r_1r_2(cos\theta + isin\theta)$$

$$\frac{Z_1}{Z_2} = \frac{r_1}{r_2}(cos(\theta_1 - \theta_2) + isin(\theta_1 - \theta_2)$$

In general. $Z_1Z_2...Z_n = r_1r_2...r_n(cos(\theta_1 + \theta_2).... + \theta_3 + isin(\theta_1 + \theta_2) + ... + sin)$
If $Z_1 = Z_2...Z_n$ then

$$Z_1Z_2...Z_n = Z_n = n(cosn\theta + isinn\theta)$$

The above equation is known as De-Mover's Theorem. Using Euler's formula, De-Mover's theorem reduces to $(re^{i\theta})^n = r^ne^{in\theta}$.

Example 3: Prove the following identities.

1. $cos5\theta = 16cos^5 - 2\theta cos^3\theta + 5cos\theta$
2. $sin3\theta = \frac{3}{4}sin\theta - \frac{1}{4}sin3\theta \Longrightarrow sin^3\theta = \frac{3}{4}sin^o - \frac{1}{4}sin^3\theta$

Solution

1. By De-Moivre's Theorem

$$(cos\theta + isin\theta)^5 = cos5\theta + isin5\theta \quad (1)$$

$= cos^5\theta + 5icos^4sin\theta - 10cos^3sin^2\theta - 10icos^2\theta sin^3\theta + 5cos\theta sin^4\theta + sin5\theta$ By Binomi

Roots of Complex Number and n^{th} Root of Unity

A number w is called the nth root of a complex number Z if $W^n = Z$ and we write $W = Z^{\frac{1}{n}}$
By De-Moivre's Theorem

$$Z = r(cos\theta + isin\theta)$$

$$W = Z^{\frac{1}{n}} = r(cos\theta + isin\theta)^{\frac{1}{n}}$$

$$= r^{\frac{1}{n}}(cos\frac{\theta}{n} + isin\frac{\theta}{n})$$

$$Z^n = r^n(cos\theta + isin\theta)^n = r^n(cosn\theta + isinn\theta)$$

$$Z^n = r^n(cos\theta + 2k\pi)^n = r^n(cosn\theta + isinn\theta)$$

$$Z^{\frac{1}{n}} = r^{\frac{1}{n}}[(cos(\frac{\theta + 2k\pi}{n}) + isin(\frac{\theta + 2k\pi}{n})$$

$k = 0, 1, 2, 3.....n - 1)$

Example 1: Find all values of Z for which $Z_5 = 32$

Solution

$$Z^{\frac{1}{n}} = r^{\frac{1}{n}}[(cos(\frac{\theta + 2k\pi}{n}) + isin(\frac{\theta + 2k\pi}{n})$$

$Z_5 = 32 \Rightarrow n = 5,$

$$Z_5 = 32^{\frac{1}{5}}[cos(\frac{\theta + 2k\pi}{5}) + isin\frac{\theta + 2k\pi)}{5})$$

$$2^{\frac{1}{5}}[cos(\frac{\theta + 2k\pi}{5}) + isin\frac{\theta + 2k\pi)}{5})$$

For

$$K = 0, \quad Z_1 = 2(cos\frac{\pi}{5} + isin\frac{\pi}{5}) = 2e^{\frac{i\pi}{5}}$$

$$K = 1, \quad Z_2 = 2(cos\frac{1}{3}\frac{\pi}{5} + \frac{3\pi}{5}) = 2e^{\frac{3i\pi}{5}}$$

$$K = 2, \quad Z_3 = 2(cos\pi_5 + isin\frac{5\pi}{5}) = -2$$

$$K = 3, \quad Z_4 = 2(cos\frac{7}{3\pi} + isin\frac{7\pi}{5}) = 2e^{\frac{7}{3i\pi}}$$

$$K = 4, \quad Z_5 = 2(cos\frac{9\pi}{5} + isin\frac{9\pi}{5}) = 2e^{\frac{9}{3i\pi}}$$

nth Roots of Units

Given that $Z^n = 1$, we seek z such that $Z^n = 1$, the solution of this equation is called the nth roots of unit. Following the same principle of the roots of complex number we see that

$$1 = cos0^o + isin0^o$$

$$1 = cos2\pi k + isin2k\pi$$

Hence, $Z^n = 1cos2\pi k + isin2k\pi$

$$\Rightarrow Z(cos2\pi k + isin2k\pi)^{\frac{1}{n}} k = 0, 1, 2...n - 1..... = 1.$$

so if $w = cos2\pi + isin2\pi = e^{\frac{i2\pi}{n}}$ then the nth roots of unity can be listed as $1, w, w^2, ...w^{n-1}$

Example 2: Find all the fourth roots of unity.

Solution

$$Z^4 = 1cos^2k\pi + isin^2k\pi$$
$$Z = cos\frac{2k\pi}{4} + isin\frac{2k\pi}{4}$$
$$cos\frac{k\pi}{2} + isin\frac{k\pi}{2}$$

for $k = 0$ $Z_1 = 1$
for $k = 1$ $Z_2 = cos\frac{\pi}{2} + isin\frac{\pi}{2} = i$
for $k = 2$ $Z_3 = cos\pi + isin\pi = -1$
for $k = 3$ $Z_4 = cos^3\frac{k\pi}{2} + isin\frac{3\pi}{2} = -i$

Exercise

1. If a and b are real, express each of the following in the form $a + ib$
   (a) $2(3 - 2i) + i(3 + i)$ (b) $(2 + 3i) - i(5 - i)$ (c) $i^5(2 + i)$ (d) $(2i - 1)^2 + (4 + 2 - i)$
   (e) $3i^{30} - i^{19}$ (f) $\frac{(4i + 3)(2 + 5i)}{(4 + 3i)(3i - 2)}$ (g) $(\sqrt{2} + 3i)^2$

2. If $z_1 = 2 + i, z_2 = 3 + 2i, z_3 = -\frac{1}{2} + \frac{\sqrt{3}}{2}$, find
   $2z_2 + z_1 - 5i^2$

3. Represent the complex numbers (i) $4 - i$ (ii) $-3 - 2i$ by points on the complex plane (Argand Diagram).

4. Find the modulus and argument of each of the following:
   (i) $\frac{(3-i)(2-i)}{4+i}$ (ii) $-3 - 6i$.

5. If $z = 2 - 3i$, mark the points (i) z (ii) iz (iii) i^3z in the complex plane.

6. Prove the following identities:
   (a) $sin3\theta = 3sin\theta - 4sin^3\theta$
   (b) $cos4\theta = 8sin^4\theta - 8sin^2\theta + 1$
   (c) $\frac{sin5\theta}{sin\theta} = 16cos^4\theta - 12cos^2\theta + 1.$

7. If $z_1 = 1 - i, z_2 = -2 + 4i, z_3 = 3 - 2i$, evaluate each of the following:
   (i) $12z_2 - 3z_1i^2$ (ii) $Im\left(z_1\frac{z_2}{z_3}\right)$.

8. Find each of the indicated roots
   (i) $(-1 + i)^{\frac{1}{3}}$ (ii) $(-23 - 2i)^{\frac{1}{4}}$ (iii) $(243)^{\frac{1}{4}}$

9. Find all the fifth roots of -1.

10. If $z = x = iy$, find the real and imaginary parts of (i) z^3 (ii) $\frac{1}{z^3}$.

11. Express the following in polar form
(i) $-2\sqrt{3} - 2i$ (ii) $3\frac{\sqrt{2}}{2} - 1$

12. If $(x + iy)^2 = a + ib^2$ show that $x^2 - y^2 = a$. $2xy = b$. hence evaluate $\sqrt{8 + ai\sqrt{8 + bi}}$

Chapter 9

CIRCLE MEASURE

A radius of a given circle measures the distance from the center of the circle to its edge. The diameter is a measure of the distance from one edge of the circle through its center to the other edge. The circumference of a given circle measures the total distance around the outside of the circle. Measuring of angle is a way of measuring lengths around the perimeter of a given circle. This leads to the use of radians as unit of measuring angles. An angle gives the measurement of the amount of rotation between two straight lines. Clockwise direction gives negative angle measure while anti-clockwise direction gives positive angle measure.

A unit circle refers to a circle with centre 0 and radius 1. A full rotation around such circle leads to a distance equal to the length of circumference, 2π, of the circle. This implies that a full turn measures 2π radians. The equation of a unit circle in the coordinate plane is $x^2 + y^2 = 1$. It can be divided into four quadrants. e, the unit circle, into 4 quadrants. The first quadrant is given in terms of coordinates as $x > 0$, $y > 0$, or, in terms of angle in degree, as $0^0 < \theta < 90^0$, or in radians as $0 < \theta < \pi/2$. The second quadrant is given in terms of coordinates as $x < 0$, $y > 0$, or, in terms of angle in degree, as $90^0 < \theta < 180^0$, or in radians as $\pi/2 < \theta < \pi$. The third quadrant is given in terms of coordinates as $x < 0$, $y < 0$, or, in terms of angle in degree, as $180^0 < \theta < 270^0$, or in radians as $\pi < \theta < 3\pi/2$. The fourth quadrant is given in terms of coordinates as $x > 0$, $y < 0$, or, in terms of angle in degree, as $270^0 < \theta < 360^0$, or in radians as $3\pi/2 < \theta < 2\pi$.

Let l be the length of arc of a circle and let θ be its central angle (measured in radians). A radian is the measure of the angle θ which intercepts the arc l equal in length to the radius r of the circle. This is written as

$$\theta = \frac{l}{r}.$$

A full turn (360^0) is equivalent to 2π radians. To convert from degrees to radians, we have

$$\text{radians} = \frac{\pi \times \text{degrees}}{180}.$$

To convert from radians to degrees, we have

$$\text{degrees} = \frac{180 \times \text{radians}}{\pi}$$

The area of a sector of a given circle is given as

$$A = \frac{1}{2}r^2\theta \text{ where } \theta \text{ is in radians}$$

The area of a segment is given as

$$\begin{aligned}\text{Area of segments} &= \text{Area of Sector} - \text{Area of triangles}\\ &= \frac{1}{2}r^2 \times (\theta \text{ radians}) - \frac{1}{2}r^2 \sin\theta^0\\ &= \frac{1}{2}r^2(\theta \text{ radians} - \sin\theta^0)\end{aligned}$$

Linear speed is given as

$$v = \frac{\text{arclength}}{\text{time}} = \frac{l}{t}.$$

Angular speed is given as

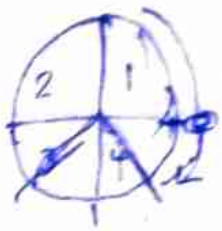
$$w = \frac{\text{central angle}}{\text{time}} = \frac{\theta}{t}$$

where θ is measured in radians.

Let α and β be two positive angles. Then α and β are said to be *complements* of each other if their sum is $\pi/2$. They are said to be *supplements* of each other if their sum is π.

Examples/Applications

1. Convert 135^0 to radians.



Solution

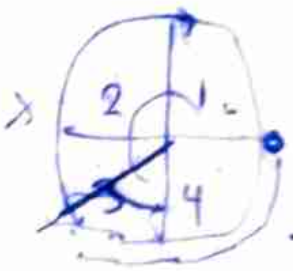
$$135^0 \times \frac{\pi \text{radians}}{180^0} = \frac{3}{4}\pi \text{ radians}.$$

2. Convert 30^0 to radians.

Solution

$$30^0 \times \frac{\pi \text{radians}}{180^0} = \frac{1}{6}\pi \text{ radians}.$$

3. Convert 45^0 to radians.



Solution

$$45^0 \times \frac{\pi \text{radians}}{180^0} = \frac{1}{4}\pi \text{ radians}.$$

4. Convert 120^0 to radians.

Solution

$$120^0 \times \frac{\pi \text{radians}}{180^0} = \frac{2}{3}\pi \text{ radians}.$$

5. Convert $\frac{\pi}{3}$ radians to degrees.

Solution

$$\frac{\pi}{3} \times \frac{180}{\pi} = 60^0$$

6. Convert $\frac{\pi}{4}$ radians to degrees.

Solution

$$\frac{\pi}{4} \times \frac{180}{\pi} = 45^0.$$

7. Convert $\frac{\pi}{180}$ radians to degrees.

Solution

$$\frac{\pi}{180} \times \frac{180}{\pi} = 1^0.$$

8. **A circle has a radius of 6 inches. Find the length of the arc intercepted by a angle of 150^0.**

Solution

Convert $150°$ to radian measure;

$$150° \times \frac{\pi \text{radians}}{180°} = \frac{5}{6}\pi \text{radians}.$$

The length of arc l is given as

$$\begin{aligned} l &= r\theta = 6\left(\tfrac{5\pi}{6}\right) \\ &= 5\pi \text{ inches} \end{aligned}$$

9. The length of the arc of a circle intercepted by a central angle 240^0 is $\frac{16\pi}{3}$cm. F radius of the circle.

Solution

$l = \theta r \Rightarrow r = \frac{l}{\theta}$

$$\theta = 240° \times \frac{\pi}{180^0} = \frac{4\pi}{3} \text{ radians}$$

$$\therefore r = \left(\frac{16\pi}{3} \div \frac{4\pi}{3}\right) = 4 \text{ cm}.$$

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Exercises

1. Express the following angles in radians. leaving your answers in terms of π:

   (a) $125°$ (b) $85°$
   (c) $75°$ (d) $155°$
   (e) $100°$ (f) $50°$
   (g) $25°$ (h) $105°$
   (i) $125°$ (j) $85°$
   (k) $750°$ (l) $930°$.

2. Express the following angles in degrees:

   (a) $\frac{2\pi}{5}$ (b) $\frac{5\pi}{7}$
   (c) $\frac{\pi}{9}$ (d) $\frac{2\pi}{3}$
   (e) $\frac{pi}{5}$ (f) $\frac{\pi}{2}$
   (g) 1.5π (h) 0.25π
   (i) $\frac{3\pi}{5}$ (j) $\frac{4\pi}{9}$.

3. The length of the arc of a circle intercepted by a central angle 210^0 is $\frac{7\pi}{3}$ inches. Find the radius of the circle.